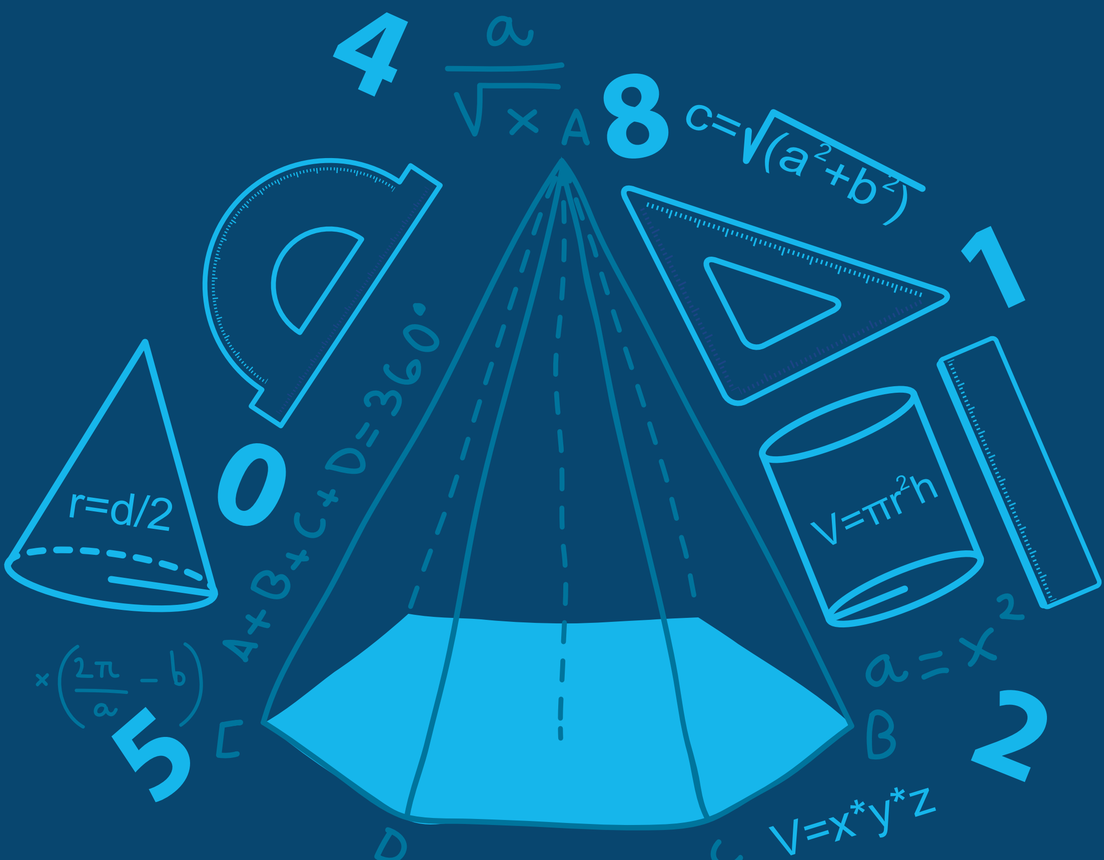
MATHEMATICS

IIT-JEE Advanced Revision Package



This study package is prepared by:

कृति
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PART # 01

ALGEBRA

EXERCISE # 01

SECTION-1 : (ONE OPTION CORRECT TYPE)

1. Number of solutions of $|z-1| + |z+1| = 4$ and $|2z-1+i| = \sqrt{14}$ is :

(A) 2 (B) 3 (C) 4 (D) none of these

2. Let X be the set of three digit numbers, which when divided by its sum of its digits give maximum value and Y be the set of all possible real values of a for which the $x^3 - 3ax^2 + 3(298a + 299)x - 2 = 0$ have a positive point of maxima, then the number of elements in $X \cap Y$, is :

(A) 0 (B) 6 (C) 7 (D) 9

3. Let $f(x) = x^2 - bx + c$, b is a odd positive integer, $f(x) = 0$ have two prime numbers as roots and $b + c = 35$. Then the global minimum value of f(x) is

(A) $-\dfrac{183}{4}$ (B) $\dfrac{173}{16}$

(C) $-\dfrac{81}{4}$ (D) data not sufficient

4. If z_1, z_2, z_3, z_4 are the reflections of the complex number z, with respect to the origin, real axis and imaginary axis respectively in an argand plane, then the value of $\arg(z_1^4 \cdot z_2^5 \cdot z_3^4 \cdot z_4^5 + 5)$ is

(A) $\dfrac{\pi}{3}$ (B) $\dfrac{\pi}{2}$ (C) $\dfrac{2\pi}{3}$ (D) none of these

5. If $a = e^{1/e}$ then the number of point of intersection of the curve $y = \log_a x$ and the line $y = x$, is

(A) three (B) zero (C) one (D) two

6. The coefficient of $a^8b^4c^9d^9$ in $(abc + abd + acd + bcd)^{10}$ is

(A) 10! (B) $\dfrac{10!}{8!4!9!9!}$

(C) 2520 (D) None of these.

7. Sum of all divisors of 5400 whose units digit is 0 is

(A) 5400 (B) 10800

(C) 16800 (D) None of these.

8. Sequence $\{t_n\}$ is a G.P. If t_6, 2, 5, t_{14} form another G.P in that order, then $t_1 . t_2 . t_3 ..... t_{19}$ is equal to

(A) 190 (B) 10^{10} (C) $10^{\frac{19}{2}}$ (D) 10^9

9. If $|\beta_k| < 3$, $1 \le k \le n$, then all the complex numbers z satisfying the equation
$1 + \beta_1 z + \beta_2 z^2 + \ldots + \beta_n z^n = 0$, $|z| < 1$

(A) lie inside the circle $|z| = \frac{1}{4}$ (B) lie in $\frac{1}{3} < |z| < \frac{1}{2}$

(C) lie on the circle $|z| = \frac{1}{4}$ (D) lie outside the circle $|z| = \frac{1}{4}$

10. The sum of the series ${}^{4n}C_0 + {}^{4n}C_4 + {}^{4n}C_8 + \ldots + {}^{4n}C_{4n}$ is

(A) $2^{4n-2} + (-1)^n 2^{2n-1}$ (B) $2^{4n-2} + (-1)^{n+1} 2^{2n-1}$

(C) $2^{4n-2} - 2^{2n-1}$ (D) $2^{4n-2} + 2^{2n-1}$

11. In a shooting competition, a man can score 5, 4, 3, 2 or 0 points for each shot. In how many ways he can achieve a score of 30 in just 7 shots

(A) 455 (B) 460 (C) 420 (D) 495

12. The product $\left(\frac{2^3-1}{2^3+1}\right)\left(\frac{3^3-1}{3^3+1}\right)\left(\frac{4^3-1}{4^3+1}\right)$ … (to infinity) is equal to

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

13. If $\alpha, \alpha^2, \ldots, \alpha^{n-1}$ be the n^{th} roots of unity, then $\left(\frac{3^n-1}{3^{n-1}}\right)\left(\sum_{r=1}^{n-1}\frac{1}{3-\alpha^r}+\frac{1}{2}\right) =$

(A) $-n$ (B) 0 (C) n (D) 1

14. If $\text{Re}\left(\frac{z-8i}{z+6}\right) = 0$, then $z = x + iy$ lies on the curve

(A) $x^2 + y^2 + 6x - 8y = 0$ (B) $4x - 3y + 24 = 0$

(C) $x^2 + y^2 - 8 = 0$ (D) none of these

15. The set of values of 'a' for which $x^3 + ax^2 + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$ has atleast one solution is

(A) $\frac{\pi}{8}+2$ (B) $\frac{\pi}{8}+1$ (C) $-\left(\frac{\pi}{8}+1\right)$ (D) $-\left(\frac{\pi}{8}+2\right)$

16. Let $a_1 = 1$, $a_n = n(a_{n-1} + 1)$ for $n = 2, 3, \ldots$

define $P_n = \left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)\ldots\left(1+\frac{1}{a_n}\right)$. Then $\lim_{n\to\infty} P_n$ must be

(A) $1 + e$ (B) e (C) 1 (D) ∞

17. If n is a positive integer then $\sum_{k=1}^{n} k^3\left(\frac{{}^nC_k}{{}^nC_k - 1}\right)^2$ equals

(A) $\frac{n}{12}(n+1)^2(n+2)$ (B) $\frac{n}{12}(n+1)(n+2)^2$

(C) $\frac{n}{12}(n+1)(n+2)$ (D) none of these

18. If $|z - 25i| \le 15$, then maximum of arg(z) – minimum of arg(z) equals

(A) $2\cos^{-1}\left(\frac{3}{5}\right)$ (B) $2\cos^{-1}\left(\frac{4}{5}\right)$ (C) $\frac{\pi}{2}+\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\sin^{-1}\left(\frac{3}{5}\right)-\cos^{-1}\left(\frac{3}{5}\right)$

19. The least value of the expression $x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$ is

(A) 0 (B) 1
(C) no least value (D) none of these

20. The largest term of the sequence $a_n = \dfrac{n}{n^2 + 10}$ is

(A) $\dfrac{3}{19}$ (B) $\dfrac{2}{13}$ (C) 1 (D) $\dfrac{1}{7}$

21. The number of positive integral solution of $x^2 + 9 < (x + 3)^2 < 8x + 25$ is

(A) 2 (B) 3 (C) 4 (D) 5

22. $n.{}^{n-2}C_{r-1} = r(k^2 - 3)\,{}^{n}C_{r-1} + {}^{n-2}C_{r-1}$, then the value of k is

(A) $(-\infty, -2]$ (B) $\left(-\infty, -\sqrt{3}\right) \cup \left(\sqrt{3}, 2\right]$
(C) $\left[-2, -\sqrt{3}\right) \cup \left(\sqrt{3}, 2\right]$ (D) $\left[-2, -\sqrt{3}\right) \cup \left(\sqrt{3}, \infty\right]$

23. If the ratio of the squares of the roots of the equation $x^2 + px + q = 0$ be equal to the ratio of the roots of the equation $x^2 + lx + m = 0$, then

(A) $q^2m^2 = (p^2 - 2q)^2 l$ (B) $(p^2 - 2q)^2 m = q^2l^2$
(C) $q^2p^2 = (l^2 - 2q)^2 q$ (D) none of these

24. The equation $|z - 2i| + |z + 2i| = k$, $k > 0$, can represent on ellipse if k is

(A) 2 (B) 3
(C) 4 (D) 5

25. If n boys and n girls sit along a line alternately in x ways and along a circle in y ways such that x = 12 y then the number of ways in which n boys can sit at a round table so that all shall not have same neighbors is

(A) 6 (B) 120
(C) 60 (D) 12

26. If [.] denotes the greatest integer function, then the domain of the real valued function $\log_{[x+1/2]} |x^2 = x - 6|$ is

(A) $\left(\dfrac{1}{2}, 1\right] \cup (1, \infty)$ (B) $\left[\dfrac{3}{2}, 2\right) \cup (2, +\infty)$
(C) $\left(0, \dfrac{3}{2}\right] \cup (2, +\infty)$ (D) $(0, 1] \cup \left(\dfrac{3}{2}, +\infty\right)$

27. Equation $\dfrac{a^2}{x-\alpha} + \dfrac{b^2}{x-\beta} + \dfrac{c^2}{x-\gamma} = m - n^2x$ (a, b, c, m, n $\in$ R) has necessarily

(A) all the roots real (B) all the roots imaginary
(C) 2 real and 2 imaginary (D) 2 rational and 2 irrational

28. The number of non-integral solutions of $\left|\,|4x - x^2| - 1\right| = 3$ is

(A) four (B) two (C) three (D) none of these

29. The coefficient of x^n in the polynomial $\left(x + {}^{2n+1}C_0\right)\left(x + {}^{2n+1}C_1\right)\left(x + {}^{2n+1}C_2\right)....\left(x + {}^{2n+1}C_n\right)$ is

(A) 2^{n+1} (B) $2^{2n+1} - 1$
(C) 2^{2n} (D) None of these

30. The inequality $(x - 3m)(x - m - 3) < 0$ is satisfied for x in [1, 3]. Determine the value of m for which this holds

(A) (1, 2) (B) (0, 1/3) (C) (1, 3) (D) none of these

31. If the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ has equal roots where a, b, c are distinct positive numbers and $n \in N$, then

(A) $a^n + c^n \geq 2b^n$ (B) $a^n + c^n > 2b^n$ (C) $a^n + c^n \leq 2b^n$ (D) $a^n + c^n < 2b^n$

32. If z_1 and z_2 are two complex numbers such that $|z_1| = |z_2| + |z_1 - z_2|$, then

(A) $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$ (B) $\text{Re}\left(\frac{z_1}{z_2}\right) = 0$

(C) $\text{Re}\left(\frac{z_1}{z_2}\right) = \text{Im}\left(\frac{z_1}{z_2}\right)$ (D) none of these

33. If α, β are roots of $2x + \frac{1}{x} = 2$ and $f(x) = \frac{8x^6 + 1}{x^3}$, then

(A) $f(\alpha) - f(\beta) = 0$ (B) $f(\alpha) + f(\beta) = 8$ (C) $f(\alpha) - f(\beta) = 2$ (D) $f(\alpha) + f(\beta) = 12$

34. If f(x) = 0 is a cubic equation with positive and distinct roots α, β, γ such that β is the H.M of the roots of $f'(x) = 0$. Then α, β, γ are in

(A) A.P. (B) G.P.

(C) H.P. (D) none of these

35. The number of ways in which the squares of a 8×8 chess board can be painted red or blue so that each 2×2 square has two red and two blue squares is

(A) 2^9 (B) $2^9 - 1$

(C) $2^9 - 2$ (D) none of these

36. All the roots of the equation $11z^{10} + 10iz^9 + 10iz - 11 = 0$ lie

(A) inside $|z| = 1$ (B) on $|z| = 1$

(C) outside $|z| = 1$ (D) can't say

37. Let f(x), g(x) and h(x) be quadratic polynomials having positive leading co-efficients and real and distinct roots. If each pair of them has a common root, then the roots of f(x) + g(x) + h(x) = 0 are

(A) always real and distinct (B) always real and may be equal

(C) may be imaginary (D) always imaginary

38. If x is positive and x – [x], [x] and x are in G.P., then {x} is equal to, (where [.] denotes the greatest integer function and {.} denotes the fractional part of x)

(A) $\frac{\sqrt{5} - 1}{2}$ (B) $\frac{\sqrt{5} + 1}{4}$ (C) $\frac{\sqrt{5} - 1}{4}$ (D) none of these

39. The value of $\sum_{r=0}^{n-1}\left(\frac{n+1}{n}\right)\left(\frac{r \cdot {}^nC_r \, {}^nC_{r+1}}{r+2}\right)$ is

(A) ${}^{2n-1}C_{n+1}$ (B) ${}^{2n}C_{n-1}$ (C) ${}^{2n-1}C_{n-1}$ (D) ${}^{2n-1}C_{n-2}$

40. The number of points in the cartesian plane with integral co-ordinates satisfying the inequation $|x| \leq 10$, $|y| \leq 10$, $|x - y| \leq 10$ is

(A) 321 (B) 331 (C) 341 (D) none of these

41. The value of $\sum_{r=1}^{n} r^4 - \sum_{r=-1}^{n+2} (n+1-r)^4$ is equal to

(A) $-[1 + n^4 + (n + 1)^4]$ (B) $-[1 + (n + 1)^4 + (n + 2)^4]$

(C) $-[1 + (n - 1)^4 + n^4]$ (D) none of these

42. The value of a (a < 0) for which least value of quadratic expression $4x^2 - 4ax + a^2 - 2a + 2$ on the interval $0 \le x \le 2$ is equal to 3 is

(A) $-\sqrt{2}$ (B) $\sqrt{2}-2$

(C) $1-\sqrt{2}$ (D) none of these

43. The perpendicular distance of line $(1 - i) z + (1 + i)\bar{z} + 3 = 0$, from $(3 + 2i)$ will be

(A) 13 (B) $\frac{13}{2}$ (C) 26 (D) none of these

44. If the complex numbers z_1, z_2 satisfying $|z_1| = 16$ and $|z_2 - 2 - 3i| = 7$ the minimum value of $|z_1 - z_2|$

(A) 0 (B) 1

(C) 7 (D) 2

45. Find the value of $\frac{1}{3.5}+\frac{1}{7.9}+\frac{1}{11.13}+....\infty$ terms :

(A) $\frac{1}{4}-\frac{\pi}{2}$ (B) $\frac{\pi}{8}-\frac{1}{9}$

(C) $\frac{1}{2}-\frac{\pi}{8}$ (D) none of these

46. If a, b > 0, a + b = 1, then the minimum value of $\left(a+\frac{1}{a}\right)^2+\left(b+\frac{1}{b}\right)^2$ is

(A) 8 (B) 16

(C) 18 (D) $\frac{25}{2}$

47. The results of 10 cricket matches (win, lose or draw) have to be predicted. How many different forecasting can contain exactly 7 correct results?

(A) 100 (B) 120

(C) 960 (D) None of these

48. Let z_1, z_2 be two distinct complex numbers with non-zero real and imaginary parts such that $\arg(z_1 + z_2) = \pi/2$, then $\arg(z_1-\bar{z}_1)-\arg(z_2+\bar{z}_2)$ is equal to

(A) $\frac{\pi}{2}$ (B) π

(C) $-\frac{\pi}{2}$ (D) None of these.

49. The number of ways in which 4 persons P_1, P_2, P_3, P_4 can be arranged in a row such that P_2 does not follow P_1, P_3 does not follow P_2 and P_4 does not follow P_3 is

(A) 24 (B) 12

(C) 11 (D) 10

50. Let $z \in$ C and if $A=\left\{z:\arg(z)=\frac{\pi}{4}\right\}$ and $B=\left\{z:\arg(z-3-3i)=\frac{2\pi}{3}\right\}$ then $n(A\cap B)$ is equal to

(A) 1 (B) 2

(C) 3 (D) 0

SECTION–2 : (MORE THAN ONE OPTION CORRECT TYPE)

51. If α, β are roots of $2x + \frac{1}{x} = 2$ and $f(x) = \frac{8x^6 + 1}{x^3}$, then

(A) $f(\alpha) - f(\beta) = 0$ (B) $f(\alpha) + f(\beta) = -8$

(C) $f(\alpha) - f(\beta) = 2$ (D) $f(\alpha) + f(\beta) = 8$

52. If z_1, z_2, z_3, z_4 be the vertices of a parallelogram taken in anticlockwise direction and $|z_1 - z_2| = |z_1 - z_4|$, then

(A) $\sum_{r=1}^{4} (-1)^r z_r = 0$ (B) $z_1 + z_2 - z_3 - z_4 = 0$

(C) $\arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$ (D) $\arg\left(\frac{z_4 - z_1}{z_2 - z_1}\right) = \frac{\pi}{2}$

53. The coefficient of 3 consecutive terms in the expansion of $(1 + x)^n$ are in the ratio 1 : 7 : 35 the

(A) n is divisible by 5

(B) n is not divisible by any number other than 1 and itself

(C) n is divisible by 35 (D) n is divisible by 23

54. Let $f(n) = 2^n + 7^n$ where n is a positive integer, then

(A) if f(n) is divisible by 5, then f(n + 1) is also divisible by 5

(B) if f(n) is not divisible by 5, then f(n + 1) is not divisible by 5

(C) f(3) is not divisible by 5 (D) f(n) is not divisible by 5 for all n

55. The number of non negative integral solutions of $x_1 + x_2 + x_3 \le n$ is

(A) ${}^{n+2}C_2$ (B) ${}^{n+3}C_3$ (C) ${}^{n+2}C_n$ (D) ${}^{n+3}C_n$

56. If a, b, c $\in$ R such that $a^2 + b^2 + c^2 < 2(ab + bc + ca)$, then

(A) either a, b, c are all positive or all negative (B) atleast two of a, b, c are equal

(C) none of a, b, c can be zero (D) a, b, c are all distinct

57. If $\dfrac{\tan\alpha - i\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right)}{1 + 2i\sin\frac{\alpha}{2}}$ is purely imaginary, then α is given by

(A) $2n\pi + \frac{\pi}{4}$ (B) $2n\pi$ (C) $n\pi + \frac{\pi}{4}$ (D) $n\pi - \frac{\pi}{4}$

58. Values of x for which the sixth term of the expansion of E = $\left(3^{\log_3 \sqrt{9^{|x-2|}}} + 7^{1/5 \log_7 [4\cdot 3^{|x-2|} - 9]}\right)^7$ is 567 are

(A) 3 (B) 1 (C) 2 (D) 4

59. For a positive integer n, let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{2^n - 1}$, then

(A) $a(n) < n$ (B) $a(n) > \frac{n}{2}$

(C) $a(2n) > n$ (D) $a(2n) < 2n$

60. The value of x, for which the 6^{th} term in the expansion of $\left[10^{\log_{10}\sqrt{9^{x-1}+7}} + \frac{1}{10^{1/5 \log_{10} 3^{x-1}+1}}\right]^7$ is 84 is equal to

(A) 1 (B) 2

(C) 3 (D) 4

61. If a + ib $\geq$ 8 – 6i, then

(A) a = 8, b = 6

(B) a = 8, b = – 6

(C) a = – 6, b = 8

(D) inequality is not defined in case of complex number

62. $\frac{1}{\sqrt{2}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{8}}+\frac{1}{\sqrt{8}+\sqrt{11}}+\cdots$ n terms value of the above expression is

(A) $\frac{\sqrt{3n+2}-\sqrt{2}}{3}$ (B) $\frac{n}{\sqrt{2+3n}+\sqrt{2}}$ (C) less than n (D) less than $\sqrt{\frac{n}{3}}$

63. If $|z_1 + z_2| = |z_1 - z_2|$ and $|z_1| = |z_2|$, then

(A) $z_1 = z_2$ (B) $z_1 = -z_2$ (C) $z_1 = \pm iz_2$ (D) $z_2 = \pm iz_1$

64. For a positive integers n, if the expansion of $\left(\frac{5}{x^2}+x^4\right)^n$ has term independent of x, then 'n' can be

(A) 18 (B) 21 (C) 27 (D) 99

65. If $x^2 + mx + 1 = 0$ and $(b - c)x^2 + (c - a)x + (a - b) = 0$ have both roots common then

(A) m = – 2 (B) m = – 1

(C) a, b, c are in A.P. (D) a, b, c are in H.P.

66. The modulus equation $||x - 3| + a| = 25$ ($a \in R$) can have real solutions for x if a lies on the interval

(A) $(-\infty, 25]$ (B) $[-25, 25]$ (C) $(-\infty, -25]$ (D) $(25, \infty)$

67. If x, y, a, b are real numbers such that $(x + iy)^{1/5} = a + ib$ and $P = \frac{x}{a} - \frac{y}{b}$, then

(A) (a – b) is a factor of P (B) (a + b) is a factor of P

(C) (a + ib) is a factor of P (D) (a – ib) is a factor of P

68. The complex numbers z_1, z_2, z_3 are the vertices of a triangle, find all the complex numbers z which make the triangle into parallelogram

(A) $z = z_1 + z_2 - z_3$ (B) $z = z_1 + z_3 - z_2$ (C) $z = z_2 + z_3 - z_1$ (D) $z = z_1 + z_2 + z_3$

69. $z_0 = \left(\frac{1-i}{2}\right)$ then the value of the product $(1 + z_0)(1 + z_0^2)\left(1+z_0^{2^2}\right)\cdots\left(1+z_0^{2^n}\right)$ must be

(A) $2^{2^{n-1}}$ (B) $(1-i)\left(1-\frac{1}{2^{2^n}}\right)$ (C) $\frac{5}{4}(1-i)$ if n = 1 (D) 0

70. If α, β are the roots of $8x^2 - 10x + 3 = 0$ then the equation whose roots are $(\alpha + i\beta)^{100} + (\alpha - i\beta)^{100}$ is

(A) $x^2 + x + 1 = 0$ (B) $x^2 - x + 1 = 0$ (C) $\frac{x^3-1}{x-1} = 0$ (D) none of these

71. All the three roots of $az^3 + bz^2 + cz + d = 0$ have negative real parts ($a, b, c \in R$), then

(A) ab > 0 (B) bc > 0 (C) ad > 0 (D) bc – ad > 0

72. If $a, b, c \in R$ such that $a^2 + b^2 + c^2 < 2(ab + bc + ca)$, then

(A) either a, b, c are all positive or all negative (B) atleast two of a, b, c are equal

(C) none of a, b, c can be zero (D) a, b, c are all distinct

73. If z_1, z_2 be two complex numbers ($z_1 \neq z_2$) satisfying $\left|z_1^2 - z_2^2\right| = \left|\bar{z}_1^2 + \bar{z}_2^2 - 2\bar{z}_1\bar{z}_2\right|$, then

(A) $|\arg z_1 - \arg z_2| = \pi$ (B) $|\arg z_1 - \arg z_2| = \frac{\pi}{2}$

(C) $\frac{z_1}{z_2}$ is purely imaginary (D) $\frac{z_1}{z_2}$ is purely real

74. a, b $\in$ I satisfies equation $a(b - 1) = 3 + b - b^2$, then a + b is equal to

(A) 2 (B) 3

(C) 1 (D) – 1

75. Let $f(x) = ax^2 + bx + 2$, such that $a + b + 2 < 0$ and $a - 2b + 8 < 0$, then

(A) $a < 0$, f(x) has one real root in (0, 2) (B) f(x) has one real root in the interval (0, 1)

(C) f(x) has one real root in the interval (1, 2) (D) f(x) has one real root in the interval $\left(-\frac{1}{2}, 0\right)$

76. A woman has 11 close friend. Number of ways in which she can invite 5 of them to dinner, if two particular of them are not on speaking terms & will not attend together is -

(A) ${}^{11}C_5 - {}^9C_3$ (B) ${}^9C_5 + 2\,{}^9C_4$

(C) $3\,{}^9C_4$ (D) None of these

77. If f(x) = 0 is a polynomial whose coefficients all $\pm$ 1 and whose roots are all real, then the degree of f(x) can be equal to

(A) 1 (B) 2

(C) 3 (D) 4

78. The diagonals of a square are along the pair represented by $2x^2 - 3xy - 2y^2 = 0$. If (2, 1) is the vertex of the square, then the other vertices are

(A) (–1, 2) (B) (1, –2)

(C) (–2,–1) (D) (1,2)

79. If p, q, r are in H.P and p, q, – 2r are in G.P; (p, q, r > 0) then

(A) p^2, q^2, r^2 are in G.P.

(B) p^2, q^2, r^2 are in A.P.

(C) 2p, q, 2r are in A.P.

(D) p + q + r = 0

80. If the equations $\bar{a}z + a\bar{z} + b = 0$ and $\bar{a}z - a\bar{z} + b_1 = 0$ represent two lines C_1 and C_2 in the complex plane then

(A) L_1 and L_2 are perpendicular (B) b is purely real

(C) b_1 is purely imaginary (D) b_1 is purely real

81. If α is a real root of $x^3 + 2x^2 + 10x - 20 = 0$, then

(A) α is rational (B) α^2 is rational

(C) α is irrational (D) α^2 is irrational

82. If z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$, then

(A) $3 \le |z_1 - 2z_2| \le 5$ (B) $1 \le |z_1 + z_2| \le 3$

(C) $|z_1 - 3z_2| \ge 5$ (D) $|z_1 - z_2| \ge 1$

83. If z_1, z_2, z_3, z_4 are root of the equation $a_0z^4 + z_1z^3 + z_2z^2 + z_3z + z_4 = 0$, where a_0, a_1, a_2, a_3 and a_4 are real, then

(A) $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$ are also roots of the equation

(B) z_1 is equal to at least one of $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$

(C) $-\bar{z}_1, -\bar{z}_2, -\bar{z}_3, -\bar{z}_4$ are also roots of the equation (D) none of these

84. If $a^3 + b^3 + 6\,abc = 8\,c^3$ & ω is a cube root of unity then :

(A) a, c, b are in A.P. (B) a, c, b are in H.P.

(C) $a + b\omega - 2\,c\omega^2 = 0$ (D) $a + b\omega^2 - 2\,c\omega = 0$

85. The points z_1, z_2, z_3 on the complex plane are the vertices of an equilateral triangle if and only if :

(A) $\Sigma\,(z_1 - z_2)\,(z_2 - z_3) = 0$ (B) $z_1^2 + z_2^2 + z_3^2 = 2\,(z_1 z_2 + z_2 z_3 + z_3 z_1)$

(C) $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$ (D) $2\,(z_1^2 + z_2^2 + z_3^2) = z_1 z_2 + z_2 z_3 + z_3 z_1$

86. If $|z_1 + z_2| = |z_1 - z_2|$ then

(A) $|\text{amp } z_1 - \text{amp } z_2| = \dfrac{\pi}{2}$ (B) $|\text{ amp } z_1 - \text{amp}_2| = \pi$

(C) $\dfrac{z_1}{z_2}$ is purely real (D) $\dfrac{z_1}{z_2}$ is purely imaginary

87. If $\cos\alpha + \cos\beta + \cos\gamma = 0$ and also $\sin\alpha + \sin\beta + \sin\gamma = 0$, then which of the following is true.

(A) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = \sin 2\alpha + \sin 2\beta + \sin 2\gamma$

(B) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha + \beta + \gamma)$

(C) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha + \beta + \gamma)$

(D) $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$

88. If $\sum_{r=1}^{n} r(r+1)\,(2r + 3) = an^4 + bn^3 + cn^2 + dn + e$, then

(A) $a + c = b + d$ (B) $e = 0$

(C) a, b – 2/3, c – 1 are in A.P. (D) c/a is an integer

89. The sides of a right triangle form a G.P. The tangent of the smallest angle is

(A) $\sqrt{\dfrac{\sqrt{5}+1}{2}}$ (B) $\sqrt{\dfrac{\sqrt{5}-1}{2}}$ (C) $\sqrt{\dfrac{2}{\sqrt{5}+1}}$ (D) $\sqrt{\dfrac{2}{\sqrt{5}-1}}$

90. Sum to n terms of the series $S = 1^2 + 2(2)^2 + 3^2 + 2(4^2) + 5^2 + 2(6^2) + \ldots$ is

(A) $\dfrac{1}{2}n(n+1)^2$ when n is even (B) $\dfrac{1}{2}n^2(n+1)$ when n is odd

(C) $\dfrac{1}{4}n^2(n+2)$ when n is odd (D) $\dfrac{1}{4}n(n+2)^2$ when n is even.

91. If a, b, c are in H.P., then:

(A) $\dfrac{a}{b+c-a}, \dfrac{b}{c+a-b}, \dfrac{c}{a+b-c}$ are in H.P.

(B) $\dfrac{2}{b} = \dfrac{1}{b-a} + \dfrac{1}{b-c}$

(C) $a - \dfrac{b}{2}, \dfrac{b}{2}, c - \dfrac{b}{2}$ are in G.P. (D) $\dfrac{a}{b+c}, \dfrac{b}{c+a}, \dfrac{c}{a+b}$ are in H.P.

92. If b_1, b_2, b_3 ($b_i > 0$) are three successive terms of a G.P. with common ratio r, the value of r for which the inequality $b_3 > 4b_2 - 3b_1$ holds is given by

(A) $r > 3$ (B) $r < 1$ (C) $r = 3.5$ (D) $r = 5.2$

93. If a, b are non-zero real numbers, and α, β the roots of $x^2 + ax + b = 0$, then

(A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$

(B) $1/\alpha, 1/\beta$ are the roots of $bx^2 + ax + 1 = 0$

(C) $\alpha/\beta, \beta/\alpha$ are the roots of $bx^2 + (2b - a^2)x + b = 0$

(D) $-\alpha, -\beta$ are the roots of $x^2 + ax - b = 0$

94. $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is ($a, b, c, d \in R$)

(A) $-d/a$ (B) d/a (C) $(b - a)/a$ (D) $(a - b)/a$

95. If $(x^2 + x + 1) + (x^2 + 2x + 3) + (x^2 + 3x + 5) + \ldots\ldots + (x^2 + 20x + 39) = 4500$, then x is equal to:

(A) 10 (B) -10 (C) 20.5 (D) -20.5

96. $\cos\alpha$ is a root of the equation $25x^2 + 5x - 12 = 0$, $-1 < x < 0$, then the value of $\sin 2\alpha$ is:

(A) 24/25 (B) $-12/25$ (C) $-24/25$ (D) 20/25

97. If the quadratic equations, $x^2 + abx + c = 0$ and $x^2 + acx + b = 0$ have a common root then the equation containing their other roots is/are:

(A) $x^2 + a(b + c)x - a^2bc = 0$ (B) $x^2 - a(b + c)x + a^2bc = 0$

(C) $a(b + c)x^2 - (b + c)x + abc = 0$ (D) $a(b + c)x^2 + (b + c)x - abc = 0$

98. ${}^{n+1}C_6 + {}^nC_4 > {}^{n+2}C_5 - {}^nC_5$ for all 'n' greater than:

(A) 8 (B) 9 (C) 10 (D) 11

99. There are 10 points $P_1, P_2, \ldots, P_{10}$ in a plane, no three of which are collinear. Number of straight lines which can be determined by these points which do not pass through the points P_1 or P_2 is:

(A) ${}^{10}C_2 - 2.\,{}^9C_1$ (B) 27 (C) ${}^8C_2$ (D) ${}^{10}C_2 - 2.\,{}^9C_1 + 1$

100. You are given 8 balls of different colour (black, white,...). The number of ways in which these balls can be arranged in a row so that the two balls of particular colour (say red & white) may never come together is:

(A) $8! - 2.7!$ (B) $6.\,7!$ (C) $2.\,6!.\,{}^7C_2$ (D) none

101. A man is dealt a poker hand (consisting of 5 cards) from an ordinary pack of 52 playing cards. The number of ways in which he can be dealt a "straight" (a straight is five consecutive values not of the same suit, eg. {Ace, 2, 3, 4, 5}, {2, 3, 4, 5, 6}.......................... & {10, J, Q, K, Ace}) is

(A) $10(4^5 - 4)$ (B) $4!\,2^{10}$ (C) $10 \cdot 2^{10}$ (D) 10200

102. Number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3,...... n is:

(A) $\left(\frac{n-1}{2}\right)^2$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd

(C) $\frac{(n-1)^2}{4}$ if n is odd (D) $\frac{n(n-2)}{4}$ if n is even

103. Consider the expansion $(a_1 + a_2 + a_3 + \ldots\ldots + a_p)^n$ where $n \in N$ and $n \le p$. The correct statement(s) is/are:

(A) number of different terms in the expansion is ${}^{n+p-1}C_n$

(B) co-efficient of any term in which none of the variables $a_1, a_2, \ldots, a_p$ occur more than once is 'n'

(C) co-efficient of any term in which none of the variables $a_1, a_2, \ldots, a_p$ occur more than once is $n!$ if $n = p$

(D) Number of terms in which none of the variables $a_1, a_2, \ldots\ldots, a_p$ occur more than once is $\binom{p}{n}$.

104. In the expansion of $(x + y + z)^{25}$

(A) every term is of the form ${}^{25}C_r.\,{}^rC_k.\,x^{25-r}.\,y^{r-k}.\,z^k$

(B) the coefficient of $x^8 y^9 z^9$ is 0

(C) the number of terms is 325 (D) none of these

SECTION - 3: (COMPREHENSION TYPE)

COMPREHENSION-1

Paragraph for Questions Nos. 105 to 107

Let t be a real number satisfying

$2t^3 - 9t^2 + 30 - a = 0$ . . . (1)

And $x + \frac{1}{x} = t$ . . . (2)

105. If equation (1) has three real and distinct roots then

(A) $a > 30$ (B) $a < 3$ (C) $3 < a < 30$ (D) $a < 3$ or $a > 30$

106. If equation (2) has two real and distinct roots then

(A) $-2 < t < 2$ (B) $-1 < t < 1$ (C) $t < -2$ or $t > 2$ (D) none of these

107. If $x + \frac{1}{x} = t$ gives six real and distinct values of x, then

(A) $3 < a < 30$ (B) $a \in \phi$ (C) $a \in (2, 5)$ (D) none of these

COMPREHENSION-2

Paragraph for Questions Nos. 108 to 110

Consider the equation $x + y - [x][y] = 0$, where $[\cdot]$ = Greatest integer function.

108. The number of integral solutions to the equation, is

(A) 0 (B) 1 (C) 2 (D) none of these

109. Equation of one of the lines on which the non integral solution of given equation, lies is

(A) $x + y = -1$ (B) $x + y = 0$ (C) $x + y = 1$ (D) $x + y = 5$

110. Number of the point of intersection between all the possible lines on which the non-integral solutions of the given equation lies, is

(A) 0 (B) 1 (C) 2 (D) 3

COMPREHENSION-3

Paragraph for Questions Nos. 111 to 113

$y = ax^2 + bx + c = 0$ is a quadratic equation which has real roots if and only if $b^2 - 4ac \geq 0$. If $f(x, y) = 0$ is a second degree equation, then using above fact we can get the range of x and y by treating it as quadratic equation in y or x. Similarly $ax^2 + bx + c \geq 0 \ \forall \ x \in R$ if $a > 0$ and $b^2 - 4ac \leq 0$.

111. If $0 < \alpha, \beta < 2\pi$, then the number of ordered pairs (α, β) satisfying $\sin^2(\alpha + \beta) - 2\sin\alpha\sin(\alpha + \beta) + \sin^2\alpha + \cos^2\beta = 0$ is:

(A) 2 (B) 0 (C) 4 (D) none of these

112. If $A + B + C = \pi$, then the maximum value of $\cos A + \cos B + k\cos C$ (where $k > 1/2$) is

(A) $\frac{1}{k} + \frac{k}{2}$ (B) $\frac{2k^2+1}{3}$ (C) $\frac{k^2+2}{2}$ (D) $\frac{1}{2k} + k$

113. A circle with radius |a| and centre on y-axis slides along it and a variable line through (a, 0) cuts the circle at points P and Q. The region in which the point of intersection of tangents to the circle at points P and Q lies is represented by

(A) $y^2 \geq 4(ax - a^2)$ (B) $y^2 \leq 4(ax - a^2)$ (C) $y \geq 4(ax - a^2)$ (D) $y \leq 4(ax - a^2)$

COMPREHENSION-4

Paragraph for Questions Nos. 114 to 116

Let N = $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$ be a natural number where p_i $(1 \leq i \leq n)$ is a prime number. The total number of divisors of N is $(\alpha_1 + 1)(\alpha_2 + 1)\ldots(\alpha_n + 1)$. The sum of all divisors is $\left(\frac{p_1^{\alpha_1+1}-1}{p_1-1}\right)\left(\frac{p_2^{\alpha_2+1}-1}{p_2-1}\right)\cdots\left(\frac{p_n^{\alpha_n+1}-1}{p_n-1}\right)$. The number of ways in which N can be resolved in two factors is $\frac{(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_n+1)+1}{2}$ or $\frac{(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_n+1)}{2}$ as N is a perfect square or not. Number of ways of resolving N in two coprime factors is 2^{n-1}.

114. The number of ways in which 420 can be factorised in two non coprime factors is

(A) 24 (B) 8 (C) 12 (D) 4

115. The number of positive integral solution of $x_1x_2x_3x_4 = 420$ is

(A) 420 (B) 240 (C) 640 (D) none of these

116. The sum of all the even divisors of 420 is

(A) 860 (B) 192 (C) 1344 (D) 1152

COMPREHENSION-5

Paragraph for Questions Nos. 117 to 119

If z_1, z_2 be the complex numbers representing two points A and B, then we define the complex slope of the line AB as $\mu = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$, it can be noted that $|\mu| = 1$ and μ remains same for any two points on the line AB, since if z_3 . z_4 be complex numbers representing some other points on the same line, then

$$\mu' = \frac{z_3 - z_4}{\bar{z}_3 - \bar{z}_4} = \frac{\lambda(z_1 - z_2)}{\bar{\lambda}(\bar{z}_1 - \bar{z}_2)} \qquad (\because z_3 - z_4 = \lambda(z_1 - z_2)\ \lambda \text{ real}) = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = \mu$$

117. The complex slope of the line $\bar{a}z + a\bar{z} + b = 0$ where a is complex and b is real is

(A) $\frac{a}{a}$ (B) $-\frac{a}{a}$ (C) $\frac{\bar{a}}{a}$ (D) $-\frac{\bar{a}}{a}$

118. If the complex slope of a line which is not parallel to y-axis is $\cos\phi + i\sin\phi$, then the line makes an angle θ with x-axis, θ must be

(A) 2ϕ (B) $90° - \phi$

(C) $\frac{\phi}{2}$ (D) ϕ

119. If μ and μ' be complex slopes of two perpendicular lines, then

(A) $\mu\mu' = 1$ (B) $\mu\mu' = -1$

(C) $\mu + \mu' = 0$ (D) none of these

COMPREHENSION-6

Paragraph for Questions Nos. 120 to 122

Let z be a complex number lying on a circle $|z| = \sqrt{2}\,a$ and $b = b_1 + ib_2$ (any complex number), then

120. The equation of tangent at the point 'b' is

(A) $z\bar{b} + \bar{z}b = a^2$ (B) $z\bar{b} + \bar{z}b = 2a^2$ (C) $z\bar{b} + \bar{z}b = 3a^2$ (D) $z\bar{b} + \bar{z}b = 4a^2$

121. The equation of straight line parallel to the tangent at the point b and passing through centre of circle is

(A) $z\bar{b} + \bar{z}b = 0$ (B) $2z\bar{b} + \bar{z}b = \lambda$ (C) $2z\bar{b} + 3\bar{z}b = 0$ (D) $z\bar{b} + \bar{z}b = \lambda$

122. The equation of lines passing through the centre of the circle and making an angle $\frac{\pi}{4}$ with the normal at 'b' are

(A) $z = \pm\frac{ib^2}{2a^2}\bar{z}$ (B) $z = \pm\frac{ib^2}{a^2}\bar{z}$ (C) $z = \pm\frac{ib^2}{3a^2}\bar{z}$ (D) $z = \pm\frac{ib^2}{4a^2}\bar{z}$

COMPREHENSION-7

Paragraph for Questions Nos. 123 to 125

Suppose $f(x) = 3x^3 - 13x^2 + 14x - 2$. It is assumed that $f(x) = 0$ have three real roots say α, β and γ where $\alpha < \beta < \gamma$.

123. $[\alpha]$, $[\beta]$, $[\gamma]$ (where [.] denotes the greatest integer function) are in

(A) A.P (B) G.P (C) H.P (D) none of these

124. $\lim_{n\to\infty} \alpha^{n!}\cdot\beta^{1/n!}$ will be equal to

(A) 1 (B) e (C) 0 (D) none of these

125. The value of $\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma$ is

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

COMPREHENSION-8

Paragraph for Questions Nos. 126 to 128

The quantities $1 + x$, $1 + x + x^2$, $1 + x + x^2 + x^3$, …, $1 + x + x^2 + \ldots + x^n$ are multiplied and terms of the product are arranged in increasing powers of x in the form $a_0 + a_1 x + a_2 x^2 + \ldots$, then

126. The number of terms in the product is

(A) n^2 (B) $n(n + 1)$ (C) $\frac{n(n+1)}{2}$ (D) $\frac{n^2+n+2}{2}$

127. The coefficients of equidistant terms from beginning and end are

(A) always equal (B) never equal
(C) sometimes equal (D) can't be said

128. The sum of odd coefficients = sum of even coefficients = ?

(A) $n!$ (B) $(n + 1)!$
(C) $\frac{(n+1)!}{2}$ (D) none of these

COMPREHENSION-9

Paragraph for Questions Nos. 129 to 131

Consider the quadratic equation $az^2 + bz + c = 0$ where a , b , c and z are complex numbers

129. The condition that the equation has both real roots is

(A) $\frac{a}{\bar{a}} = -\frac{b}{\bar{b}} = \frac{c}{\bar{c}}$ (B) $\frac{a}{\bar{a}} = \frac{b}{\bar{b}} = \frac{c}{\bar{c}}$ (C) $\frac{a}{\bar{a}} = \frac{b}{\bar{b}} = -\frac{c}{\bar{c}}$ (D) none of these

130. The condition that equation has both roots purely imaginary

(A) $\frac{a}{\bar{a}} = -\frac{b}{\bar{b}} = -\frac{c}{\bar{c}}$ (B) $\frac{a}{\bar{a}} = -\frac{b}{\bar{b}} = \frac{c}{\bar{c}}$ (C) $\frac{a}{\bar{a}} = \frac{b}{\bar{b}} = -\frac{c}{\bar{c}}$ (D) none of these

131. Condition that equation has one complex root m such that |m| = 1

(A) $\frac{\bar{b}c - b\bar{a}}{a\bar{a} - c\bar{c}} = \frac{a\bar{a} + c\bar{c}}{\bar{c}b + a\bar{b}}$ (B) $\frac{\bar{b}c + b\bar{a}}{a\bar{a} + c\bar{c}} = \frac{a\bar{a} + c\bar{c}}{\bar{c}b + a\bar{b}}$

(C) $(\bar{b}c - b\bar{a})(\bar{c}b - a\bar{b}) = (a\bar{a} - c\bar{c})^2$ (D) none of these

COMPREHENSION-10

Paragraph for Questions Nos. 132 to 134

8 players compete in a tournament, everyone plays everyone else just once. The winner of a game gets 1, the loser 0 or each gets $\frac{1}{2}$ if the game is drawn. The final result is that everyone gets a different score and the player playing placing second gets the same score as the total of four bottom players.

Now answer the following questions:

132. The total of the player placing II was

(A) 6 (B) $6\frac{1}{2}$ (C) $5\frac{1}{2}$ (D) can't say

133. The result of the game between player placing III and player placing VII was

(A) player III was the winner (B) player VII was the winner

(C) the game ended in a drawn (D) can't say

134. The total score of the top four players was

(A) 22 (B) 21 (C) 20 (D) 19

COMPREHENSION-11

Paragraph for Questions Nos. 135 to 137

Let z_1, z_2, z_3 be the complex number associated with vertices A, B, C of a triangle ABC which is circumscribed by the circle |z| = 1. Altitude through A meets the side BC and D and circum-circle at E. Let P be the image of E about BC and F be the image of E about origin.

Now answer the following questions:

135. The complex number of point P is

(A) $\frac{z_1 + z_2 + z_3}{3}$ (B) $\frac{2(z_1 + z_2 + z_3)}{3}$

(C) $z_1 + z_2 + z_3$ (D) none of these

136. The complex number of point E is

(A) $\frac{z_1z_2}{z_3}$ (B) $\frac{z_2z_3}{z_1}$

(C) $-\frac{z_2z_3}{z_1}$ (D) $-\frac{z_1z_2}{z_3}$

137. The distance of point C from F i.e. CF is equal to

(A) $|z_1 - z_2|$ (B) $|z_1 + z_2|$

(C) $\frac{|z_1 - z_3|}{2}$ (D) $\frac{|z_1 + z_3|}{2}$

COMPREHENSION-12

Paragraph for Questions Nos. 138 to 140

A complex number z = x + iy satisfies $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$. Then-

138. Locus of z is

(A) major arc of the circle (B) minor arc of the circle

(C) circle having centre at origin (D) none of these

139. Radius of the circle given by above equation is

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$

(C) $\sqrt{3}$ (D) 1

140. Maximum value of |z| satisfying the given equation is

(A) $\frac{2}{\sqrt{3}} + 1$ (B) $\frac{1}{\sqrt{3}} + 1$

(C) $\sqrt{3}$ (D) $\frac{4}{\sqrt{3}}$

SECTION-4: (MATRIX MATCH TYPE)

141. Match the following

	Column I		*Column II*
(A)	The number of integral solutions of the equation x + 2y = 2xy is	(p)	1
(B)	The number of real solutions of the system of equations $x = \frac{2z^2}{1+z^2}, y = \frac{2x^2}{1+x^2}, z = \frac{2y^2}{1+y^2}$	(q)	2
(C)	If a(y + z) = x, b(z + x) = y, c(x + y) = z, where $a \neq -1$, $b \neq -1$, $c \neq -1$ admit non-trivial solutions, then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$ is	(r)	0
(D)	The number of solutions of the equation $\sqrt{3x^2 + 6x + 7} + \sqrt{5x^2 + 10x + 14} \leq 4 - 2x - x^2$ is	(s)	Infinite

142. Match the following

List – I	List – II
(A) $\arg\left(\dfrac{z+1}{z-1}\right) = \dfrac{\pi}{4}$	(i) parabola
(B) $z = \dfrac{3i-t}{2+it}$ $(t \in R)$	(ii) part of a circle
(C) $\arg z = \dfrac{\pi}{4}$	(iii) full circle
(D) $z = t + it^2$ $(t \in R)$	(iv) line

143. Match the following:

List – I	List – II
(A) If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then a is	(i) $(0, 1]$
(B) Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let m(b) be the minimum value of f(x). As b varies, then m(b) is	(ii) $\left[\dfrac{2+\sqrt{3}}{2}, \infty\right)$
(C) If a, b, c $\in$ R and equation $px^2 + qx + r = 0$ has two real roots α and β such that $\alpha < -1$ and $\beta > 1$, then $x^2 + \left\lvert\dfrac{q}{p}\right\rvert x + \dfrac{r}{p}$ is	(iii) < 2
(D) The set of values of a for which both the roots of the equation $x^2 + (2a - 1)x + a = 0$ are positive is	(iv) < 0

144. Match the following:

List – I	List – I
(A) If $\lvert x^2 - x\rvert \ge x^2 + x$, then x	(i) $[0, \infty)$
(B) $\lvert x + y\rvert > x - y$, where $x > 0$, then y =	(ii) $(-\infty, 0]$
(C) If $\log_2 x \ge \log_3^{(x^2)}$, then x =	(iii) $[-1, \infty)$
(D) $[x] + 2 \ge \lvert x\rvert$ (where [.] denotes the greatest integer function)	(iv) $(0, 1]$

145. Match the following:

List – I	List – II
(A) If inequation $ax^2 - ax + 1 < 0$ $\forall$ x $\in$ R, then a belongs to	(i) $[0, 4)$
(B) If $x^3 - 3x + \dfrac{a}{2} = 0$ has three real and distinct root, then $\lvert a\rvert$ belongs to	(ii) $[0, 3]$
(C) If $x^3 + ax^2 + x + 1 = 0$ has exactly one real root, then a^2 may belongs to	(iii) $(0, 4)$
(D) If quadratic equation $x^2 - 3ax + a^2 - 9 = 0$ has roots of opposite sign then a belongs to	(iv) $(-3, 3)$

146. Match the following:

List – I	List – II
(A) $\underbrace{1\ 1\ 1 \ldots 1}_{91 \text{ times}}$	(i) is a prime
(B) 1.2.3. … n(n + 1) (n …3.2.1)	(ii) is not a prime
(C) $10^{4n} + 10^{4(n-1)} \ldots + 10^{8} + 10^{4} + 1, n \in N$	(iii) is a perfect square integer
(D) $\underbrace{4\ 4\ 4 \ldots 4}_{n \text{ times}}\ \underbrace{8\ 8\ 8 \ldots 9}_{(n-1) \text{ times}}$	(iv) is a perfect square of odd integer.

147. Match the following:

List – I	List – II
(A) The least value of $2\log_{100}^{a} - \log_{a}^{0.0001}, a > 1$ is	(i) 0
(B) If α, β are the roots of $6x^2 - 2x + 1 = 0$ and $S_n = \alpha^n + \beta^n$, the $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} S_r$ is	(ii) 1
(C) If $x^2 - x + 1 = 0$, then the value of x^{3n} where n is even	(iii) 2
(D) The number of roots of the equation $x - \frac{10^2}{x-1} = 1 - \frac{10^2}{x-1}$ is	(iv) 3
	(v) 4

148. Match the following:

List –I	List-II
(A) $(x - 1)(x - 3) + k(x - 2)(x - 4) = 0$ $(k \in R)$ has real roots for $k \in$	(i) $(-5, -1)$
(B) Range of the function $\frac{x-1}{x^2 - k + 1}$ does not contain any value in the interval $[-1, 1]$ for $k \in$	(ii) ϕ
(C) The equation $x \in \left(0, \frac{\pi}{2}\right)$, secx + cosecx = k has real roots if $k \in$	(iii) $(-\infty, \infty)$
(D) The equation $x^2 + 2(k - 1)x + k + 5 = 0$ has roots positive and distinct if $k \in$	(iv) $[2\sqrt{2}, \infty)$

149. Match the following:

List – I	List – II
(A) Given positive rational numbers a, b, c such that $a + b + c = 1$, then $a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$	(i) is equal to $-\frac{1}{2}(n-1)$
(B) If n is a positive integer ≥ 1, then $\frac{3^n}{2^n + n \cdot 6^{\frac{n-1}{2}}}$	(ii) is equal to $\frac{2}{n}$
(C) If $n \in N > 1$, then sum of real part of roots of $z^n = (z + 1)^n$	(iii) ≤ 1
(D) If the quadratic equations $2x^2 + bx + 1 = 0$ and $4x^2 + ax + 1 = 0$ have a common root, then the value of $\frac{a^2 + 2b^2 - 3ab + 4}{n}$, when $n \in N$	(iv) ≥ 1

150. Match the following:

List – I	List – II
(A) If a – b, ax – by, $ax^2 - by^2$ (a, b $\neq$ 0) are in G.P., then x, y, $\dfrac{ax - by}{a - b}$ are in	(i) A.P.
(B) If the slope of one of the lines represented by $a^3x^2 - 2hxy + b^3y^2 = 0$ be the square of the other, then ab^2, h, a^2b are in	(ii) G.P.
(C) a, b, c, d are distinct positive numbers, then $\dfrac{a^n}{b^n} > \dfrac{c^n}{d^n}$ for	(iii) H.P.
(D) If $a_1, a_2, a_3, \ldots$ are in H.P. and $f(k) = \sum_{r=1}^{n} a_r - a_k$, then $\dfrac{a_1}{f(1)}, \dfrac{a_2}{f(2)}, \dfrac{a_3}{f(3)} \ldots \dfrac{a_n}{f(n)}$ are in	

151. Match the following:

List – I		List – II	
(A)	If x, y, z $\in$ N then number of ordered triplet (x, y, z) satisfying xyz = 243 is	(i)	19
(B)	The number of terms in the expansion of $(x + y + z)^6$ is	(ii)	28
(C)	If x $\in$ N, then number of solutions of $x^2 + x - 400 \leq 0$ is	(iii)	21
(D)	If x, y, z $\in$ N, then number of solution of x + y + z = 10	(iv)	36

152. If $f(x) = x^3 + ax^2 + bx + c = 0$ has three distinct integral roots and $(x^2 + 2x + 2)^3 + a(x^2 + 2x + 2)^2 + b(x^2 + 2x + 2) + c = 0$ has no real roots then

(A) a = (1) 0
(B) b = (2) 2
(C) c = (3) 3
(D) If the roots of $f'(x) = k$ are equal then k = (4) –1

153. Match the following:

(A) The coefficient of x^{-15} in $\left(3x^2 + \dfrac{3^{-4/7}}{x^3}\right)^{10}$ is — (i) 41

(B) If $(1 - x + x^2)^4 = a_0 + a_1x + a_2x^2 + \ldots + a_8x^8$, then $a_0 + a_2 + a_4 + a_6 + a_8$ is equal to — (ii) 34

(C) $\left(\sqrt{2}+1\right)^4 + \left(\sqrt{2}-1\right)^4$ is equal to — (iii) 40

(D) Coefficient of x^{11} in the expansion of $\dfrac{1}{6}(2x^2 + x - 3)^6$ is equal to — (iv) 32

154. Match the following:

(A) Locus of $\left|2z - \left(\sqrt{3}i + 1\right)\right| + \left|2z - \left(\sqrt{3}i - 1\right)\right| = 2$ — (i) two infinite line segments

(B) Locus of |z + i| + |z – i| = 4 — (ii) ellipse

(C) Locus of ||z – i| – |z + i|| = 4 — (iii) line segment

(D) Locus of ||z + i| – |z – i|| = 2 — (iv) nothing on the plane

155. Match the following :

	Column – I	Column – II
(a)	$\dfrac{5x+1}{(x+1)^2} < 1$	(P) $x \in (-\infty, 0) \cup (0, 2) \cup (2, \infty)$
(b)	$\lvert x\rvert + \lvert x-3\rvert > 3$	(Q) $x \in (-\infty, -5) \cup (-3, 3) \cup (5, \infty)$
(c)	$\dfrac{1}{\lvert x\rvert - 3} < \dfrac{1}{2}$	(R) $x \in (-\infty, -1) \cup (-1, 0) \cup (3, \infty)$
(d)	$\dfrac{x^4}{(x-2)^2} > 0$	(S) $x \in (-\infty, 0) \cup (3, \infty)$

156. Match the following :

	Column - I	Column - II
(a)	If $\dfrac{x(y+z-x)}{\log x} = \dfrac{y(z+x-y)}{\log y} = \dfrac{z(x+y-z)}{\log z}$ and a. $x^y \cdot y^x$ = b.$y^z \cdot z^y$ = c$z^x \cdot x^z$, then $\dfrac{a+b}{c}$ equals	(P) 1
(b)	$y = 10^{\frac{1}{1-\log_{10} x}}$, $z = 10^{\frac{1}{1-\log_{10} y}}$ implies $x = 10^{\frac{1}{a+b\log_{10} z}}$, than a – b equals	(Q) 2
(c)	If $a^2 + b^2 = c^2 \Rightarrow \log_{c+b} a + \log_{c-b} a = k \log_{c+b} a \log_{c-b} a$, then k equals	(R) 3
(d)	If $b = \sqrt{ac}$ where a > 0, c > 0 & b ≠ 1 and if $\dfrac{\log_a N - \log_b N}{\log_b N - \log_c N} = k.\log_a c$, then k equals	(S) 0

157. Match the following

	Column – I	Column – II
(A)	If $\log_{\sin x} \log_3 \log_{0.2} x < 0$, then	(p) $x \in [-1, 1]$
(B)	If $\dfrac{(e^x-1)(2x-3)(x^2+x+2)}{(\sin x-2)\,x(x+1)} \le 0$, then	(q) $x \in [-3, 6)$
(C)	If $\lvert 2 - \lvert [x] - 1\rvert \rvert \le 2$, then [.] represents greatest integer function.	(r) $x \in \left(0, \dfrac{1}{125}\right)$
(D)	If $\lvert \sin^{-1}(3x-4x^3)\rvert \le \dfrac{\pi}{2}$, then	(s) $x \in (-\infty, -1) \cup \left[\dfrac{3}{2}, \infty\right)$

158 Match the column

Column – I		Column – II	
(A)	$x\|x\| =$	(p)	$\begin{cases} -2x & : \quad x < -1 \\ 2 & : \quad -1 \le x \le 1 \\ 2x & : \quad x \ge 1 \end{cases}$
(B)	$\|x - 1\| + \|x + 1\| =$	(q)	$\begin{cases} -x^2 & : \quad x \le 0 \\ x^2 & : \quad x > 0 \end{cases}$
(C)	If $-1 \le x \le 2$, then $2x - \{x\} =$	(q)	$\begin{cases} -x & : \quad -1 \le x < 0 \\ 0 & : \quad 0 \le x < 1 \\ x & : \quad 1 \le x < 2 \end{cases}$
(D)	If $-1 \le x \le 2$, then $x[x] =$	(s)	$\begin{cases} x-1 & : \quad -1 \le x < 0 \\ x & : \quad 0 \le x < 1 \\ x+1 & : \quad 1 \le x < 2 \end{cases}$

159 **Match The column**

Column – I		Column – II	
(A)	Number of real solution of $a^2 + b^2 + c^2 = x^2$ is	(p)	2
(B)	The number of non-negative real roots of $2^x - x - 1 = 0$, equals	(q)	∞
(C)	Let p and q be the roots of the quadratic equation $x^2 - (\alpha - 2)x - \alpha - 1 = 0$. What is the minimum possible value of $p^2 + q^2$?	(r)	6
(D)	The value of 'c' for which $\|\alpha^2 - \beta^2\| = \frac{7}{4}$, where α and β are the roots of $2x^2 + 7x + c = 0$, is	(s)	5

160. **Match the column**

Column – I		Column – II	
(A)	Find all possible values of k for which every solution of the inequation $x^2 - (3k - 1)x + 2k^2 - 3k - 2 \ge 0$ is also a solution of the inequation $x^2 - 1 \ge 0$.	(p)	16
(B)	If a, b, c and d are four positive real numbers such that abcd = 1, the minimum value of $(1 + a)(1 + b)(1 + c)(1 + d)$ is	(q)	[0, 1]
(C)	Solution set of the inequality $5^{x+2} > \left(\frac{1}{25}\right)^{1/x}$ is	(r)	$\frac{1}{6}$
(D)	Let $f(x) = x^3 + 3x + 1$ if $g(x)$ is the inverse function of $f(x)$. Then $g'(5)$ equal to	(s)	$(0, \infty)$

161 **Match the column**

Column – I		**Column – II**	
(A)	Suppose that $F(n + 1) = \frac{2F(n)+1}{2}$ for $n = 1, 2, 3, \ldots$ and $F(1) = 2$. Then $F(101)$ equals	(p)	42
(B)	If $a_1, a_2, a_3, \ldots\ldots\ldots a_{21}$ are in A.P. and $a_3 + a_5 + a_{11} + a_{17} + a_{19} = 10$ then the value of $\sum_{i=1}^{21} a_i$ is	(q)	1620
(C)	10^{th} term of the sequence $S = 1 + 5 + 13 + 29 + \ldots$, is	(r)	52
(D)	The sum of all two digit numbers which are not divisible by 2 or 3 is	(s)	2045

162. **Match the column**

Column – I		**Column – II**	
(A)	The arithmetic mean of two numbers is 6 and their geometric mean G and harmonic mean H satisfy the relation $G^2 + 3H = 48$. Find the two numbers.	(p)	$\frac{240}{77}$
(B)	The sum of the series $\frac{5}{1^2.4^2} + \frac{11}{4^2.7^2} + \frac{17}{7^2.10^2} + \ldots$ is.	(q)	(4,8)
(C)	If the first two terms of a Harmonic Progression be $\frac{1}{2}$ and $\frac{1}{3}$, then the Harmonic Mean of the first four terms is	(r)	$\frac{1}{3}$

163. Match the column

Column – I			**Column – II**
(A)	${}^mC_1\,{}^nC_m - {}^mC_2\,{}^{2n}C_m + {}^mC_3\,{}^{3n}C_m \ldots\ldots\ldots$	(p)	coefficient of x^m in the expansion of $((1+)^n - 1)^m$
(B)	${}^nC_m + {}^{n-1}C_m + {}^{n-2}C_m + \ldots\ldots + {}^mC_m$	(q)	coefficient of x^m in $\frac{(1+x)^{n+1}}{x}$
(C)	$C_0\,C_n + C_1\,C_{n-1} + \ldots\ldots\ldots + C_n\,C_0$	(r)	coefficient of x^n in $(1 + x)^{2n}$
(D)	$2^k\,{}^nC_0 - 2^{k-1}\,{}^nC_1\,{}^{n-1}C_{k-1} \ldots\ldots\ldots (-1)^k\,{}^nC_k\,{}^{n-k}C_0$	(s)	coefficient of x^k in the exp. of $(1 + x)^n$

164. Match the column

Column – I		**Column – II**	
(A)	The number of distinct terms in the expansion of $(x_1 + x_2 + x_3 + ... + x_n)^3$ is	(p)	${}^{n+3}C_n$
(B)	The number of Integral terms in ther expansion of $[5^{1/2} + 7^{1/8}]^{1024}$	(q)	${}^{n+2}C_3$
(C)	Degree of polynomial $[x + (x^3 - 1)^{1/2}]^5 + [x - (x^3 - 1)^{1/2}]^5$ is	(r)	129
(D)	Coefficients of the second, third and fourth terms in the expansion of $(1 + x)^n$ are in A.P. the n is equal to	(s)	7

165.

Column – I		**Column – II**	
(A)	If (r + 1) th term is the first negative term in the expansion of $(1 + x)^{7/2}$, then the value of r (where $\|x\| < 1$) is	(p)	divisible by 2
(B)	The coefficient of y in the expansion of $(y^2 + 1/y)^5$ is	(q)	divisible by 5
(C)	If the second term in the expansion $\left(a^{\frac{1}{13}} + \frac{a}{\sqrt{a^{-1}}}\right)^n$ is $14a^{5/2}$, then the value of n is	(r)	divisible by 10
(D)	The coefficient of x^4 in the expression $(1 + 2x + 3x^2 + 4x^3 + ...\text{up to } \infty)^{1/2}$ is c, $(c \in N)$, then c + 1 (where $\|x\| < 1$) is	(s)	a prime number

166. **Match the following :**

Column - I		**Column - II**	
(a)	The number of cubes with the six faces numbered 1 to 6 can be made, if the sum of the number in each pair of opposite faces is 7, is equal to	(P)	2
(b)	A citizen is expected to vote for atleast one of three positions mayor, secretary and attorney. The number of ways he/she can vote if there are 3 candidates each for three position, is 9 k where k is	(Q)	4
(c)	The number of ways in which 4 married couples can be seated at a round table if no husband and wife as well as no two men are to seat together is 3 k where k is	(R)	3
(d)	The sum of all numbers of the form $\frac{12!}{a!b!c!}$ where a, b, c $\in$ W, satisfy a + b + c = 12, is 3^{3k} where k is	(S)	7

167. Match the following :

	Column - I	Column - II	
(a)	The number of five - digit numbers having the product of digits 20 is	(P)	77
(b)	A man took 5 space plays out of an engine to clean them. The number of ways in which he can place atleast two plays in the engine from where they came out is	(Q)	31
(c)	The number of integer between 1 & 1000 inclusive in which atleast two consecutive digits are equal is	(R)	50
(d)	The value of $\frac{1}{15}\sum_{1\le i\le j\le 9}\sum i\cdot j$	(S)	181

168. Match the following :

	Column - I	Column - II	
(A)	The number of arrangements that can be made taking 4 letters, at a time, out of the letters of the word "PASSPORT" is :	(p)	2454
(B)	The numberof ways of forming a 4 letter word using the letters of the word MATHEMATICS, is	(q)	606
(C)	The number of selections of four letters from the letters of the word ASSASSINATION is	(r)	72
(D)	The total number of ways of selecting five letters from the letter of the words INDEPENDENT is	(s)	2424

169.

	Column – I	Column – II	
(A)	The total number of selections of fruits which can be made from, 3 bananas, 4 apples and 2 oranges is	(p)	Greater than 50
(B)	If 7 points out of 12 are in the same straight line, then the number of triangles formed is	(q)	Greater than 100
(C)	The number of ways of selecting 10 balls from unlimited number of red, black, white and green balls is	(r)	Greater than 150
(D)	The total number of proper divisors of 38808 is	(s)	Greater than 200

170.

	Column – I		**Column – II**
(A)	Number of 4 letter words that can be formed using the letter of the words 'RESONANCE' is	(p)	$\frac{11!}{3!}$
(B)	Number of ways of selecting 3 persons out of 12 sitting in a row, if no two selected persons were sitting together, is	(q)	1206
(C)	Number of solutions of the equation $x + y + z = 20$, where $1 \le x < y < z$ and $x, y, z \in I$, is	(r)	24
(D)	Number of ways in which indian team can bat, if Yuvraj wants to bat before Dhoni and Pathan wants to bat after Dhoni is (assume all the batsman bat)	(s)	120

SECTION5: (INTEGER TYPE)

171. In a class tournament where the participants were to play one game with another, two class players fell ill, having played 3 games each. If the total number of games played is 84, then the number of participants at the beginning is _______________

172. If $|z|^2 + (3 - 4i)z + (3 + 4i)\bar{z} + 75 = 0$ and $(1 - i)z + (1 + i)\bar{z} - 16 = 0$ intersect at z_1 and z_2, then the integral part of the sum of the areas of the quadrilaterals having $(z_1 + z_2)$ and $(z_1 - z_2)$ as diagonals passing through origin is ____________ (Two vertices of 1st quadrilateral are z_1 and z_2 and of 2nd quadrilateral are z_1 and $- z_2$).

173. If $x = 1.2 (2^2 - 1^2) + 2.3 (3^2 - 2^2) + 3.4 (4^2 - 3^2) + \ldots$ upto 50 terms, then the value of $\frac{x}{51^3}$ is ________

174. If α is the absolute maximum value of the expression $\frac{3x^2 + 2x - 1}{x^2 + x + 1}$ $\forall$ x $\in$ R, then $[\alpha]$is _______, (where [.] denotes the greatest integer function)

175. The number of solutions of the equation $e^{|x|} = |x| + 1$ is __________

176. The value of x satisfying the equations $\log_3(\log_2 x) + \log_{1/3}(\log_{1/2} y) = 1$; $xy^2 = 9$is________

177. If there are six letters $L_1, L_2, L_3, L_4, L_5, L_6$ and their corresponding six envelopes $E_1, E_2, E_3, E_4, E_5, E_6$. Letters having odd value can be put into odd value envelopes and even value letter can be put into even value envelopes, so that no letter go into the right envelopes, the number of arrangement will be equal to ________

178. The number of integral values of a ; a $\in$ (6, 100) for which the equation $[\tan x]^2 + \tan x - a = 0$ has real roots; where [.] denotes greatest integer function is _________

179. If the co-efficient of rth, (r+1)th and (r + 2)th terms in the expansion of $(1 + x)^{14}$ are in A.P. then the greatest possible value of r is ___________

180. The remainder when $(3m + (-1)^n)^{18}$ is divided by 9 is __________ (m, n are natural numbers).

181. The numbers of five digits that can be made with the digits 1, 2, 3 each of which can be used at most thrice in a number, is ______________

182. The number of TIMES the digit 0 will be written when listing of the integers from 1 to 100 is _____________

183. If 6-digit number abcdef is multiplied with 6 and the resulting number is defabc. The number is ____________.

184. If $f : \{a, b, c, d, e\} \to \{a, b, c, d, e\}$ f is onto and $f(x) + x$ for each $x \in \{a, b, c, d, e\}$ is equal to ____.

185. The number of real solutions of the equation $x^6 - x^5 + x^4 - x^3 + x^2 - x + \frac{3}{4} = 0$ is ________

186. The number of ordered triplets (a, b, c) such that L.C.M (a, b) = 1000, L.C.M (b, c) = 2000 and L.C.M (c, a) = 2000 is _______________

187. If the number of ordered pairs of (x, y) satisfying the system of equations $5x\left(1 + \frac{1}{x^2 + y^2}\right) = 12$ and $5y\left(1 - \frac{1}{x^2 + y^2}\right) = 4$ is n, then n is ——————

188. Consider the sequence a_n given by $a_1 = \frac{1}{3}$, $a_{n+1} = a_n^2 + a_n$. Let $S = \frac{1}{a_2} + \frac{1}{a_3} + \cdots + \frac{1}{a_{2008}}$, then [S] is equal to ___________ (where [ ] represents greatest integer function)

189. Let (x_i, y_i) where i = 1, 2, 3, 4 are the integral solutions of equation $2x^2y^2 + y^2 - 6x^2 - 12 = 0$. The area of quadrilateral whose vertices are (x_i, y_i), i = 1, 2, 3, 4 is _______

190. In the expansion of $(a^{1/3} + b^{1/9})^{6561}$, where a, b are distinct prime numbers, then the number of irrational terms are ________

191. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1x + a_2x^2 + ... + a_{20}x^{20}$, then calculate $a_1 + a_2 + a_4$

192. Given that a, g are roots of the equation, A x^2 - 4 x + 1 = 0 and b, d the roots of the equation, B x^2 - 6 x + 1 = 0, find values of A *B, such that a, b, g & d are in H.P.

193. In maths paper there is a question on "Match the column" in which column A contains 6 entries & each entry of column A corresponds to exactly one of the 6 entries given in column B written randomly. 2 marks are awarded for each correct matching & 1 mark is deducted from each incorrect matching. A student having no subjective knowledge decides to match all the 6 entries randomly. Find the number of ways in which he can answer, to get atleast 25 % marks in this question.

194. Find the number of positive integral solutions of, $x^2 - y^2 = 352706$

195. Find the nonzero value of ' x ' for which the fourth term in the expansion $\left(5^{\frac{2}{5}\log_5\sqrt{4^x + 44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1} + 7}}}\right)^8$, is 336.

196. In the binomial expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7th term from the beginning to the 7th term from the end is 1 : 6 ; find n.

197. Find the coefficient of $a^5 b^4 c^7$ in the expansion of $(bc + ca + ab)^8$.

198. Find the number of positive integral solutions of xyz = 21600

199. Find the value of 8k for which the expression $3x^2 + 2xy + y^2 + 4x + y + k$ can be resolved into two linear factors.

200. How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if, each digit is to be used atmost one.

END OF EXERCISE # 01

EXERCISE # 02

SECTION-1 : (ONE OPTION CORRECT TYPE)

201. $\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ is a square root of the unit matrix of second order if δ is equal to

(A) α (B) β

(C) γ (D) none of these

202. Let $\vec{a}$ and $\vec{b}$ are unit vectors such that $|\vec{a} + \vec{b}| = \sqrt{3}$, then the value of $(2\vec{a}+5\vec{b})\cdot(3\vec{a}+\vec{b}+\vec{a}\times\vec{b})$ is equal to

(A) $\frac{11}{2}$ (B) $\frac{13}{2}$

(C) $\frac{39}{2}$ (D) none of these

203. Out of 40 consecutive integers two are chosen at random, the probability that their sum is odd is

(A) $\frac{14}{29}$ (B) $\frac{20}{39}$

(C) $\frac{1}{2}$ (D) none of these

204. 5 different games are to be distributed among 4 children randomly. The probability that each child get atleast one game is

(A) $\frac{1}{4}$ (B) $\frac{15}{64}$

(C) $\frac{21}{64}$ (D) None of these

205. If the shortest distance between lines $\vec{r} = \hat{i}+2\hat{j}+3\hat{k}+\lambda_1(2\hat{i}+3\hat{j}+4\hat{k})$ and $\vec{r} = 2\hat{i}+4\hat{j}+5\hat{k}+\lambda_2(3\hat{i}+4\hat{j}+5\hat{k})$ is x, then $\cos^{-1}\cos\sqrt{6}x$ is equal to

(A) $\frac{1}{2}$ (B) 0

(C) 1 (D) π

206. Let $|\vec{a}| = 3,\ |\vec{b}| = 4,\ |\vec{c}| = 5$ and $\vec{a}\perp(\vec{b}+\vec{c}),\ \vec{b}\perp(\vec{c}+\vec{a})$ and $\vec{c}\perp(\vec{a}+\vec{b})$. Then $|\vec{a}+\vec{b}+\vec{c}|$ is

(A) $\sqrt{14}$ (B) $\sqrt{6}$

(C) $\sqrt{12}$ (D) None of these

207. $\frac{d}{dx}\begin{vmatrix} x^2 & x+1 & 3 \\ 1 & 2x-1 & x^3 \\ 0 & x & -2 \end{vmatrix} = -6x^5 - 3$. Number of possible solution of the given equation is

(A) 5 (B) 3

(C) 2 (D) 1

208. A pair of dice is rolled till a sum of either 5 or 7 is obtained. Then the probability that 5 comes before 7 is

(A) $\frac{1}{5}$ (B) $\frac{2}{3}$

(C) $\frac{4}{7}$ (D) $\frac{2}{5}$

209. Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and $a_{pq} = (i^p)^q$, where $i = \sqrt{-1}$. The value of Δ is

(A) real and positive (B) real and negative

(C) 0 (D) imaginary

210. If $f(x) = \begin{cases} x\left[\dfrac{e^{(1/x)^n} - e^{-(1/x)^n}}{e^{(1/x)^n} + e^{-(1/x)^n}}\right] & x \neq 0 \\ 0 & x = 0 \end{cases}$ where, $n \in \{1, 2, 3, \dots, 15\}$, then the probability that f(x) is differentiable is

(A) $\frac{1}{2}$ (B) $\frac{8}{15}$

(C) $\frac{7}{15}$ (D) $\frac{1}{3}$

211. $\vec{b} = \left(\sqrt{4\cos\frac{\alpha}{2}},\ -5,\ \tan\alpha\right)$, $\vec{c} = \left(\dfrac{3}{\sqrt{\cos\frac{\alpha}{2}}},\ \tan\alpha,\ \tan\alpha\right)$ are perpendicular to each other, $\vec{a}$ = (sin 2α, 1, – 2) makes an obtuse angle with x-axis, then α is equal to

(A) $2n\pi + \tan^{-1} 2$ (B) $n\pi + \tan^{-1} 2$

(C) $2n\pi + \pi + \tan^{-1} 3$ (D) none of these

212. The probability that quadratic equation $x^2 + ax + b = 0$ does not have real distinct roots if a, b are selected at random from the set {1, 2, 3, 4} is

(A) $\frac{9}{16}$ (B) $\frac{11}{16}$

(C) $\frac{13}{16}$ (D) None of these

213. Let, $\vec{a} = \hat{i} + \hat{j} + \hat{k}; \vec{b} = \hat{i} - \hat{j} + \hat{k}$, then the point of intersection of lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is

(A) (1, 2, 2) (B) (2, 1, 2)

(C) (2, 1, 1) (D) (2, 0, 2)

214. If A and B are squares matrices such that $A^{2006} = 0$ and AB = A + B, then det(B) =

(A) 0 (B) 1

(C) – 1 (D) None of these

215. Given 2006 vectors in the plane. The sum of every 2005 vectors is a multiple of the vector. Not all the vectors all multiple of each other. The sum of all the vectors is

(A) necessarily a zero vector (B) may be a zero vector

(C) can never be a zero vector (D) can't say

216. Box A contains black balls and box B contains white balls take a certain number of balls from A and place them in B, then take same number of balls from B and place them in A. The probability that number of white balls in A is equal to number of black balls in B is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) 1 (D) None of these

217. The number of planes that are equidistant from four non-coplanar points is

(A) 3 (B) 4

(C) 7 (D) 9

218. A plane passing through (1, 1, 1) cuts positive direction of co-ordinate axes at A, B and C the volume of tetrahedron OABC satisfies

(A) $V \le \frac{9}{2}$ (B) $V \ge \frac{9}{2}$

(C) $V = \frac{9}{2}$ (D) None of these

219. A unit vector is orthogonal to $5\hat{i}+2\hat{j}+6\hat{k}$ and is coplanar to $2\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$, then the vector is

(A) $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$ (B) $\frac{2\hat{i}+5\hat{j}}{\sqrt{29}}$

(C) $\frac{6\hat{i}-5\hat{k}}{\sqrt{61}}$ (D) $\frac{2\hat{i}+2\hat{j}-\hat{k}}{3}$

220. A cubical die with faces marked 1, 2, 3, …, 6 is loaded such that the probability of throwing the number t is proportional to t^2. The probability that the number 5 has appeared given that when the die is rolled the number turned up is not even, is

(A) $\frac{1}{7}$ (B) $\frac{3}{7}$

(C) $\frac{5}{7}$ (D) $\frac{2}{3}$

221. Let X be a set containing n elements. If two subsets A and B of X are picked at random, the probability that A and B have the same number of elements is

(A) $\frac{{}^{2n}C_n}{2^n}$ (B) $\frac{1}{{}^{2n}C_n}$

(C) $\frac{1\cdot 3\cdot 5..(2n-1)}{2^n\cdot n!}$ (D) $\frac{3^n}{4^n}$

222. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors such that $\vec{b}\times\vec{c}=\vec{a}$, $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{c}\times\vec{a}=\vec{b}$ then

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 1$ (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] \neq 1$

(C) $|\vec{a}|+|\vec{b}|+|\vec{c}|=3$ (D) None of these

223. Value of $\Delta = \begin{vmatrix} \sin(2\alpha) & \sin(\alpha+\beta) & \sin(\alpha+\gamma) \\ \sin(\beta+\alpha) & \sin(2\beta) & \sin(\gamma+\beta) \\ \sin(\gamma+\alpha) & \sin(\gamma+\beta) & \sin(2\gamma) \end{vmatrix}$ is

(A) $\Delta = 0$ (B) $\Delta = \sin^2\alpha + \sin^2\beta + \sin^2\gamma$
(C) $\Delta = 3/2$ (D) None of these

224. $\Delta = \begin{vmatrix} 1 & \frac{4\sin B}{b} & \cos A \\ 2a & 8\sin A & 1 \\ 3a & 12\sin A & \cos B \end{vmatrix}$ is (where a, b, c are the sides opposite to angles A, B, C respectively in a triangle)

(A) $\frac{1}{2}\cos 2A$ (B) 0 (C) $\frac{1}{2}\sin 2A$ (D) $\frac{1}{2}(\cos^2 A + \cos^2 B)$

225. Let m be a positive integer & $D_r = \begin{vmatrix} 2r-1 & {}^mC_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$ $(0 \le r \le m)$, then the value of $\sum_{r=0}^{m} D_r$ is given by:
(A) 0 (B) $m^2 - 1$ (C) 2^m (D) $2^m \sin^2(2^m)$

226. If a, b, c, are real numbers, and $D = \begin{vmatrix} a & 1+2i & 3-5i \\ 1-2i & b & -7-3i \\ 3+5i & -7+3i & c \end{vmatrix}$ then D is

(A) purely real (B) purely imaginary
(C) non real (D) integer

227. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$ then f(100) is equal to:

(A) 0 (B) 1 (C) 100 (D) – 100

228. Identity the correct statement
(A) If system of n simultaneous linear equations has a unique solution, then coefficient matrix is singular
(B) If system of n simultaneous linear equations has a unique solution, then coefficient matrix is non singular
(C) If A^{-1} exists, $(\text{adj } A)^{-1}$ may or may not exist

(D) $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then $F(x) \,.\, F(y) = F(x-y)$

229. Matrix A = $\begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if x y z = 60 and 8x + 4y + 3z = 20, then A(adj A) is equal to

(A) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ (B) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$

(C) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ (D) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$

230. $\left[\dfrac{\vec{a}}{|\vec{a}|^2} - \dfrac{\vec{b}}{|\vec{b}|^2}\right]^2$ =

(A) $|\vec{a}|^2 - |\vec{b}|^2$ (B) $\left[\dfrac{\vec{a} - \vec{b}}{|\vec{a}|\,|\vec{b}|}\right]^2$

(C) $\left[\dfrac{\vec{a}\,|\vec{a}| - \vec{b}\,|\vec{b}|}{|\vec{a}|\,|\vec{b}|}\right]^2$ (D) None

231. A, B, C & D are four points in a plane with pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ respectively such that $(\vec{a} - \vec{d}).(\vec{b} - \vec{c}) = (\vec{b} - \vec{d}).(\vec{c} - \vec{a})$ = 0. Then for the triangle ABC, D is its:

(A) incentre (B) circumcentre
(C) orthocentre (D) centroid

232. Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \dfrac{2\pi}{3}$. If $|\vec{a}|$ = 1, $|\vec{b}|$ = 2 then $\left\{(\vec{a} + 3\vec{b}) \text{ x } (3\vec{a} - \vec{b})\right\}^2$ =

(A) 225 (B) 250
(C) 275 (D) 300

233. Consider a tetrahedron with faces f_1, f_2, f_3, f_4. Let $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ be the vectors whose magnitudes are respectively equal to the areas of f_1, f_2, f_3, f_4 & whose directions are perpendicular to these faces in the outward direction. Then,

(A) $\vec{a}_1 + \vec{a}_2 + \vec{a}_3 + \vec{a}_4 = 0$ (B) $\vec{a}_1 + \vec{a}_3 = \vec{a}_2 + \vec{a}_4$

(C) $\vec{a}_1 + \vec{a}_2 = \vec{a}_3 + \vec{a}_4$ (D) None

234. For non–zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\left|\vec{a} \times \vec{b}.\vec{c}\right| = \left|\vec{a}\right| \left|\vec{b}\right| \left|\vec{c}\right|$ holds if and only if;

(A) $\vec{a}.\vec{b} = 0, \vec{b}.\vec{c} = 0$ (B) $\vec{c}.\vec{a} = 0, \vec{a}.\vec{b} = 0$

(C) $\vec{a}.\vec{c} = 0, \vec{b}.\vec{c} = 0$ (D) $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$

235. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}, \vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} =$

(A) 2 (B) 4 (C) 16 (D) 64

236. A point taken on each median of a triangle divides the median in the ratio 1 : 3, reckoning from the vertex. Then the ratio of the area of the triangle with vertices at these points to that of the original triangle is:

(A) 5 : 13 (B) 25 : 64 (C) 13 : 32 (D) Nnone

237. Let the centre of the parallelopiped formed by $\vec{PA} = \hat{i} + 2\hat{j} + 2\hat{k}; \vec{PB} = 4\hat{i} - 3\hat{j} + \hat{k};$

$\vec{PC} = 3\hat{i} + 5\hat{j} - \hat{k}$ is given by the position vector (7, 6, 2). Then the position vector of the point P is:

(A) (3, 4, 1) (B) (6, 8, 2) (C) (1, 3, 4) (D) (2, 6, 8)

238. Taken on side $\vec{AC}$ of a triangle ABC, a point M such that $\vec{AM} = \frac{1}{3}\vec{AC}$. A point N is taken on the side $\vec{CB}$ such that $\vec{BN} = \vec{CB}$ then, for the point of intersection X of $\vec{AB}$ & $\vec{MN}$ which of the following holds good?

(A) $\vec{XB} = \frac{1}{3}\vec{AB}$ (B) $\vec{AX} = \frac{1}{3}\vec{AB}$ (C) $\vec{XN} = \frac{3}{4}\vec{MN}$ (D) $\vec{XM} = 3\vec{XN}$

239. If the acute angle that the vector, $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ makes with the plane of the two vectors $2\hat{i} + 3\hat{j} - \hat{k}$ & $\hat{i} - \hat{j} + 2\hat{k}$ is $\cot^{-1}\sqrt{2}$ then:

(A) $\alpha(\beta + \gamma) = \beta\gamma$ (B) $\beta(\gamma + \alpha) = \gamma\alpha$

(C) $\gamma(\alpha + \beta) = \alpha\beta$ (D) $\alpha\beta + \beta\gamma + \gamma\alpha = 0$

240. Locus of the point P for which $\vec{OP}$ represents a vector with direction cosine $\cos\alpha = \frac{1}{2}$ ('O' is the origin) is:

(A) A circle parallel to y z plane with centre on the x – axis

(B) a cone concentric with positive x – axis having vertex at the origin and the slant height equal to the magnitude of the vector

(C) a ray emanating from the origin and making an angle of 60º with x – axis

(D) a disc parallel to y z plane with centre on x – axis & radius equal to $\left|\vec{OP}\right| \sin 60º$

241. There are 4 urns. The first urn contains 1 white & 1 black ball, the second urn contains 2 white & 3 black balls, the third urn contains 3 white & 5 black balls & the fourth urn contains 4 white & 7 black balls. The selection of each urn is not equally likely. The probability of selecting i^{th} urn is $\frac{i^2+1}{34}$ (i = 1, 2, 3, 4). If we randomly select one of the urns & draw a ball, then the probability of ball being white is :

(A) $\frac{569}{1496}$ (B) $\frac{27}{56}$

(C) $\frac{8}{73}$ (D) none of these

242. $2/3^{rd}$ of the students in a class are boys & the rest girls. It is known that probability of a girl getting a first class is 0.25 & that of a boy is 0.28. The probability that a student chosen at random will get a first class is:

(A) 0.26 (B) 0.265

(C) 0.27 (D) 0.275

243. The contents of urn I and II are as follows,

Urn I: 4 white and 5 black balls

Urn II: 3 white and 6 black balls

One urn is chosen at random and a ball is drawn and its colour is noted and replaced back to the urn. Again a ball is drawn from the same urn, colour is noted and replaced. The process is repeated 4 times and as a result one ball of white colour and 3 of black colour are noted. Find the probability the chosen urn was I.

(A) $\frac{125}{287}$ (B) $\frac{64}{127}$

(C) $\frac{25}{287}$ (D) $\frac{79}{192}$

244. The sides of a rectangle are chosen at random, each less than 10 cm, all such lengths being equally likely. The chance that the diagonal of the rectangle is less than 10 cm is

(A) 1/10 (B) 1/20

(C) $\pi/4$ (D) $\pi/8$

245. The sum of two positive quantities is equal to 2n. The probability that their product is not less than 3/4 times their greatest product is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) None of these

SECTION–2 : (MORE THAN ONE OPTION CORRECT TYPE)

246. A plane through the line $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z}{3}$ has the equation

(A) $x + y - z = 0$ (B) $2x - 7y + z + 9 = 0$

(C) $x + 4y - 2z - 3 = 0$ (D) None of these

247. Let $\hat{\alpha}$, $\hat{\beta}$ and $\hat{\gamma}$ be the unit vectors such that $\hat{\alpha}$ and $\hat{\beta}$ are mutually perpendicular and $\hat{\gamma}$ is equally inclined to $\hat{\alpha}$ and $\hat{\beta}$ at an angle θ. If $\hat{\gamma} = x\hat{\alpha} + y\hat{\beta} + z\left(\hat{\alpha} \times \hat{\beta}\right)$, then

(A) $z^2 = 1 - 2x^2$ (B) $z^2 = 1 - 2y^2$ (C) $z^2 = 1 - x^2 - y^2$ (D) $x^2 = y^2$

248. If a, b, c form an A.P. with common difference d ($\neq 0$) and x, y, z form a G.P. with common ratio r ($\neq 1$), then the area of the triangle with vertices; (a, x), (b, y) and (c, z) is independent of

(A) a (B) b (C) x (D) r

249. The digits A, B, C are such that the three digit numbers A88, 6B8, 86C are divisible by 72, then the determinant $\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$ is divisible by

(A) 216 (B) 72

(C) 144 (D) 288

250. If $\vec{a} + 2\vec{b} + 3\vec{c} = 0$, then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} =$

(A) $2\left(\vec{a} \times \vec{b}\right)$ (B) $6\left(\vec{b} \times \vec{c}\right)$

(C) $3\left(\vec{a} \times \vec{b}\right)$ (D) 0

251. Which of the following is/are orthogonal matrix

(A) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ (B) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} \sqrt{3} & 1 & 0 \\ -1 & \sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

252. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}| = 0$, then which of the following case(s) is/are true

(A) $\vec{a}$ is perpendicular to $\vec{b}$ (B) either $\vec{a}$ or $\vec{b}$ is $\vec{0}$

(C) $\vec{a} + \vec{b}$ must be equal to $\vec{a} - \vec{b}$ (D) None of these

253. The det $\Delta = \begin{vmatrix} d^2 + r & de & df \\ de & e^2 + r & ef \\ df & ef & f^2 + r \end{vmatrix}$ is divisible by

(A) r^2 (B) $(d + e^2 + f^2 + r)$

(C) $(d^2 + e^2 + f^2 + r)$ (D) $(d^2 + e + f^2 + r^2)$

254. Let $\phi_1(x) = x + a_1$, $\phi_2(x) = x^2 + b_1x + b_2$ and $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \phi_1(x_1) & \phi_1(x_2) & \phi_1(x_3) \\ \phi_2(x_1) & \phi_2(x_2) & \phi_2(x_3) \end{vmatrix}$, then

(A) Δ is independent of a_1 (B) Δ is independent of b_1 and b_2
(C) Δ is independent of x_1, x_2 and x_3 (D) None of these

255. If $\Delta = \begin{vmatrix} x & 2y-z & -z \\ y & 2x-z & -z \\ y & 2y-z & 2x-2y-z \end{vmatrix}$, then

(A) $x - y$ is a factor of Δ (B) $(x - y)^2$ is a factor of Δ
(C) $(x - y)^3$ is a factor of Δ (D) Δ is independent of z

256. Let $\Delta = \begin{vmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\theta\sin\phi & \sin\theta\cos\phi & 0 \end{vmatrix}$, then

(A) Δ is independent of θ (B) Δ is indepedent of ϕ

(C) Δ is a constant (D) $\left.\frac{d\Delta}{d\theta}\right]_{\theta=\pi/2} = 0$

257. Let $\Delta = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, then

(A) $x + a$ is a factor of Δ (B) $(x + a)^2$ is a factor of Δ
(C) $(x + a)^3$ is a factor of Δ (D) $(x + a)^4$ is not a factor of Δ

258. Let $\Delta = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$, then

(A) $1 - x^3$ is a factor of Δ (B) $(1 - x^3)^2$ is factor of Δ
(C) $\Delta(x) = 0$ has 4 real roots (D) $\Delta'(1) = 0$

259. The determinent $\Delta = \begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ b\alpha + c & c\alpha + d & a\alpha^3 - c\alpha \end{vmatrix}$ is equal to zero if

(A) b, c, d are in A.P. (B) b, c, d are in G.P.
(C) b, c, d are in H.P. (D) α is a root of $ax^3 - bx^2 - 3cx - d = 0$

260. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is :

(A) 2 if a = 6 (B) 2 if a = 1 (C) 1 if a = 2 (D) 1 if a = – 6

261. Which of the following statement is always true

(A) Adjoint of a symmetric matrix is a symmetric matrix

(B) Adjoint of a unit matrix is unit matrix

(C) A (adj A) = (adj A) A

(D) Adjoint of a diagonal matrix is diagonal matrix

262. Matrix $\begin{bmatrix} a & b & (a\alpha - b) \\ b & c & (b\alpha - c) \\ 2 & 1 & 0 \end{bmatrix}$ is non invertible if

(A) $\alpha = 1/2$ (B) a, b, c are in A.P.

(C) a, b, c are in G.P. (D) a, b, c are in H.P.

263. The singularity of matrix $\begin{bmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{bmatrix}$ depends upon which of the following parameter

(A) a (B) p (C) x (D) d

264. Which of the following statement is true

(A) Every skew symmetric matrix of odd order is non singular

(B) If determinant of a square matrix is nonzero, then it non singular

(C) Rank of a matrix is equal or higher than the order of the matrix

(D) Adjoint of a singular matrix is always singular

265. If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (where $bc \neq 0$) satisfies the equations $x^2 + k = 0$, then

(A) a + d = 0 (B) k = – |A| (C) k = |A| (D) none of these

266 If $A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, then

(A) | A | = 2 (B) A is non-singular

(C) Adj. A = $\begin{bmatrix} 1/2 & -1/2 & 0 \\ 0 & -1 & 1/2 \\ 0 & 0 & -1/2 \end{bmatrix}$ (D) A is skew symmetric matrix

267. If $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are linearly independent set of vectors & $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$ then:

(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) None of these

268. Given three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ such that they are non – zero, non – coplanar vectors, then which of the following are coplanar.

(A) $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ (B) $\vec{a} - \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$

(C) $\vec{a} + \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} + \vec{a}$ (D) $\vec{a} + \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} - \vec{a}$

269. Let $\vec{p} = 2\hat{i} + 3\hat{j} + a\hat{k}$, $\vec{q} = b\hat{i} + 5\hat{j} - \hat{k}$ & $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}, \vec{q}, \vec{r}$ are coplanar and $\vec{p}\cdot\vec{q}$ = 20, a & b have the values:

(A) 1, 3 (B) 9, 7 (C) 5, 5 (D) 13, 9

270. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ & $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ & $\hat{j}$ system where $|\vec{z}_1| = |\vec{z}_2|$ = r & $\vec{z}_1 . \vec{z}_2$ = 0 then $\vec{w}_1 = a\hat{i} + c\hat{j}$ & $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy:

(A) $|\vec{w}_1|$ = r (B) $|\vec{w}_2|$ = r (C) $\vec{w}_1 . \vec{w}_2$ = 0 (D) None of these

271. If $\vec{a}$ & $\vec{b}$ are two non colinear unit vectors & $\vec{a}, \vec{b}, x\vec{a} - y\vec{b}$ form a triangle, then:

(A) x = – 1; y = 1 & $|\vec{a} + \vec{b}| = 2\cos\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)$

(B) x = – 1; y = 1 & $\cos\left(\vec{a}\,\hat{}\,\vec{b}\right) + |\vec{a} + \vec{b}|\cos\left[\vec{a}, \hat{-}(\vec{a} + \vec{b})\right] = -1$

(C) $|\vec{a} + \vec{b}| = -2\cot\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)\cos\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)$ & x = – 1, y = 1

(D) none

272. The value(s) of $\alpha \in [0, 2\pi]$ for which vector $\vec{a} = \hat{i} + 3\hat{j} + (\sin 2\alpha)\hat{k}$ makes an obtuse angle with the Z-axis and the vectors $\vec{b} = (\tan\alpha)\hat{i} - \hat{j} + 2\sqrt{\sin\frac{\alpha}{2}}\,\hat{k}$ and $\vec{c} = (\tan\alpha)\hat{i} + (\tan\alpha)\hat{j} - 3\sqrt{\operatorname{cosec}\frac{\alpha}{2}}\,\hat{k}$ are orthogonal, is/are:

(A) $\tan^{-1} 3$ (B) $\pi - \tan^{-1} 2$ (C) $\pi + \tan^{-1} 3$ (D) $2\pi - \tan^{-1} 2$

273. A parallelogram is constructed on the vectors $\vec{p}$ & $\vec{q}$. A vector which coincides with the altitude of the parallelogram & perpendicular to the side $\vec{p}$ expressed in terms of the vectors $\vec{p}$ & $\vec{q}$ is:

(A) $\vec{q} - \frac{\vec{q}\cdot\vec{p}}{(\vec{p})^2}\vec{p}$ (B) $\frac{(\vec{p} \times \vec{q}) \times \vec{p}}{\vec{p}^2}$ (C) $\frac{\vec{q}\cdot\vec{p}}{\vec{p}^2}\vec{p} - \vec{q}$ (D) $\frac{\vec{p} \times (\vec{p} \times \vec{q})}{\vec{p}^2}$

274. Identify the statement(s) which is/are incorrect?

(A) $\vec{a} \times \left[\vec{a} \times (\vec{a} \times \vec{b})\right] = (\vec{a} \times \vec{b})(\vec{a}^2)$

(B) If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and $\vec{v}\cdot\vec{a} = \vec{v}\cdot\vec{b} = \vec{v}\cdot\vec{c} = 0$ then $\vec{v}$ must be a null vector

(C) If $\vec{a}$ and $\vec{b}$ lie in a plane normal to the plane containing the vectors $\vec{c}$ and $\vec{d}$ then $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0$

(D) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then $\vec{a}\cdot\vec{b}' + \vec{b}\cdot\vec{c}' + \vec{c}\cdot\vec{a}' = 3$

275. If $a = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 4\hat{j} - 4\hat{k}$, then the vector $\vec{a} \times (\vec{b} \times \vec{c})$ is orthogonal to:

(A) $\vec{a}$ (B) $\vec{b}$ (C) $\vec{c}$ (D) $\vec{a} + \vec{b} + \vec{c}$

276. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-zero, non-collinear vectors such that a vector $\vec{p}$ = a b $\cos\left(2\pi - \left(\vec{a} \wedge \vec{b}\right)\right) \vec{c}$ and a vector $\vec{q}$ = a c $\cos\left(\pi - \left(\vec{a} \wedge \vec{c}\right)\right) \vec{b}$ then $\vec{p} + \vec{q}$ is

(A) parallel to $\vec{a}$ (B) perpendicular to $\vec{a}$

(C) coplanar with $\vec{b}$ & $\vec{c}$ (D) none of these

277. Which of the following statement(s) is/are true?

(A) If $\vec{n} \cdot \vec{a} = 0$, $\vec{n} \cdot \vec{b} = 0$ & $\vec{n} \cdot \vec{c} = 0$ for some non zero vector $\vec{n}$, then $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$

(B) there exist a vector having direction angles $\alpha = 30°$ & $\beta = 45°$

(C) locus of point for which x = 3 & y = 4 is a line parallel to the z - axis whose distance from the z axis is 5

(D) the vertices of a regular tetrahedron are OABC where 'O' is the origin. The vector $\vec{OA} + \vec{OB} + \vec{OC}$ is perpendicular to the plane ABC.

278. In a ΔABC, let M be the mid point of segment AB and let D be the foot of the bisector of $\angle$C. Then the ratio $\frac{\text{Area } \Delta \text{ CDM}}{\text{Area } \Delta \text{ ABC}}$ is:

(A) $\frac{1}{4}\frac{a-b}{a+b}$ (B) $\frac{1}{2}\frac{a-b}{a+b}$

(C) $\frac{1}{2}\tan\frac{A-B}{2}\cot\frac{A+B}{2}$ (D) $\frac{1}{4}\cot\frac{A-B}{2}\tan\frac{A+B}{2}$

279. The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are of the same length & pairwise form equal angles. If $\vec{a} = \hat{i} + \hat{j}$ & $\vec{b} = \hat{j} + \hat{k}$ the pv's of $\vec{c}$ can be:

(A) (1, 0, 1) (B) $\left(-\frac{4}{3}, \frac{1}{3}, -\frac{4}{3}\right)$ (C) $\left(\frac{1}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (D) $\left(-\frac{1}{3}, \frac{4}{3}, -\frac{1}{3}\right)$

280. Equation of the plane passing through $A(x_1, y_1, z_1)$ and containing the line $\frac{x - x_2}{d_1} = \frac{y - y_2}{d_2} = \frac{z - z_2}{d_3}$ is

(A) $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$ (B) $\begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$

(C) $\begin{vmatrix} x - d_1 & y - d_2 & z - d_3 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0$ (D) $\begin{vmatrix} x & y & z \\ x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$

281. The equations of the line of shortest distance between the lines

$\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z-2}{2}$ are

(A) $3(x-21) = 3y + 92 = 3z - 32$

(B) $\frac{x-(62/3)}{1/3} = \frac{y+31}{1/3} = \frac{z-(31/3)}{1/3}$

(C) $\frac{x-21}{1/3} = \frac{y+(92/3)}{1/3} = \frac{z-(32/3)}{1/3}$

(D) $\frac{x-2}{1/3} = \frac{y+3}{1/3} = \frac{z-1}{1/3}$

282. A line passes through a point A with p.v. $3\hat{i} + \hat{j} - \hat{k}$ & is parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$. If P is a point on this line such that AP = 15 units, then the p.v. of the point P is:

(A) $13\hat{i} + 4\hat{j} - 9\hat{k}$

(B) $13\hat{i} - 4\hat{j} + 9\hat{k}$

(C) $7\hat{i} - 6\hat{j} + 11\hat{k}$

(D) $-7\hat{i} + 6\hat{j} - 11\hat{k}$

283. The equations of the planes through the origin which are parallel to the line

$\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z+1}{-2}$ and at a distance $\frac{5}{3}$ from it are

(A) $2x + 2y + z = 0$

(B) $x + 2y + 2z = 0$

(C) $2x - 2y + z = 0$

(D) $x - 2y + 2z = 0$

284. The value(s) of k for which the equation $x^2 + 2y^2 - 5z^2 + 2kyz + 2zx + 4xy = 0$ represents a pair of planes passing through origin is/are

(A) 2 (B) – 2 (C) 6 (D) – 6

285. The equation of lines AB is $\frac{x}{2} = \frac{y}{-3} = \frac{2}{6}$. Through a point P(1, 2, 5), line PN is drawn perpendicular to AB and line PQ is drawn parallel to the plane $3x + 4y + 5z = 0$ to meet AB is Q. Then

(A) coordinate of N is $\left(\frac{52}{49}, -\frac{78}{49}, \frac{156}{49}\right)$

(B) the coordinates of Q is $\left(3, -\frac{9}{2}, 9\right)$

(C) the equation of PN is $\frac{x-1}{3} = \frac{y-2}{-176} = \frac{z-5}{-89}$

(D) the equation of PQ is $\frac{x-1}{4} = \frac{y-2}{-13} = \frac{z-5}{8}$

286. Let a perpendicular PQ be drawn from P (5, 7, 3) to the line $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$ when Q is the foot. Then

(A) Q is (9, 13, – 15)

(B) PQ = 14

(C) the equation of plane containing PQ and the given line is $9x - 4y - z - 14 = 0$

(D) none of these

287. In throwing a die let A be the event 'coming up of an odd number', B be the event 'coming up of an even number', C be the event 'coming up of a number ≥ 4' and D be the event 'coming up of a number < 3', then

(A) A and B are mutually exclusive and exhautive

(B) A and C are mutually exclusive and exhautive

(C) A, C and D form an exhautive system

(D) B, C and D form an exhautive system

288. Let $0 < P(A) < 1$, $0 < P(B) < 1$ & $P(A \cup B) = P(A) + P(B) - P(A).\ P(B)$, then:

(A) $P(B/A) = P(B) - P(A)$

(B) $P(A^C \cup B^C) = P(A^C) + P(B^C)$

(C) $P((A \cup B)^C) = P(A^C).\ P(B^C)$

(D) $P(A/B) = P(A)$

289. For any two events A & B defined on a sample space,

(A) $P\left(A/B\right) \geq \dfrac{P(A) + P(B) - 1}{P(B)}$, $P(B) \neq 0$ is always true

(B) $P\left(A \cup \overline{B}\right) = P(A) - P(A \cap B)$

(C) $P(A \cup B) = 1 - P(A^c).\ P(B^c)$, if A & B are independent

(D) $P(A \cup B) = 1 - P(A^c).\ P(B^c)$, if A & B are disjoint

290. If A, B & C are three events, then the probability that none of them occurs is given by:

(A) $P\left(\overline{A}\right) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

(B) $P\left(\overline{A}\right) + P\left(\overline{B}\right) + P\left(\overline{C}\right)$

(C) $P\left(\overline{A}\right) - P(B) - P(C) + P(A \cap B) + P(B \cap C) + P(C \cap A) - P(A \cap B \cap C)$

(D) $P\left(\overline{A} \cup \overline{B} \cup \overline{C}\right) - P(A) - P(B) - P(C) + P(A \cap B) + P(B \cap C) + P(C \cap A)$

291. A student appears for tests I, II & III. The student is successful if he passes either in tests I & II or tests I & III. The probabilities of the student passing in the tests I, II & III are p, q & 1/2 respectively. If the probability that the student is successful is 1/2, then:

(A) $p = 1,\ q = 0$

(B) $p = 2/3,\ q = 1/2$

(C) $p = 3/5,\ q = 2/3$

(D) there are infinitely many values of p & q.

SECTION–3 : (COMPREHENSION TYPE)

COMPREHENSION-1

Paragraph for Questions Nos. 292 to 294

Letters of word TITANIC are arranged to form all the possible anagrams. What is the probability that anagrams (words) will have

292. Both T together

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{5}{7}$ (D) $\frac{6}{7}$

293. Starting letter T and ending with A

(A) $\frac{2}{15}$ (B) $\frac{3}{16}$ (C) $\frac{1}{21}$ (D) None of these

294. Starting letter as either T or vowel

(A) $\frac{4}{7}$ (B) $\frac{5}{7}$ (C) $\frac{6}{7}$ (D) $\frac{3}{7}$

COMPREHENSION-2

Paragraph for Questions Nos. 295 to 297

A and B are square matrices of order 3 given by $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & k & 5 \\ 4 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$.

295. If matrix A is singular matrix, then value of k is

(A) 5 (B) – 8 (C) 8 (D) – 5

296. If k = 2, then tr(AB) is equal to

(A) 66 (B) 42 (C) 84 (D) 63

297. If C = A – B and tr(C) = 0, then k is equal to

(A) 5 (B) – 5 (C) 7 (D) – 7

COMPREHENSION-3

Paragraph for Questions Nos. 298 to 300

Two lines whose equations are $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{\lambda}$ and $\frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3}$ lie in the same plane, then

298. The value of $\sin^{-1}\sin\lambda$ is equal to

(A) 3 (B) $\pi - 3$ (C) 4 (D) $\pi - 4$

299. Point of intersection of the lines lies on

(A) $3x + y + z = 20$ (B) $3x + y + z = 25$

(C) $3x + 2y + z = 24$ (D) None of these

300. Equation of plane containing both lines

(A) $x + 5y - 3z = 10$ (B) $x + 6y + 5z = 20$

(C) $x + 6y - 5z = 10$ (D) None of these

COMPREHENSION-4

Paragraph for Questions Nos. 301 to 303

Each question contains 4 statements, each statement is either true or false. You have to tick the correct order of sequence. If you tick the alternative marked as TFFT it would mean that 1^{st} is true, 2^{nd} and 3^{rd} false and 4^{th} is true.

301. A and B are symmetric matrices of same order then
Statement (1): A + B is skew–symmetric
Statement (2): AB – BA is skew–symmetric
Statement (3): AB + BA is skew–symmetric
Statement (4): A – B is skew–symmetric
(A) TFTF (B) FTFT (C) TTTT (D) FTFF

302. Statement (1): if A and B are symmetric then AB is symmetric $\Leftrightarrow$ A and B commute
Statement (2): if A is symmetric, then B^TAB is symmetric
Statement (3): All positive odd integral power of skew-symmetric matrix are symmetric
Statement (4): All positive even integral power of skew-symmetric matrix are symmetric
(A) TFTF (B) FFFF (C) TTFT (D) TFFF

303. The matrix which commute with A = $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
Statement (1): Are always singular
Statement (2): Are always non-singular
Statement (3): Are always symmetric
Statement (4): Are always of the form $\begin{bmatrix} x & y \\ 0 & x \end{bmatrix}$, where x and y are variable,
(A) FFFF (B) TFTF (C) FTFT (D) FFFT

COMPREHENSION-5

Paragraph for Questions Nos. 304 to 306

T is the region of the plane x + y + z = 1 with x, y, z > 0. S is the set of points (a, b, c) in T such that just two of the following three inequalities hold: $a \le \frac{1}{2}$, $b \le \frac{1}{3}$, $c \le \frac{1}{6}$.

304. Area of the region T is
(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) none of these

305. Area of the region S is
(A) $\frac{\sqrt{3}}{72}$ (B) $\frac{7\sqrt{3}}{36}$ (C) $\frac{\sqrt{3}}{4}$ (D) None of these

306. The difference of the region T and region S consists of
(A) three parallelograms (B) three equilateral triangles
(C) three rectangles (D) none of these

COMPREHENSION # 06

Paragraph for Questions Nos. 307 to 309

There are four boxes A_1, A_2, A_3 and A_4. Box A_i has i cards and on each card a number is printed, the numbers are from 1 to i. A box is selected randomly, the probability of selection of box A_i is $\frac{i}{10}$ and then a card is drawn. Let E_i represents the event that a card with number 'i' is drawn.

307. $P(E_1)$ is equal to

(A) $\frac{1}{5}$ (B) $\frac{1}{10}$ (C) $\frac{2}{5}$ (D) $\frac{1}{4}$

308. $P(A_3/E_2)$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

309. Expectation of the number on the card is

(A) 2 (B) 2.5

(C) 3 (D) 3.5

COMPREHENSION # 07

Paragraph for Questions Nos. 310 to 312

Sania Mirza is to play with Sharapova in a three set match. For a particular set, the probability of Sania winning the set is y and if she wins probability of her winning the next set becomes $\sqrt{y}$ else the probability that she wins the next one becomes y^2. There is no possibility that a set is to be abondoned. R is probability that Sania wins the first set.

310. If R = $\frac{1}{2}$ then the probability that match will end in first two sets is nearly equal to

(A) 0.73 (B) 0.95

(C) 0.51 (D) 0.36

311. If R = $\frac{1}{2}$ and Sania wins the second set probability that she has won first set as well is equal to

(A) 0.74 (B) 0.46

(C) 0.26 (D) 0.54

312. If Sania looses the first set then the values of R such that her probability of winning the match is still larger than that of her loosing is given by

(A) $R \in \left(\frac{1}{2}, 1\right)$ (B) $R \in \left(\left(\frac{1}{2}\right)^{\frac{1}{3}}, 1\right)$

(C) $R \in \left(\left(\frac{1}{2}\right)^{3/2}, 1\right)$ (D) no values of R

COMPREHENSION # 08

Paragraph for Questions Nos. 313 to 315

If a pair of fair and ubaised dice are rolled randomly. The events A, B, C, D, E, F are as follows :

A : getting an even number on the first die.
B : getting an odd number on the first die.
C : getting the sum of numbers on the dice ≤ 5.
D : getting the sum of numbers on the dice > 5 and less than 10.
E : getting the sum of numbers on the dice ≥ 10.
F : getting an odd number on exactly one of the dice.

Make your choice to the most appropriate answer on the basics of above information.

313. Which one of the following is **CORRECT**?
(A) A and C are mutually exclusive
(B) A, B, F are mutually exclusive and exchausive events
(C) A and F are mutually exclusive
(D) $B \subset F$

314. P(E/F) equals
(A) $\frac{1}{6}$ (B) $\frac{2}{27}$ (C) $\frac{1}{9}$ (D) $\frac{2}{9}$

315. P(A/C) =
(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{3}{4}$

COMPREHENSION # 09

Paragraph for Questions Nos. 316 to 318

Consider the experiment of distribution of balls among urns. Suppose we are given M urns, numbered 1 to M, among which we are to distribute n balls (n < M). Let P(A) denote the probability that each of the urns numbered 1 to n will contain exactly one ball. Then answer the following questions.

316. If the balls are different and any number of balls can go to any urns, then P(A) is equal to
(A) $\frac{M!}{n^M}$ (B) $\frac{n!}{M^n}$
(C) $\frac{n!}{{}^MP_n}$ (D) $\frac{1}{M^n}$

317. If the balls are identical and any number of balls can go to any urns, then P(A) equals
(A) $\frac{1}{M^n}$ (B) $\frac{1}{{}^{M+n-1}C_{M-1}}$
(C) $\frac{1}{{}^{M+n-1}C_{n-1}}$ (D) $\frac{1}{{}^{M+n-1}P_{M-1}}$

318. If the balls are identical but atmost one ball can be put in any box, then P(A) is equal to
(A) $\frac{1}{{}^MP_n}$ (B) $\frac{n!}{{}^nC_M}$ (C) $\frac{n!}{{}^MC_n}$ (D) $\frac{1}{{}^MC_n}$

COMPREHENSION # 10

Paragraph for Questions Nos. 319 to 321

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-zero and non-coplanar vectors and $\vec{a}' = \dfrac{\vec{b}\times\vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]}$, $\vec{b}' = \dfrac{\vec{c}\times\vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]}$ and $\vec{c}' = \dfrac{\vec{a}\times\vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$, then $\vec{a}'$, $\vec{b}'$, $\vec{c}'$ are said to form the reciprocal system of the system of vectors $\vec{a}$, $\vec{b}$, $\vec{c}$.

319. $[\vec{a}\ \vec{b}\ \vec{c}'] =$

(A) 0 (B) 1 (C) $\dfrac{|\vec{a}\times\vec{b}|}{[\vec{a}\ \vec{b}\ \vec{c}]}$ (D) $\dfrac{|\vec{a}\times\vec{b}|^2}{[\vec{a}\ \vec{b}\ \vec{c}]}$

320. $[\vec{a}\ \vec{b}'\ \vec{c}'] =$

(A) 0 (B) 1 (C) $\dfrac{\vec{a}^2}{[\vec{a}\ \vec{b}\ \vec{c}]}$ (D) $\dfrac{|\vec{a}|}{[\vec{a}\ \vec{b}\ \vec{c}]}$

321. $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}] =$

(A) $[\vec{a}\ \vec{b}\ \vec{c}]$ (B) $[\vec{a}'\ \vec{b}'\ \vec{c}']$ (C) $\dfrac{2}{[\vec{a}'\ \vec{b}'\ \vec{c}']}$ (D) $2[\vec{a}'\ \vec{b}'\ \vec{c}']$

COMPREHENSION # 11

Paragraph for Questions Nos. 322 to 324

Let $\vec{a} = 2\hat{i}+3\hat{j}-6\hat{k}$, $\vec{b} = 2\hat{i}-3\hat{j}+6\hat{k}$ and $\vec{c} = -2\hat{i}+3\hat{j}+6\hat{k}$. Let $\vec{a}_1$ be projection of $\vec{a}$ on $\vec{b}$ and $\vec{a}_2$ be the projection of $\vec{a}_1$ on $\vec{c}$. then

322. $\vec{a}_2 =$

(A) $\dfrac{943}{49}(2\hat{i}-3\hat{j}-6\hat{k})$ (B) $\dfrac{943}{49^2}(2\hat{i}-3\hat{j}-6\hat{k})$

(C) $\dfrac{943}{49}(-2\hat{i}+3\hat{j}+6\hat{k})$ (D) $\dfrac{943}{49^2}(-2\hat{i}+3\hat{j}+6\hat{k})$

323. $\vec{a}_1 . \vec{b} =$

(A) -41 (B) $-\dfrac{41}{7}$

(C) 41 (D) 287

324. Which of the following is true.

(A) $\vec{a}$ and $\vec{a}_2$ are collinear (B) $\vec{a}_1$ and $\vec{c}$ are collinear

(C) $\vec{a}$, $\vec{a}_1$, $\vec{b}$ are coplanar (D) $\vec{a}$, $\vec{a}_1$, $\vec{a}_2$ are coplanar

COMPREHENSION # 12

Paragraph for Questions Nos. 325 to 327

ABCD is a parallelogram . L is a point on BC which divides BC in the ratio 1 : 2. AL intersects BD at P. M is a point on DC which divides DC in the ratio 1 : 2 and AM intesects BD in Q

325. Point P divides AL in the ratio

(A) 1 : 2 (B) 1 : 3

(C) 3 : 1 (D) 2 : 1

326. Point Q divides DB in the ratio

(A) 1 : 2 (B) 1 : 3

(C) 3 : 1 (D) 2 : 1

327. PQ : DB =

(A) 2/3 (B) 1/3

(C) 1/2 (D) 3/4

COMPREHENSION # 13

Paragraph for Questions Nos. 328 to 330

Three vector $\vec{a}$, $\vec{b}$ and $\vec{c}$ are forming a right handed system, if $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$, $\vec{c}\times\vec{a}=\vec{b}$. If vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are forming a right handed system, then answer the following question

328. If vector $3\vec{a}-2\vec{b}+2\vec{c}$ and $-\vec{a}-2\vec{c}$ are adjacent sides of a parallelogram the angle between the diagonal is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$ (D) $\frac{2\pi}{3}$

329. If $\vec{x}=\vec{a}+\vec{b}-\vec{c}$, $\vec{y}=-\vec{a}+\vec{b}-2\vec{c}$, $\vec{z}=-\vec{a}+2\vec{b}-\vec{c}$, then a unit vector normal to the vectors $\vec{x}+\vec{y}$ and $\vec{y}+\vec{z}$ is

(A) $\vec{a}$ (B) $\vec{b}$

(C) $\vec{c}$ (D) none of these

330. Vectors $2\vec{a}-3\vec{b}+4\vec{c}$, $\vec{a}+2\vec{b}-\vec{c}$ and $x\vec{a}-\vec{b}+2\vec{c}$ are coplanar, then x =

(A) $\frac{8}{5}$ (B) $\frac{5}{8}$

(C) 0 (D) 1

COMPREHENSION # 13

Paragraph for Questions Nos. 331 to 333

Let a point P whose position vector is $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is called Lattice point if x, y, z $\in$ N. If atleast two of x, y, z are equal then this Lattice point is called isosceles Lattice point. If all x, y, z are equal then this Lattice point is called equilateral Lattice point.

331. The number of Lattice points on the plane $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) = 10$ are

(A) 36 (B) 45 (C) 84 (D) 120

332. If a Lattice point is selected at random from Lattice points which satisfy $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) \le 11$, then the probability that the selected Lattice point is equilateral given that it is isosceles Lattice point is

(A) $\frac{1}{22}$ (B) $\frac{1}{23}$

(C) $\frac{2}{33}$ (D) $\frac{5}{22}$

333. Area of triangle formed by the isosceles Lattice points lying on the plane $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) = 4$ is

(A) $2\sqrt{2}$ (B) $\sqrt{2}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{3}{2}\sqrt{2}$

COMPREHENSION # 14

Paragraph for Questions Nos. 334 to 336

Let AB be the st.line $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6}$. From the point P(1, 2, 5) perpendicular PN is drawn to AB, where N is the foot of perpendicular. A st.line PQ is drawn parallel to the plane 3x + 4y + 5z = 0 to meet AB in Q. Then.

334. Coordinates of N are

(A) $\left(\frac{52}{49}, \frac{78}{49}, \frac{156}{49}\right)$ (B) $\left(-\frac{52}{49}, \frac{78}{49}, \frac{156}{49}\right)$

(C) $\left(\frac{52}{49}, \frac{-78}{49}, \frac{156}{49}\right)$ (D) $\left(\frac{52}{49}, \frac{78}{49}, \frac{-156}{49}\right)$

335. Coordinates of Q are

(A) (3, –9/2, 9) (B) (–3, 9/2, 9)

(C) (3, 9/2, –9) (D) None

336. Equation of PQ is

(A) $\frac{x-1}{4} = \frac{2-y}{13} = \frac{z-5}{8}$ (B) $\frac{x-1}{4} = \frac{y-2}{13} = \frac{z-5}{8}$

(C) $\frac{x-1}{4} = \frac{y-2}{13} = \frac{5-z}{8}$ (D) $\frac{x-1}{-4} = \frac{y-2}{13} = \frac{z-5}{8}$

COMPREHENSION # 15

Paragraph for Questions Nos. 337 to 339

Intersection of a sphere by a plane is called circular section.

(i) If the plane intersects the sphere in more than one different points, than the section is called a circle.

(ii) If the circle of section is of greatest possible radius , then the circle is called great circle.

(iii) If the radius of circular section is zero, then the section is a point circle.

(iv) If the plane does not meet the sphere at all, then the section is an imaginary circle.

337. Sphere $x^2 + y^2 + z^2 = 4$ intersected by the plane $2x + 3y + 6z + 7 = 0$ is

(A) a great circle (B) a real circle but not great

(C) a point circle (D) an imaginary circle

338. Sphere $x^2 + y^2 + z^2 – 2x + 4y + 6z – 17 = 0$ interected by the plane $3x – 4y + 2z – 5 = 0$ is

(A) a great circle

(B) a real circle but not great

(C) a point circle

(D) an imaginary circle

339. The sphere $x^2 + y^2 + z^2 + 2x + 6y – 8z – 1 = 0$ intersected by the plane $x + 2y – 3z – 7 = 0$ is

(A) a great circle

(B) a real circle but not great

(C) a point circle

(D) an imaginary circle

COMPREHENSION # 16

Paragraph for Questions Nos. 340 to 342

Let $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ be two planes, where $d_1, d_2 > 0$. Then origin lies in acute angle if $a_1a_2 + b_1b_2 + c_1c_2 < 0$ and origin lies in obtuse angle if $a_1a_2 + b_1b_2 + c_1c_2 > 0$.
Further point (x_1, y_1, z_1) and origin both lie either in acute angle or in obtuse angle , if one of (x_1, y_1, z_1) and origin lie in acute angle and the other in obtuse angle, if

$$(a_1x_1 + b_1y_1 + c_1z_1 + d_1)(a_2x_1 + b_2y_1 + c_2z_1 + d_2) < 0$$

340. Given the planes $2x + 3y – 4z + 7 = 0$ and $x – 2y + 3z – 5 = 0$, if a point P is $(1, – 2, 3)$, then

(A) O and P both lie in acute angle between the planes

(B) O and P both lie in obtuse angle

(C) O lies in acute angle, P lies in obtuse angle.

(D) O lies in obtue angle, P lies an acute angle.

341. Given the planes $x + 2y – 3z + 5 = 0$ and $2x + y + 3z + 1 = 0$. If a point P is $(2, –1, 2)$, then

(A) O and P both lie in acute angle between the planes

(B) O and P both lie in obtuse angle

(C) O lies in acute angle, P lies in obtuse angle.

(D) O lies in obtue angle, P lies an acute angle.

342. Given the planes $x + 2y – 3z + 2 = 0$ and $x – 2y + 3z + 7 = 0$, if the point P is $(1, 2, 2)$, then

(A) O and P both lie in acute angle between the planes

(B) O and P both lie in obtuse angle

(C) O lies in acute angle, P lies in obtuse angle.

(D) O lies in obtue angle, P lies an acute angle.

COMPREHENSION # 17

Paragraph for Questions Nos. 343 to 345

A tetrahedron is a triangular pyramid. If position vector of all the vertices of tetrahedron are $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$, then position vector of centroid of $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$. If $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ are adjacent sides of tetrahedron, then volume of tetrahedron is $\frac{1}{6}\left|[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}]\right|$

343. In a regular tetrahedron angle between two opposite edges is

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{2}$

344. In a regular tetrahedron if the distance between centroid and midpoint of any age of tetrahedron is equal to

(A) $\frac{1}{3}$ (edge of tetrahedron)
(B) $\frac{1}{2\sqrt{2}}$ (edge of tetrahedron)
(C) $\frac{1}{2}$ (edge of tetrahedron)
(D) $\frac{1}{3\sqrt{2}}$ (edge of tetrahedron)

345. If vector $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four vectors whose magnitudes are equal to area of the faces of a tetrahedron and directions perpendicular and outward directions to the faces respectively then

(A) $\vec{a}+\vec{b}+\vec{c}+\vec{d} = 0$
(B) $\vec{a}+\vec{b} = \vec{c}+\vec{d}$
(C) $\vec{a}+\vec{c} = \vec{b}+\vec{d}$
(D) None of these

COMPREHENSION # 18

Paragraph for Questions Nos. 346 to 348

Consider the determinant

$$\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$$

M_{ij} = Minor of the element of i^{th} row and j^{th} column
C_{ij} = Cofactor of the element of i^{th} row and j^{th} column

346. Value of $b_1 . C_{31} + b_2 . C_{32} + b_3 . C_{33}$ is
(A) 0 (B) Δ (C) 2Δ (D) Δ^2

347. If all the elements of the determinant are multiplied by 2, then the value of new determinant is
(A) 0 (B) 8Δ (C) 2Δ (D) $2^9 . \Delta$

348. $a_3 M_{13} - b_3 . M_{23} + d_3 . M_{33}$ is equal to
(A) 0 (B) 4Δ
(C) 2Δ (D) Δ

COMPREHENSION # 19

Paragraph for Questions Nos. 349 to 351

$\begin{vmatrix} 0 & 2 & 3 \\ -2 & p & 5 \\ q & -5 & 0 \end{vmatrix}$ is a skew symmetric determinant then

349. Value of p is
(A) – 3 (B) 0 (C) 3 (D) 1

350. Value of determinant is
(A) 2 (B) 0 (C) – 2 (D) 1

351. Value of p – 2q is
(A) 3 (B) – 3 (C) – 6 (D) 6

COMPREHENSION # 20

Paragraph for Questions Nos. 352 to 354

Let A be a m × n matrix. If there exists a matrix L of type n × m such that LA = I_n, then L is called left inverse of A. Similarly, if there exists a matrix R of type n × m such that AR = I_m, then R is called right inverse of A.
For example to find right inverse of matrix

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \text{ we take } R = \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$$

and solve AR = I_3 i.e.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ x – u = 1 y – v = 0 z – w = 0
x + u = 0 y + v = 1 z + w = 0
2x + 3u = 0 2y + 3v = 0 2z + 3w = 1

As this system of equations is inconsistent, we say there is no right inverse for matrix A.

352. Which of the following matrices is NOT left inverse of matrix $\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$

(A) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & -7 & 3 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (C*) $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 3 & -1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$

353. The number of right inverses for the matrix $\begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$

(A) 0 (B) 1 (C) 2 (D*) infinite

354. For which of the following matrices number of left inverses is greater than the number of right inverses

(A) $\begin{bmatrix} 1 & 2 & 4 \\ -3 & 2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & 3 \\ 1 & 1 \\ 4 & 4 \end{bmatrix}$

COMPREHENSION # 21

Paragraph for Questions Nos. 355 to 357

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ \ell & m & n \end{vmatrix}$

M_{ij} denotes the minor of an element in i^{th} row and j^{th} column
C_{ij} denotes the cofactor of an element in i^{th} row and j^{th} column

355. The value of $p \,.\, C_{21} + q \,.\, C_{22} + r \,.\, C_{23}$ is

(A) 0 (B) $-\Delta$
(C) Δ (D) Δ^2

356. The value of $x \,.\, C_{21} + y \,.\, C_{22} + z \,.\, C_{23}$ is

(A) 0
(B) $-\Delta$
(C) Δ
(D) Δ^2

357. The value of $q \,.\, M_{12} - y \,.\, M_{22} + m \,.\, M_{32}$ is

(A) 0
(B) $-\Delta$
(C) Δ
(D) Δ^2

COMPREHENSION # 22

Paragraph for Questions Nos. 358 to 359

Read the following write up carefully and answer the following questions:

Let S = set of triplets (A, B, C) where A, B, C are subsets of {1,2,3,......, n}. $E_1 =$ events that a selected triplet at random from set S will satisfy $A \cap B \cap C = \phi, A \cap B = \phi, B \cap C = \phi$. $E_2 =$ events that a selected triplet at random from set S will satisfy $A \cap B \cap C = \phi, A \cap B = \phi, B \cap C = \phi, A \cap C = \phi$. $P(\in)$ represents probability of an event E then -

358. $P(E_1)$ is equal to -

(A) $\dfrac{7^n - 6^n + 5^n}{8^n}$

(B) $\dfrac{7^n - 2\times6^n + 5^n}{8^n}$

(C) $\dfrac{7^n - 2\times6^n}{8^n}$

(D) $\dfrac{7^n - 2\times6^n + 5^n}{8^n}$

359. $P(E_2)$ is equal to-

(A) $\dfrac{7^n - 3\times6^n + 5^n}{8^n}$

(B) $\dfrac{7^n - 3\times6^n + 3\times5^n - 4^n}{8^n}$

(C) $\dfrac{7^n - 2\times6^n + 2\times5^n - 4^n}{8^n}$

(D) $\dfrac{7^n - 6^n + 5^n - 4^n}{8^n}$

SECTION-4: (MATRIX MATCH TYPE)

360. Match the followings -

	Column - I		Column - II
(A)	Sum of square of the direction cosines of line is	(P)	0
(B)	All the points on the z-axis have their x and y coordinate equal to	(Q)	1
(C)	Distance between the points (1,3,2) and (2, 3, 1) is	(R)	0
(D)	Shortest distance between the lines $\dfrac{x-6}{1}=\dfrac{y-2}{-2}=\dfrac{z-2}{2}$ and $\dfrac{x+4}{3}=\dfrac{y}{-2}=\dfrac{z+1}{-2}$ is	(S)	$\sqrt{2}$

361. Match the following:

	Column – I		Column – II
(A)	If a, b > 0 a + b = 1, then minimum value of $\left(a^2+\frac{1}{a^2}\right)^2+\left(b^2+\frac{1}{b^2}\right)^2$ is	(p)	$\frac{3}{2}$
(B)	The perpendicular distance of the image of the point (3, 4 –12) in the xy-plane from the z-axis is	(q)	5
(C)	The area of the quadrilateral whose vertices are 1, i, ω, iω is	(r)	$\frac{27}{4}$
(D)	The minimum value of $(\sin^2 x + \cos^2 x + \text{cosec}^2 2x)^3$ is	(s)	$\frac{289}{8}$

362. Match the following:

List – I	List – II
(A) Lines with direction ratios (1, – c, – b), (– c, – 1, – a) and (–b, – a, 1) are coplanar then $a^2 + b^2 + c^2 + 2abc$ is	(i) –1
(B) If the lines x = ay + 1, x = by + 2 and x = cy + 3, z = dy + 4 are perpendicular then ac + bd is equal to	(ii) 1
(C) If (a, b, c) lies on a plane which form ΔABC with axes whose centroid lies on (α, β, γ) then $\frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\gamma}$ is equal to	(iii) 3
(D) Let [.] denotes the greatest integer less than or equal to x, then f(x) = [x sinπx] is not differentiable if x =	(iv) 0
	(v) 2

363. Match the following:

List – I	List – II
(A) If the line $\frac{x-1}{1}=\frac{y+1}{-2}=\frac{z+1}{\lambda}$ lies in the plane 3x – 2y + 5z = 0, then x is equal to	(i) $\sin^{-1}\sqrt{\frac{6}{25}}$
(B) If (3, λ, μ) is a point on the line 2x + y + z – 3 = 0 = x – 2y + z – 1, then $\lambda + \mu$ is equal to	(ii) $-\frac{7}{5}$
(C) The angle between the line x = y = z and the plane 4x – 3y + 5z = 2 is	(iii) – 3
(D) The angle between the planes x + y + z = 0 and 3x – 4y + 5z = 0 is	(iv) $\cos^{-1}\sqrt{\frac{8}{75}}$

364. Match the following:

List – I		List – II	
(A)	$\vec{a}, \vec{b}$ unit vectors and $\vec{a}+2\vec{b} \perp 5\vec{a}-4\vec{b}$, then $2(\vec{a}\cdot\vec{b})$ is equal to	(i)	0
(B)	The points (1, 0, 3), (–1, 3, 4), (1, 2, 1), (k, 2, 5) are coplanar, then k =	(ii)	– 1
(C)	The vectors (1, 1, m), (1, 1, m + 1), (1, –1, m) are coplanar, then number of values of m is	(iii)	1
(D)	$\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{i}\times(\vec{a}\times\vec{b})$ is equal to	(iv)	2

365. **Match the following**

Column - I		Column - II	
(a)	The remainder when $(81)^{8^8}$ is divided by 26 is equal to	(P)	6
(b)	If $\sum_{r=1}^{8} \frac{r\,.\,2^r}{(r+2)!} = 1 - \frac{2^{p+1}}{q!}$, then $\frac{p+q}{6}$ is equal to	(Q)	5
(c)	If the number of 3 digit natural numbers in which no digit is smaller than a digit to its left is 33 k, then value of k is	(R)	4
(d)	If ten different things are distributed among 3 persons, the chance that a particular person received more then 7 things is $\frac{67k}{2.3^{10}}$, then value of k is	(S)	3

366. **Match the following**

Column - I		Column - II	
(a)	One ball is drawn from a bag containing 4 balls and is found to be white. The events that the bag contains "1 white", "2 white", "3 white" and "4 white" are equaly likely. If the probability that all the balls are white is $\frac{p}{15}$, then the value of p is	(P)	9
(b)	From a set of 12 persons, if the number of different selection of a committee, its chair person and its secretary (possibly same as chair person) is 13.2^{10} m, then value of m is	(Q)	3
(c)	If x, y, z > 0 and x + y + z = 1, then the least value of $\frac{5x}{2-x} + \frac{5y}{2-y} + \frac{5z}{2-z}$ is	(R)	12
(d)	If $\sum_{k=1}^{12} 12k.\,{}^{12}C_k.{}^{11}C_{k-1}$ is equal to $\frac{12\times 21\times 19\times 17\times ...\times 3}{11!} \times 2^{12} \times p$, then the value of p is	(S)	6

367. Match the following

	Column - I	Column - II	
(a)	The number of five - digit numbers having the product of digits 20 is	(P)	77
(b)	A man took 5 space plays out of an engine to clean them. The number of ways in which he can place atleast two plays in the engine from where they came out is	(Q)	31
(c)	The number of integer between 1 & 1000 inclusive in which atleast two consecutive digits are equal is	(R)	50
(d)	The value of $\frac{1}{15}\sum_{1\le i\le j\le 9}\sum i\,.\,j$	(S)	181

368. Match the following

	Column - I	Column - II	
(a)	A is a real skew symmetric matrix such that $A^2 + I = 0$. Then	(P)	BA – AB
(b)	A is a matrix such that $A^2 = A$. If $(I + A)^n = I + \lambda A$, then λ equals	(Q)	A is of even order
(c)	If for a matrix A, $A^2 = A$, and $B = I - A$, then $AB + BA + I - (I - A)^2$ equals	(R)	A
(d)	A is a matrix with complex entries and A^* stands for transpose of complex conjugate of A. If $A^* = A$ & $B^* = B$, then $(AB - BA)^*$ equals	(S)	$2^n - 1$

369. Match the following

	Column - I	Column - II	
(a)	Let $\lvert A\rvert = \lvert a_{ij}\rvert_{3\times 3} \neq 0$. Each element a_{ij} is multiplied by k^{i-j}. Let $\lvert B\rvert$ the resulting determinant, where $k_1\lvert A\rvert + k_2\lvert B\rvert = 0$. Then $k_1 + k_2 =$	(P)	0
(b)	The maximum value of a third order determinant each of its entries are ± 1 equals	(Q)	4
(c)	$\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\beta \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$ if $\cos^2\alpha + \cos^2\beta + \cos^2\gamma =$	(R)	1
(d)	$\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$ where A and B are determinants of order 3. Then A + 2B =	(S)	2

370. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} - \hat{k}$ if $\vec{a} = 1\vec{b} + \mu\,\hat{c}$ where $\hat{c}$ is perpendicular to $\vec{b}$, then

	Column - I	**Column - II**
(A)	Magnitude of projection of $\vec{a}$ on $\vec{b}$ is	(p) $\frac{16}{7}$
(B)	Magnitude of projection of $\vec{b}$ on $\vec{a}$ is	(q) $\frac{16}{3}$
(C)	Value of $\lvert\lambda\rvert$ is	(r) $\frac{\sqrt{185}}{3}$
(D)	Value of $\lvert\mu\rvert$ is	(s) $\frac{16}{9}$

371. **Match the column**

	Column - I		**Column - II**
(A)	If $\vec{a} + \vec{b} = \hat{j}$ and $2\vec{a} - \vec{b} = 3\hat{i} + \frac{\hat{j}}{2}$, then cosine of the angle between $\vec{a}$ and $\vec{b}$ is	(p)	1
(B)	If $\lvert\vec{a}\rvert = \lvert\vec{b}\rvert = \lvert\vec{c}\rvert$, angle between each pair of vectors is $\frac{\pi}{3}$ and $\lvert\vec{a} + \vec{b} + \vec{c}\rvert = \sqrt{6}$, then $\lvert\vec{a}\rvert =$	(q)	$5\sqrt{3}$
(C)	Area of the parallelogram whose diagonals represent the vectors $3\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + 4\hat{k}$ is	(r)	7
(D)	If $\vec{a}$ is perpendicular to $\vec{b} + \vec{c}$, $\vec{b}$ is perpendicular to $\vec{c} + \vec{a}$, $\vec{c}$ is perpendicular to $\vec{a} + \vec{b}$, $\lvert\vec{a}\rvert = 2$, $\lvert\vec{b}\rvert = 3$ and $\lvert\vec{c}\rvert = 6$, then $\lvert\vec{a} + \vec{b} + \vec{c}\rvert =$	(s)	$-\frac{3}{5}$

372. **Match the column**

	Column – I		**Column – II**
(A)	The area of the triangle whose vertices are the points with ractangular cartesian coordinates (1, 2, 3), (–2, 1, –4), (3, 4, –2) is	(p)	0
(B)	The value of $(\vec{a} \times \vec{b}) . (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) . (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) . (\vec{b} \times \vec{d})$ is	(q)	1
(C)	A square PQRS of side length P is folded along the diagonal PR so that planes PRQ and PRS are perpendicular to one another, the shortest distance between PQ and RS is $\frac{P}{k\sqrt{2}}$, then k =	(r)	$\frac{\sqrt{1218}}{2}$
(D)	$\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{d} = 3\hat{i} + 2\hat{j} + \hat{k}$ then $(\vec{a} \times \vec{b}) . (\vec{c} \times \vec{d}) =$	(s)	21

373. **Column – I** | **Column – II**

The lines

(A) $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-1}{3} = \frac{y-3}{4} = \frac{z-5}{5}$ are

(B) $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$ are

(C) $\frac{x-2}{5} = \frac{y+3}{4} = \frac{5-z}{2}$ and $\frac{x-7}{5} = \frac{y-1}{4} = \frac{z-2}{-2}$ are

(D) $\frac{x-3}{2} = \frac{y+2}{3} = \frac{z-4}{5}$ and $\frac{x-3}{3} = \frac{y-2}{2} = \frac{z-7}{5}$ are

(p) coincident

(q) Parallel and different

(r) skew

(s) Intersecting in a point

374. **Column – I** | **Column – II**

(A) Foot of perp. drawn for point (1, 2, 3) to the line $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-2}{4}$ is

(B) Image of line point (1, 2, 3) in the line $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-2}{4}$ is

(C) Foot of perpendicular from the point (2, 3, 5) to the plane 2x + 3y – 4z + 17 = 0 is

(D) Image of the point (2, 5, 1) in the plane 3x – 2y + 4z – 5 = 0 is

(p) $\left(\frac{107}{29}, \frac{30}{29}, \frac{69}{29}\right)$

(q) $\left(\frac{88}{29}, \frac{125}{29}, \frac{69}{29}\right)$

(r) $\left(\frac{68}{29}, \frac{44}{29}, \frac{78}{29}\right)$

(s) $\left(\frac{38}{29}, \frac{57}{29}, \frac{185}{29}\right)$

375. Find the rank of the following matrices:

Column – I		**Column – II**	
(i)	$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	**(p)**	**1**
(ii)	$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$	(q)	2
(iii)	$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$	**(r)**	**3**
(iv)	$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	(s)	4

SECTION-5: (INTEGER TYPE)

376. A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all black, If P is the probability that the missed one is red. Then the value of 3P is _________

377. A is a 4 × 4 matrix with $a_{11} = 1 + x_1$, $a_{22} = 1 + x_2$, $a_{33} = 1 + x_3$, $a_{44} = 1 + x_4$ and all other entries 1, where x_i are the roots of $n^4 - n^2 + 1 = 0$. The value of det(A) is ___________

378. The number of diagonal matrices of order 3 satisfying $A^2 = A$ is _________

379. The distance between the image of (8, –8, 2) in the plane 3x – y + 4z = 1 and the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ with the plane x – y + z = 5 is ___________

380. If $|\vec{a}| = 2, |\vec{b}| = 3,$ and $|\vec{c}| = 4$ then $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2$ cannot exceed ________

381. If $\vec{\alpha} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{\beta} = 2\hat{i} - \hat{j} + \hat{k}$; $\vec{\gamma} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{\alpha} \times (\vec{\beta} \times \vec{\gamma}) = p\vec{\alpha} + q\vec{\beta} + r\vec{\gamma}$ then find the value of p+ q-r

382. If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{c} = 2\hat{i} + \hat{j} - \hat{k}$ & $\vec{d} = 3\hat{i} - \hat{j} - 2\hat{k}$ then find the absolute value of $(\vec{a} \times \vec{b}) \times (\vec{a} \times \vec{c}).\vec{d}$

383. It is given that $\vec{x} = \frac{\vec{b} x \vec{c}}{[\vec{a}\vec{b}\vec{c}]}$; $\vec{y} = \frac{\vec{c} x \vec{a}}{[\vec{a}\vec{b}\vec{c}]}$; $\vec{z} = \frac{\vec{a} x \vec{b}}{[\vec{a}\vec{b}\vec{c}]}$ where $\vec{a}, \vec{b}, \vec{c}$ are non – coplanar vectors. Find the value of $\vec{x}.(\vec{a}+\vec{b}) + \vec{y}.(\vec{b}+\vec{c}) + \vec{z}.(\vec{c}+\vec{a})$:

384. A letter is known to have come either from London or Clifton; on the postmark only the two consecutive letters ON are legible; the chance that it came from London is p. Find 1001p?

385. A speaks the truth 3 out of 4 times, and B 5 out of 6 times; the probability that they will contradict each other in stating the same fact is p,find 120p?

386. If on a straight line 10 cm. two length of 6 cm and 4 cm are measured at random, the probability that their common part does not exceed 3 cms is p. Find 48p?

387. A car is parked by an owner amongst 25 cars in a row, not at either end. On his return he finds that exactly 15 placed are still occupied. the probability that both the neighbouring places are emptyis p find 92p.

388. A gambler has one rupee in his pocket. He tosses an unbiased normal coin unless either he is ruined or unless the coin has been tossed for a maximum of five times. If for each head he wins a rupee and for each tail he looses a rupee, then the probability that the gambler is ruined is p find 80p.

389. Mr. Dupont is a professional wine taster. When given a French wine, he will identify it with probability 0.9 correctly as French, and will mistake it for a Californian wine with probability 0.1. When given a Californian wine, he will identify it with probability 0.8 correctly as Californian, and will mistake it for a French wine with probability 0.2. Suppose that Mr. Dupont is given ten unlabelled glasses of wine, three with French and seven with Californian wines. He randomly picks a glass, tries the wine and solemnly says. “French”. the probability that the wine he tasted was Californian is p/q(where p,q are relatively prime).find p+q

390. In ten trials of an experiment, if the probability of getting '4 successes’ is maximum, then the probability of failure in each trial can be equal to p/q(where p,q are relatively prime).find p+q

391. In a Nigerian hotel, among the english speaking people 40% are English & 60% Americans. The English & American spellings are **"RIGOUR" & "RIGOR"** respectively. An English speaking person in the hotel writes this word. A letter from this word is chosen at random & found to be a vowel.the probability that the writer is an Englishman is p/q(where p,q are relatively prime),find p+q.

392. The odds that a book will be favorably reviewed by three independent critics are 5 to 2, 4 to 3, and 3 to 4 respectively : the probability that of the three reviews a majority will be favourable is p/q(where p,q are relatively prime),find q-p?

393. If A, B, C are angles of a triangle ABC, then $8\begin{vmatrix} \sin\frac{A}{2} & \sin\frac{B}{2} & \sin\frac{C}{2} \\ \sin(A+B+C) & \sin\frac{B}{2} & \cos\frac{A}{2} \\ \cos\frac{(A+B+C)}{2} & \tan(A+B+C) & \sin\frac{C}{2} \end{vmatrix}$ is less than or equal to :

394. If $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = k\,abc\,(a+b+c)^3$ then the value of k is

395. Find The distance of the point of intersection of the line $x - 3 = (1/2)(y-4) = (1/2)(z-5)$ and the plane $x + y + z = 17$ from the point (3, 4, 5)

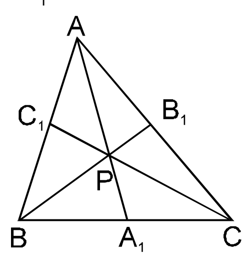
396. A, B are two inaccurate arithmeticians whose chance of solving a given question correctly are $\frac{1}{8}$ and $\frac{1}{12}$ respectively; if they obtain the same result, and if it is 1000 to 1 against their making the same mistake, the chance that the result is correct is p/q.Find p+q?

397. The value of $\left[\vec{d}\,\vec{b}\,\vec{c}\right]\vec{a} + \left[\vec{d}\,\vec{c}\,\vec{a}\right]\vec{b} + \left[\vec{d}\,\vec{a}\,\vec{b}\right]\vec{c} - \vec{d}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ is equal to:

398. The system of linear equations $x + y - z = 6$, $x + 2y - 3z = 14$ and $2x + 5y - \lambda z = 9$ $(\lambda \in R)$ has a unique solution if $\lambda \neq$

399. Let $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$ then $f = \left(\frac{\pi}{6}\right)$:

400. In the adjacent figure 'P' is any arbitrary interior point of the triangle ABC such that the lines AA_1, BB_1 and CC_1 are concurrent at P. Value of $\frac{PA_1}{AA_1} + \frac{PB_1}{BB_1} + \frac{PC_1}{CC_1}$ is always equal to



END OF EXERCISE # 02

PART # 02

TRIGONOMETRY

EXERCISE # 01

SECTION-1 : (ONE OPTION CORRECT TYPE)

401. The difference between the greatest and least values of the function $f(x) = \cos x + \frac{1}{2}\cos 2x - \frac{1}{3}\cos 3x$ is

(A) $\frac{2}{3}$ (B) $\frac{8}{7}$ (C) $\frac{9}{4}$ (D) $\frac{3}{8}$

402. If in a ΔABC $\cos A + 2\cos B + \cos C = 2$, then a, b, c are in

(A) A.P. (B) H.P. (C) G.P. (D) A.G.P.

403. If $f(x) = \sin^{4n}x - \cos^{4n}x$ and $g(x) = \sin x + \cos x$, then general solution of $f(x) = \left[g\left(\frac{\pi}{10}\right)\right]$ is (where [.] is greatest integer less than equal to x)

(A) $2n\pi + \frac{\pi}{3}, n \in I$ (B) $n\pi + \frac{\pi}{2}, n \in I$ (C) $n\pi + \frac{\pi}{4}, n \in I$ (D) none of these

404. The maximum value of $(\sin\alpha_1)(\sin\alpha_2) \ldots (\sin\alpha_n)$ under the restrictions $0 \le \alpha_1, \alpha_2, \ldots \alpha_n \le \frac{\pi}{2}$ and $(\tan\alpha_1)(\tan\alpha_2) \ldots (\tan\alpha_n) = 1$ is

(A) $\frac{1}{2^n}$ (B) $\frac{1}{2n}$ (C) $\frac{1}{2^{n/2}}$ (D) 1

405. In a ΔABC, $(a + b + c)(b + c - a) = kbc$ if

(A) $k < 0$ (B) $k > 0$ (C) $0 < k < 4$ (D) $k > 4$

406. Least value of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$ is

(A) $-\frac{\pi^3}{8}$ (B) $-\frac{\pi^3}{32}$ (C) $\frac{\pi^3}{32}$ (D) $\frac{\pi^3}{8}$

407. If r, r_0 be the in–radius and ex-radius of equilateral triangles having sides 2 and 3 respectively, then $r : r_0$ is equal to

(A) 2 : 3 (B) 1 : 3 (C) 1 : 9 (D) 2 : 9

408. In a Δ ABC, given that tan A : tan B : tan C = 3 : 4 : 5, then the value of sin A sin B sin C is

(A) $\frac{2}{\sqrt{5}}$ (B) $\frac{2\sqrt{5}}{7}$ (C) $\frac{2\sqrt{5}}{9}$ (D) $\frac{2}{3\sqrt{5}}$

409. The equation $\sin^{-1}x = |x - a|$ will have atleast one solution if

(A) $a \in [-1, 1]$ (B) $a \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(C) $a \in \left[1-\frac{\pi}{2}, 1+\frac{\pi}{2}\right]$ (D) None of these

410. The solution set of x for which min(sinx, cosx) > min(–sinx, – cosx) where $x \in (0, 2\pi)$

(A) $\left(0, \frac{3\pi}{4}\right) \cup \left(\frac{7\pi}{4}, 2\pi\right)$ (B) $(0, \pi)$

(C) $\left(\frac{3\pi}{4}, 2\pi\right)$ (D) None of these

411. The value of $\tan(\sin^{-1} \cos \sin^{-1}x) \tan(\cos^{-1} \sin \cos^{-1}x)\ \forall\ x \in \left(0, \frac{\pi}{2}\right)$ is

(A) 0 (B) 1

(C) –1 (D) none of these

412. If sin A = sin B and cos A = cos B, then

(A) $A = n\pi + (-1)^n B$ (B) $A = 2n\pi \pm B$

(C) $A = 2n\pi + B$ (D) none of these

413. If in a triangle ABC, tan A + tan B + tan C = 6 and tan A · tan B = 2, then the triangle is

(A) equilateral (B) obtuse angled

(C) acute angled (D) right angled isosceles

414. Inside a big circle exactly n small circles each of radius r can be drawn in such a way that each small circle touches the big circle and two small circles. If $n \geq 3$, then the radius of the bigger circle is

(A) $r\,\text{cosec}\left(\frac{\pi}{n}\right)$ (B) $r\left\{1+\text{cosec}\left(\frac{\pi}{n}\right)\right\}$

(C) $r\left\{1+\text{cosec}\left(\frac{2\pi}{n}\right)\right\}$ (D) $r\left\{1+\text{cosec}\left(\frac{\pi}{2n}\right)\right\}$

415. In a right angled triangle ABC, if $\angle C = \frac{\pi}{2}$ and $\angle A = 2\angle B$, then $\frac{R}{r}$ is

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(C) $\frac{2}{\sqrt{3}-1}$ (D) $\frac{2\sqrt{2}}{\sqrt{3}+1}$

416. If in a triangle $\sin^4A + \sin^4B + \sin^4C = \sin^2B \sin^2C + 2 \sin^2C \sin^2A + 2 \sin^2A \sin^2B$, then angle A can be equal to

(A) 120° (B) 50°

(C) 30° (D) 45°

417. If the hypotenuse of the right angled triangle is twice the length of perpendicular drawn from opposite vertex to it, then the difference of two acute angles is

(A) 75 (B) 0

(C) 30 (D) 60

418. The area of the circle and area of a regular pentagon having perimeter equal to that of the circle are in the ratio

(A) $\tan\left(\frac{\pi}{5}\right) : \frac{\pi}{5}$ (B) $\cot\left(\frac{\pi}{5}\right) : \frac{\pi}{5}$ (C) $\sin\left(\frac{\pi}{5}\right) : \frac{\pi}{5}$ (D) $\cos\left(\frac{\pi}{5}\right) : \frac{\pi}{5}$

419. Let α, β, γ be the altitudes on the sides a, b, c respectively of a ΔABC. If α, β, γ are the roots of the equation $x^3 - 12x^2 + 44x - 48$, then the inradius of the ΔABC is :

(A) $\frac{11}{12}$ (B) $\frac{12}{11}$

(C) 5 (D) 3

420. The maximum value of the function f (x) = $(\sin^{-1}(\sin x))^2 - \sin^{-1}(\sin x)$ is

(A) $\frac{\pi}{4}[\pi + 2]$ (B) $\frac{\pi}{4}[\pi - 2]$

(C) $\frac{\pi}{2}[\pi + 2]$ (D) $\frac{\pi}{2}[\pi - 2]$

421. The value of $\tan^4\frac{\pi}{16} + 4\tan^3\frac{\pi}{16} - 6\tan^2\frac{\pi}{16} - 4\tan\frac{\pi}{16}$ is equal to

(A) 0 (B) 1

(C) –1 (D) 2

422. If (sinθ, cosθ) and (3, 2) lie on the same side of the line x + y = 1, then θ lies between

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $(0, \pi)$

(c) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{4}\right)$

423. The number of possible real solutions of $\tan^{-1}(x^2 + x + 1) + \cos^{-1}(x^2 + 2x + 9) = \frac{3\pi}{2}$ is:

(A) 0 (B) 1

(C) 2 (D) 4

424. Sides of a ΔABC are in A.P. If a < min{b, c} and c > max{a, b}, then cosA is :

(A) $\frac{3c - 4b}{2b}$ (B) $\frac{3c - 4b}{2c}$

(C) $\frac{4c - 3b}{2a}$ (D) $\frac{4c - 3b}{2b}$

425. The number of ordered pairs (x, y) satisfying the system of equations given by sinx + siny = sin(x + y) and |x| + |y| = 1 is

(A) 2 (B) 4

(C) 6 (D) none of these

Section-2 (MORE THAN ONE option correct type)

426. The equation $2\sin\frac{x}{2}\cos^2 x - 2\sin\frac{x}{2}\sin^2 x = \cos^2 x - \sin^2 x$ has a root for which

(A) sin2x = 1 (B) $\cos 2x = -\frac{1}{2}$ (C) sin2x = – 1 (D) $\cos x = \frac{1}{2}$

427. The side of ΔABC satisfy the equation $2a^2 + 4b^2 + c^2 = 4ab + 2ac$. Then -

(A) the triangle is isosceles (B) the triangle is obtuse

(C) $B = \cos^{-1}\frac{7}{8}$ (D) $A = \cos^{-1}\frac{1}{4}$

428. If $\left(\cos^2 x + \frac{1}{\cos^2 x}\right)(1+\tan^2 2y)(3+\sin 3z) = 4$, then

(A) x may be a multiple of π (B) z can be a multiple of π

(C) y can be a multiple of $\frac{\pi}{2}$ (D) x cannot be an even multiple of π

429. If $\cos^{-1}\frac{x^2-1}{x^2+1} + \tan^{-1}\frac{2x}{x^2-1} = \frac{2\pi}{3}$, then x is equal to

(A) $\sqrt{3}$, when x > 1 (B) $2-\sqrt{3}$, when 0 < x < 1

(C) $2+\sqrt{3}$, when 0 < x < 1 (D) $-\frac{1}{\sqrt{3}}$, when x > 2

430. If inside a big circle exactly 24 small circles, each of radius 2, can be drawn in such a way that each small circle touches the big circle and also touch both its adjacent small circles, then radius of the big circle is

(A) $2\left(1+\operatorname{cosec}\frac{\pi}{24}\right)$ (B) $\left(\frac{1+\tan\frac{\pi}{24}}{\cos\frac{\pi}{24}}\right)$ (C) $2\left(1+\operatorname{cosec}\frac{\pi}{12}\right)$ (D) $\frac{2\left(\sin\frac{\pi}{48}+\cos\frac{\pi}{48}\right)^2}{\sin\frac{\pi}{24}}$

431. If $\tan^{-1}\left(x^2+3|x|-4\right) + \cot^{-1}\left(4\pi + \sin^{-1}\sin 14\right) = \frac{\pi}{2}$, then the value of $\sin^{-1}\sin 2x$ is equal to

(A) $6 - 2\pi$ (B) $2\pi - 6$ (C) $\pi - 3$ (D) $3 - \pi$

432. Let f(x) = ab sinx + $b\sqrt{1-a^2}$ cosx + c, where |a| < 1, b > 0 then

(A) maximum value of f(x) is b if c = 0

(B) difference of maximum and minimum value of f(x) is 2b

(C) f(x) = c if x = $-\cos^{-1}a$ (D) f(x) = c if x = $\cos^{-1}a$

433. If in a triangle ABC, $A \le B \le C$ and $\sin A \le \sin B \le \sin C$, then the triangle may be

(A) equilateral (B) isosceles (C) obtuse angled (D) right angled

434. If $\cos\alpha = \frac{1}{2}\left(x+\frac{1}{x}\right)$ and $\cos\beta = \frac{1}{2}\left(y+\frac{1}{y}\right)$, (xy > 0) x, y, α, $\beta \in$ R then

(A) $\sin(\alpha+\beta+\gamma) = \sin\gamma\ \forall\ \gamma \in R$ (B) $\cos\alpha\cos\beta = 1\ \forall\ \alpha, \beta \in R$

(C) $(\cos\alpha + \cos\beta)^2 = 4\ \forall\ \alpha, \beta \in R$ (D) $\sin(\alpha+\beta+\gamma) = \sin\alpha + \sin\beta + \sin\gamma\ \forall\ \alpha, \beta, \gamma \in R$

435. If $(a\cos\theta_1, a\sin\theta_1)$, $(a\cos\theta_2, a\sin\theta_2)$, $(a\cos\theta_3, a\sin\theta_3)$ represents the vertices of an equilateral triangle inscribed in a circle, then

(A) $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$ (B) $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 0$

(C) $\tan\theta_1 + \tan\theta_2 + \tan\theta_3 = 0$ (D) $\cot\theta_1 + \cot\theta_2 + \cot\theta_3 = 0$

436. $\theta = \tan^{-1}(2\tan^2\theta) - \tan^{-1}\left(\frac{1}{3}\tan\theta\right)$, then $\tan\theta$ is

(A) -2 (B) 1 (C) $\frac{2}{3}$ (D) 2

437. If $0 < \alpha, \beta < \pi$ and $\cos\alpha + \cos\beta - \cos(\alpha + \beta) = \frac{3}{2}$ then

(A) $\alpha = \frac{\pi}{3}$ (B) $\beta = \frac{\pi}{3}$ (C) $\alpha = \beta$ (D) $\alpha + \beta = \frac{\pi}{3}$

438. Let ABC be an isosceles triangle with base BC. If $'r'$ is the radius of the circle inscribed in ΔABC and $'\rho'$ be the radius of the circle described opposite to the angle A, then the product $'\rho r'$ can be equal to

(A) $R^2 \sin^2 A$ (B) $R^2 \sin^2 2B$ (C) $\frac{1}{2}a^2$ (D) $\frac{a^2}{4}$

439. Which of the following functions have maximum value unity?

(A) $\sin^2 x - \cos^2 x$ (B) $\sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x\right)$

(C) $\cos^6 x + \sin^6 x$ (D) $\cos^2 x + \sin^4 x$

440. Let $S_n = \tan^{-1}\frac{4}{7} + \tan^{-1}\frac{4}{19} + ... + \tan^{-1}\frac{4}{4n^2+3}$, then

(A) $S_n = \tan^{-1}\left(\frac{2n+5}{4n}\right)$ (B) $S_n = \cot^{-1}\left(\frac{2n+5}{4n}\right)$

(C) $S_\infty = \tan^{-1}2$ (D) $S_\infty = \cot^{-1}2$

441. If $\cos\alpha = \frac{1}{2}\left(x + \frac{1}{x}\right)$ and $\cos\beta = \frac{1}{2}\left(y + \frac{1}{y}\right)$, $(xy > 0)$ $x, y, \alpha, \beta \in R$ then

(A) $\sin(\alpha + \beta + \gamma) = \sin\gamma \ \forall \ \gamma \in R$ (B) $\cos\alpha \cos\beta = 1 \ \forall \ \alpha, \beta \in R$

(C) $(\cos\alpha + \cos\beta)^2 = 4 \ \forall \ \alpha, \beta \in R$ (D) $\sin(\alpha + \beta + \gamma) = \sin\alpha + \sin\beta + \sin\gamma \ \forall \ \alpha, \beta, \gamma \in R$

442. If [x] represents the greatest integer less than or equal to x, then which of the following statement is true

(A) sin[x] = cos[x] has no solution (B) sin[x] = tan[x] has infinitely many solutions

(C) sin[x] = cos[x] possess unique solution (D) sin[x] = tan[x] for no value of x

443. Triangles $A_1A_2A_3$ and $B_1B_2B_3$ have side lengths a_1, a_2, a_3 and b_1, b_2, b_3 respectively satisfying the relation $\sqrt{a_1 + a_2 + a_3}\sqrt{b_1 + b_2 + b_3} = \sqrt{a_1b_1} + \sqrt{a_2b_2} + \sqrt{a_3b_3}$, then which one of the following statements is/are true?

(A) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$ (B) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = 1$

(C) $\Delta A_1A_2A_3$ and $\Delta B_1B_2B_3$ are similar (D) $\Delta A_1A_2A_3$ and $\Delta B_1B_2B_3$ are congruent

444. If $\theta_R \in [0, \pi]$ for $1 \le k \le 10$, then the maximum value of $\prod_{R=1}^{10}(1 + \sin^2\theta_R)(1 + \cos^2\theta_R)$ is -

(A) $\left(\frac{3}{2}\right)^{10}$ (B) $\left(\frac{9}{4}\right)^{10}$ (C) $\left(\frac{3}{2}\right)^{20}$ (D) $\left(\frac{9}{4}\right)^{5}$

445. If $0 \le \alpha, \beta \le \frac{\pi}{2}$ and $\cos\alpha + \cos\beta = 1$, then -

(A) $\alpha + \beta \ge \frac{\pi}{2}$ (B) $\cos(\alpha + \beta) \le 0$ (C) $\alpha + \beta \le \frac{\pi}{2}$ (D) $\cos(\alpha + \beta) \ge 0$

446. If 3 sin β = sin $(2\alpha + \beta)$, then tan $(\alpha + \beta)$ – 2 tan α is :

(A) independent of α (B) independent of β

(C) dependent of both α and β (D) independent of α but dependent of β

447. If x = sec ϕ – tan ϕ & y = cosec ϕ + cot ϕ then:

(A) $x = \frac{y+1}{y-1}$ (B) $y = \frac{1+x}{1-x}$

(C) $x = \frac{y-1}{y+1}$ (D) $xy + x - y + 1 = 0$

448. $(a + 2) \sin\alpha + (2a - 1)\cos\alpha = (2a + 1)$ if $\tan\alpha$ =

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{2a}{a^2+1}$ (D) $\frac{2a}{a^2-1}$

449. $\sin x - \cos^2 x - 1$ assumes the least value for the set of values of x given by:

(A) $x = n\pi + (-1)^{n+1}(\pi/6)$, $n \in I$ (B) $x = n\pi + (-1)^n(\pi/6)$, $n \in I$

(C) $x = n\pi + (-1)^n(\pi/3)$, $n \in I$ (D) $x = n\pi - (-1)^n(\pi/6)$, $n \in I$

450. If the numerical value of $\tan(\cos^{-1}(4/5) + \tan^{-1}(2/3))$ is a/b then

(A) a + b = 23 (B) a – b = 11 (C) 3b = a + 1 (D) 2a = 3b

451. It is known that $\sin\beta = \frac{4}{5}$ & $0 < \beta < \pi$ then the value of $\frac{\sqrt{3}\sin(\alpha+\beta) - \frac{2}{\cos\frac{\pi}{6}}\cos(\alpha+\beta)}{\sin\alpha}$ is:

(A) independent of α for all β in $(0, \pi)$ (B) $\frac{5}{\sqrt{3}}$ for $\tan\beta > 0$

(C) $\frac{\sqrt{3}(7+24\cot\alpha)}{15}$ for $\tan\beta < 0$ (D) None

452. If the sides of a right angled triangle are $\{\cos2\alpha + \cos2\beta + 2\cos(\alpha + \beta)\}$ and $\{\sin2\alpha + \sin2\beta + 2\sin(\alpha + \beta)\}$, then the length of the hypotenuse is:

(A) $2[1+\cos(\alpha-\beta)]$ (B) $2[1-\cos(\alpha+\beta)]$ (C) $4\cos^2\frac{\alpha-\beta}{2}$ (D) $4\sin^2\frac{\alpha+\beta}{2}$

453. If $\tan x = \frac{2b}{a-c}$, $(a \neq c)$

$y = a\cos^2x + 2b\sin x\cos x + c\sin^2x$

$z = a\sin^2x - 2b\sin x\cos x + c\cos^2x$, then

(A) y = z (B) y + z = a + c (C) y – z = a – c (D) $y - z = (a - c)^2 + 4b^2$

454. $\left(\frac{\cos A+\cos B}{\sin A-\sin B}\right)^n + \left(\frac{\sin A+\sin B}{\cos A-\cos B}\right)^n$

(A) $2\tan^n\frac{A-B}{2}$ (B) $2\cot^n\frac{A-B}{2}$: n is even

(C) 0 : n is odd (D) none

455. The equation $\sin^6x + \cos^6x = a^2$ has real solution if

(A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$ (C) $a \in \left(-\frac{1}{2}, \frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$

456. cos4x cos8x – cos5x cos9x = 0 if

(A) cos12x = cos 14 x (B) sin13 x = 0

(C) sinx = 0 (D) cosx = 0

457. In a ΔABC, following relations hold good. In which case(s) the triangle is a right angled triangle?

(A) $r_2 + r_3 = r_1 - r$ (B) $a^2 + b^2 + c^2 = 8R^2$

(C) $r_1 = s$ (D) $2R = r_1 - r$

458. In a triangle ABC, with usual notations the length of the bisector of angle A is :

(A) $\dfrac{2bc\cos\frac{A}{2}}{b+c}$ (B) $\dfrac{2bc\sin\frac{A}{2}}{b+c}$ (C) $\dfrac{abc\operatorname{cosec}\frac{A}{2}}{2R(b+c)}$ (D) $\dfrac{2\Delta}{b+c}\cdot\operatorname{cosec}\dfrac{A}{2}$

459. AD, BE and CF are the perpendiculars from the angular points of a Δ ABC upon the opposite sides, then :

(A) $\dfrac{\text{Perimeter of } \Delta DEF}{\text{Perimeter of } \Delta ABC} = \dfrac{r}{R}$ (B) Area of ΔDEF = 2 Δ cosA cosB cosC

(C) Area of ΔAEF = $\Delta \cos^2 A$ (D) Circum radius of ΔDEF = $\dfrac{R}{2}$

460. In a triangle ABC, points D and E are taken on side BC such that BD = DE = EC. If angle ADE = angle AED = θ, then:

(A) $\tan\theta = 3\tan B$ (B) $3\tan\theta = \tan C$

(C) $\dfrac{6\tan\theta}{\tan^2\theta - 9} = \tan A$ (D) angle B = angle C

461. With usual notation, in a Δ ABC the value of $\Pi(r_1 - r)$ can be simplified as:

(A) $abc\,\Pi\tan\dfrac{A}{2}$ (B) $4rR^2$ (C) $\dfrac{(abc)^2}{R(a+b+c)^2}$ (D) $4Rr^2$

462. α, β and γ are three angles given by

$\alpha = 2\tan^{-1}(\sqrt{2} - 1)$, $\beta = 3\sin^{-1}\dfrac{1}{\sqrt{2}} + \sin^{-1}\left(-\dfrac{1}{2}\right)$ and $\gamma = \cos^{-1}\dfrac{1}{3}$. Then

(A) $\alpha > \beta$ (B) $\beta > \gamma$ (C) $\alpha < \gamma$ (D) $\alpha > \gamma$

463. $\cos^{-1}x = \tan^{-1}x$ then

(A) $x^2 = \left(\dfrac{\sqrt{5}-1}{2}\right)$ (B) $x^2 = \left(\dfrac{\sqrt{5}+1}{2}\right)$

(C) $\sin(\cos^{-1}x) = \left(\dfrac{\sqrt{5}-1}{2}\right)$ (D) $\tan(\cos^{-1}x) = \left(\dfrac{\sqrt{5}-1}{2}\right)$

464. For the equation $2x = \tan(2\tan^{-1}a) + 2\tan(\tan^{-1}a + \tan^{-1}a^3)$, which of the following is invalid?

(A) $a^2x + 2a = x$ (B) $a^2 + 2ax + 1 = 0$

(C) $a \neq 0$ (D) $a \neq -1, 1$

465. The sum $\sum\limits_{n=1}^{\infty} \tan^{-1}\dfrac{4n}{n^4 - 2n^2 + 2}$ is equal to:

(A) $\tan^{-1}2 + \tan^{-1}3$ (B) $4\tan^{-1}1$ (C) $\pi/2$ (D) $\sec^{-1}\left(-\sqrt{2}\right)$

SECTION - 3: (COMPREHENSION TYPE)

COMPREHENSION-1

Paragraph for Questions Nos. 466 to 468

The functions $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, $\cot^{-1}x$, $\text{cosec}^{-1}x$ and $\sec^{-1}x$ are called inverse circular or inverse trigonometric functions which are defined as follows

$\sin^{-1}x$	$-1 \le x \le 1$	$\frac{\pi}{2} \le \sin^{-1}x \le \frac{3\pi}{2}$	
$\cos^{-1}x$	$-1 \le x \le 1$	$-\pi \le \cos^{-1}x \le 0$	
$\tan^{-1}x$	$x \in R$	$-\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2}$	
$\text{cosec}^{-1}x$	$\lvert x\rvert \ge 1$	$\frac{\pi}{2} \le \text{cosec}^{-1}x \le \frac{3\pi}{2}$	$\ne \pi$
$\sec^{-1}x$	$\lvert x\rvert \ge 1$	$-\pi \le \sec^{-1}x \le 0$	$\ne -\frac{\pi}{2}$
$\cot^{-1}x$	$x \in R$	$0 < \cot^{-1}x < \pi$	

466. For $x \in [0, 1]$, $\sin^{-1}x$ is equal to

(A) $\cos^{-1}\sqrt{1-x^2}+\pi$ (B) $\cos^{-1}\sqrt{1-x^2}+\frac{\pi}{2}$ (C) $-\cos^{-1}\sqrt{1-x^2}$ (D) None of these

467. Number of solutions of $\tan^{-1}|x| - \cos^{-1}x = 0$ is/ are

(A) 2 (B) 1

(C) 0 (D) None of these

468. $\lim_{x\to 0} \tan\left(\frac{\sin^{-1}x^4+\sin^{-1}x^9}{4}\right)$

(A) Does not exist as L.H.L. and R.H.L. both are finite and unequal

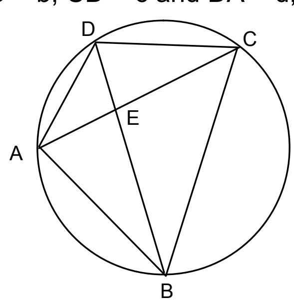
(B) Exist as L.H.L. = R.H.L.

(C) R.H.L. and L.H.L. are unequal (D) None of these

COMPREHENSION-2

Paragraph for Questions Nos. 469 to 471

ABCD be a cyclic quadrilateral and AB = a, BC = b, CD = c and DA = d; AC = x and BD = y,



469. If a = 2, b = 6, c = 4, d = 3 and y = 5, then the value of x will be

(A) 26 (B) $\frac{18}{5}$ (C) $\frac{8}{5}$ (D) $\frac{26}{5}$

470. If a, b, c and d have above values, then the value of $\angle B$ will be

(A) $\cos^{-1}\left(-\frac{15}{48}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(-\frac{1}{2}\right)$ (D) $\cos^{-1}\left(\frac{15}{48}\right)$

471. The minimum value of $\frac{(a^2+b^2+c^2)}{d^2}$ in any quadrilateral, where a, b, c and d are sides of quadrilateral, will be

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

COMPREHENSION-3

Paragraph for Questions Nos. 472 to 474

Consider the equation $\frac{4}{\sin x}+\frac{1}{1-\sin x}=k,$ where x and k are real.

472. The values of x for which the equation is defined

(A) $x \neq n\pi$, $x \neq (2n-1)\frac{\pi}{2},\ n \in I$ (B) $x \neq n\pi,\ x \neq (2n+1)\frac{\pi}{2},\ n \in I$

(C) $x \neq n\pi,\ x \neq (4n+1)\frac{\pi}{2}, n \in I$ (D) none of these

473. The least value of 'k' for which the given equation has a solution in $\left(0, \frac{\pi}{2}\right)$ must be

(A) 3 (B) 6 (C) 9 (D) 5

474. If K = 10, then the number of solution in $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$ must be

(A) 0 (B) 1 (C) 2 (D) none of these

COMPREHENSION-4

Paragraph for Questions Nos. 475 to 477

ΔABC is inscribed in a circle and AL, BM and CN are diameters (2R) of circumcircle of ΔABC, then

475. The area of ΔBLC is

(A) $R^2 \sin A \sin B \sin C$ (B) $2R^2 \sin A \cos B \cos C$

(C) $2R^2 \sin A \sin B \sin C$ (D) $R^2 \sin A \sin B \cos C$

476. Area of ΔANB is

(A) $2R^2 \sin A \sin B \sin C$ (B) $2R^2 \sin A \sin B \cos C$

(C) $2R^2 \sin A \cos B \cos C$ (D) $2R^2 \sin C \cos A \cos B$

477. Area of ΔBLC + area of ΔCMA + area of ΔANB is equal to

(A) $\frac{abc}{R}$ (B) $\frac{abc}{2R}$

(C) $\frac{abc}{3R}$ (D) None of these

COMPREHENSION-5

Paragraph for Questions Nos. 478 to 480

Let Δ ABC be an equilateral triangle of sides of length a. On side AB produced, a point P is chosen such that PA = AB.

478. Inradius of ΔAPC is

(A) $\frac{a\sqrt{3}}{2}$ (B) $\frac{a}{2}$

(C) $\frac{a\sqrt{3}}{2(2+\sqrt{3})}$ (D) None of these

479. Circumradius of ΔPBC is

(A) 2a (B) a

(C) $\frac{a}{2}$ (D) $\frac{a\sqrt{3}}{2}$

480. Let the excircle of ΔPBC w.r.t. side BC touch PC produced at E, then CE is equal to

(A) $\frac{3a+a\sqrt{3}}{2}$ (B) $\frac{3a-a\sqrt{3}}{2}$

(C) $a\sqrt{3}$ (D) None of these

COMPREHENSION-6

Paragraph for Questions Nos. 481 to 483

In a triangle ABC, the equation of the side BC is $2x - y = 3$ and its circumcentre and orthocentre are at (2, 4) and(1, 2) respectively.

481. Circumradius of triangle ABC is

(A) $\sqrt{\frac{61}{5}}$ (B) $\sqrt{\frac{51}{5}}$

(C) $\sqrt{\frac{41}{5}}$ (D) $\sqrt{\frac{43}{5}}$

482. The value of sin B sin C is equal to

(A) $\frac{9}{2\sqrt{61}}$ (B) $\frac{9}{4\sqrt{61}}$

(C) $\frac{9}{\sqrt{61}}$ (D) $\frac{9}{3\sqrt{61}}$

483. The distance of orthocentre to vertex A is equal to

(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{6}{\sqrt{5}}$

(C) $\frac{3}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{5}}$

COMPREHENSION-7

Paragraph for Questions Nos. 484 to 486

Let S_1 be the set of all those solutions of the equation $(1 + a)\cos\theta \cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$ which are independent of a and b and S_2 be the set of all such solutions which are dependent on a and b. Then

484. The set S_1 and S_2 are

(A) $\{n\pi, n \in Z\}$ and $\frac{1}{2}\{n\pi + (-1)^n \sin^{-1}(a \sin b) + b;\ n \in Z\}$

(B) $\{n\frac{\pi}{2}, n \in Z\}$ and $\{n\pi + (-1)^n \sin^{-1}(a \sin b);\ n \in Z\}$

(C) $\{n\frac{\pi}{2}, n \in Z\}$ and $\{n\pi + (-1)^n \sin^{-1}(\frac{a}{2}\sin b);\ n \in Z\}$

(D) None of these

485. Conditions that should be imposed on a and b such that S_2 is non-empty

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a \sin b| \le 1$ (D) None of these

486. All the permissible values of b if a = 0 and S_2 is a subset of $(0, \pi)$:

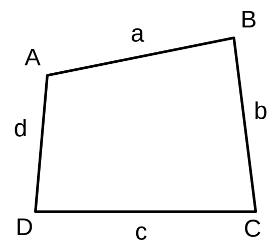
(A) $b \in (-n\pi, 2n\pi), n \in Z$ (B) $b \in (-n\pi, 2\pi - n\pi), n \in Z$

(C) $b \in (-n\pi, n\pi), n \in Z$ (D) None of these

COMPREHENSION-8

Paragraph for Questions Nos. 487 to 489

A quadrilateral ABCD is such that a circle can be inscribed in it and a circle can be circumscribed about it.



487. If $\frac{a}{b} = \frac{c}{d}$, then

(A) $\angle A = 90°$ (B) $\angle A = 90°$ (C) $\angle B = 90°$ (D) $\angle C = 90°$

488. $\tan^2\left(\frac{A}{2}\right)$ is

(A) $\frac{ab}{cd}$ (B) $\frac{bc}{ad}$

(C) $\frac{ac}{bd}$ (D) $\frac{bd}{ac}$

489. Let P_1 and P_2 be the points of contact of AB and AD respectively with the incircle of quadrilateral ABCD. Then $\cos A + \cos\angle P_1OP_2$ (where O is incentre of quadrilateral ABCD)

(A) 2cosB (B) 0

(C) 1 (D) can't be determined

COMPREHENSION-9

Paragraph for Questions Nos. 490 to 492

If $\theta = (2n + 1)\dfrac{\pi}{7}$ and n = 0, 1, 2, 3, 4, 5, 6, then

490. The value of $\sec\frac{\pi}{7}+\sec\frac{3\pi}{7}+\sec\frac{5\pi}{7}$ is

(A) 4 (B) – 3 (C) 3 (D) None of these

491. The value of $\sec^2\frac{\pi}{7}\sec^2\frac{3\pi}{7}+\sec^2\frac{3\pi}{7}\sec^2\frac{5\pi}{7}+\sec^2\frac{5\pi}{7}\sec^2\frac{\pi}{7}$ is

(A) – 80 (B) 80 (C) 24 (D) – 24

492. The value of $\tan^2\frac{\pi}{7}\tan^2\frac{3\pi}{7}\tan^2\frac{5\pi}{7}$ is

(A) 6 (B) 7 (C) 8 (D) None of these

COMPREHENSION-10

Paragraph for Questions Nos. 493 to 495

Consider $\left(1+\sin\theta+\sin^2\theta\right)^n = \sum_{r=0}^{2n} a_r(\sin\theta)^r$; $\theta \in R$.

493. $a_{n+1} + a_{n+2} + \ldots + a_{2n-1}$ equals

(A) $\dfrac{3^n}{2}$ (B) $\dfrac{3^n - a_n}{2}$

(C) $2(3^n - a_n)$ (D) $3^n - a_n$

494. $a_0^2 - a_1^2 + a_2^2 - \ldots a_{2n}^2$ is equal to

(A) a_n (B) a_n^2

(C) $2a_n^2$ (D) $\dfrac{a_n}{2}$

495. The value of $a_0 + 2a_1 + 3a_2 + \ldots + (2n + 1)a_{2n}$ is

(A) $n\,3^{n-1}$ (B) $n\,3^n$

(C) $(n + 1)3^n$ (D) None of these

SECTION - 4 (MATRIX MATCH Type)

496. Match the following:

List – I	List – II
(A) $\sin x \cos^3 x > \cos x \sin^3 x$, $0 \le x \le 2\pi$ is	(i) $\left[-\pi, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$
(B) $4\sin^2 x - 8\sin x + 3 \le 0$, $0 \le x \le 2\pi$, is	(ii) $\left[\frac{3\pi}{2}, 2\pi\right] \cup \{0\}$
(C) $\lvert\tan x\rvert \le 1$ and $x \in [-\pi, \pi]$ is	(iii) $\left(0, \frac{\pi}{4}\right)$
(D) $\cos x - \sin x \ge 1$ and $0 \le x \le 2\pi$	(iv) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

497. Match the following :

List – I	List – II
(A) The number of pairs (x, y) satisfying the equation $\sin x + \sin y = \sin(x + y)$ $\lvert x\rvert + \lvert y\rvert = 1$ is	(i) 3
(B) The number of values of x for which f (x) is valid $f(x) = \sqrt{\sec^{-1}\left(\frac{1-\lvert x\rvert}{2}\right)}$	(ii) 8
(C) If $x, y \in [0, 2\pi]$, then total number of ordered pairs (x, y) satisfying $\sin x \cos y = 1$	(iii) ∞
(D) $f(x) = \sin x - \cos x - kx + b$ decreases for all values of real values of x when $4\sqrt{2}k$ is always greater than	(iv) 6

498. Match the following

List – I	List – II
(A) If $y = \cos^{-1}(\cos x)$ then for $-\pi \le x \le 0$, value of y is	(i) $x - \pi$
(B) For $x \in (-\infty, -1] \cup (1, \infty)$ if $y = \sec(\sec^{-1} x)$, then value of y is equal t	(ii) $x + \pi$
(C) For $\frac{\pi}{2} < x < \frac{3\pi}{2}$ $y = \tan^{-1}(\tan x)$, then value of y is equal to	(iii) x
(D) For $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$ if $y = \tan^{-1}(\tan x)$, then value of y is equal to	(iv) $-x$

499. Match the following :

List – I	List – II
(A) $f(x) = \int_0^{\sin x} t^2 dt$, then period of $f'(x)$ is	(i) $\frac{\pi}{14}$
(B) If area of ellipse $b^2x^2 + a^2y^2 = a^2b^2$ ($a > b$), enclosed by x-axis and the ordinates $x = 0$ and $x = b$ be $\frac{1}{8}$th the area of entire ellipse, then $e\sqrt{1-e^2} + \sin^{-1}\sqrt{1-e^2} =$	(ii) $\frac{\pi}{2}$
(C) Let $f(x) = \frac{\operatorname{cosec}^{-1}x + \cos^{-1}\left(\frac{1}{x}\right)}{\operatorname{cosec}x}$, then greatest value is	(iii) $\frac{\pi}{4}$
(D) $\cos^{-1}\left(\sin\left(\frac{46\pi}{7}\right)\right)$ is	(iv) 2π

500. Match the following :

List – I	List – II
(A) Period of $\tan\frac{\pi}{2}[x]$ (where [.] denotes the greatest integer function)	(i) 2π
(B) Period of $\sin\left(2\pi x + \frac{\pi}{3}\right) + 2\sin\left(3\pi x + \frac{\pi}{4}\right) + 3\sin 5\pi x$	(ii) $\frac{\pi}{2}$
(C) Period of $\sin^4 x + \cos^4 x$	(iii) 2
(D) Period of $1 + \sin^{10} x$	(iv) π

501. Match the following :

List – I	List – II
(A) The number of roots of equation $2\cos x - 2x + 1 = 0$ in the interval $\left[\frac{\pi}{2}, \pi\right]$ is	(i) 2
(B) The number of solutions of $10[\ln x] + 10[2^x] = 31 + 10[\sin x]$ (where [.] denotes the greatest integer function) is	(ii) 3
(C) The number of solutions of $e^{-x^2} = \cos x$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is	(iii) 0
(D) If tangents to the parabola $y^2 = 4x$ are normal to $x^2 = 4by$, $\lvert b\rvert < \frac{1}{\sqrt{k}}$, then the numerical quantity k should be	(iv) 8

502. Match the following:

List – I	List – II
(A) Fundamental period of $f(x) = \sec^2 x - \tan^2 x$ is	(i) no fundamental period
(B) Fundamental period of $f(x) = \sin^2 x + \cos^2 x$ is	(ii) π
(C) Fundamental period of $f(x) = \tan x.\cot x$ is	(iii) $\pi/2$
(D) Fundamental period of $f(x) = \text{cosec}^2 x - \cot^2 x + \{x\}$	(iv) non-periodic

503. Match the following:

List – I	List – II
(A) The value of 'a' for which the equation $4\,\text{cosec}^2\pi(a + x) + a^2 - 4a = 0$ has real solution is	(i) 1
(B) The number of solutions of equation $\tan^2 x - \sec^{10} x + 1 = 0$ in $(0, 10)$ is	(ii) 2
(C) $\sum_{n=1}^{\infty} \sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} = \frac{\pi}{a}$	(iii) 3

504. Match the list:

List – I	List – II
(A) In a ΔABC if area $\Delta = a^2 - (b - c)^2$, then $15\tan A$ is	(i) $\frac{1}{3}$
(B) In a ΔABC $a = 6$, $b = 3$ and $\cos(A - B) = \frac{4}{5}$, then area of ΔABC is	(ii) 2
(C) If a, b, c, d are the sides of quadrilateral, then the minimum value of $\frac{a^2+b^2+c^2}{d^2}$ is	(iii) 3
(D) ΔPRQ is right angled triangle where $P(3, 1)$, $Q(6, 5)$ and $R(x, y)$ and area of $\Delta PRQ = 7$, then number of such point R is	(iv) 8
	(v) 9

505. Match the following:

List – I	List – II
(A) $\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{3}\right)+...+\infty$	(i) $\frac{\pi}{2}$
(B) $\sin^{-1}\left(\frac{12}{13}\right)+\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{63}{16}\right)$	(ii) $\frac{\pi}{4}$
(C) $\sin^{-1}\left(\frac{4}{5}\right)+2\tan^{-1}\left(\frac{1}{3}\right)$	(iii) π
(D) $\cot^{-1}9+\text{cosec}^{-1}\left(\frac{\sqrt{41}}{4}\right)$	(iv) $\frac{\pi}{3}$

506. Match the following with their minimum values, ($x \in R$)

(A) $\sin x + \cos x$ (i) -1

(B) $\sin x + |\cos x|$ (ii) $-\sqrt{2}$

(C) $|\sin x| + \cos x$ (iii) 1

(D) $|\sin x| + |\cos x|$ (iv) none

507. Match the following:

List I	**List – II**
(A) $\sin^{-1}x - \cos^{-1}x = 0$, then $\cos(5\cos^{-1}x + \sin^{-1}x)$ is equal to	(i) 3
(B) $\tan\left(\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right) = \frac{p}{q}$, (where p and q are coprime), then $3q - p$ is equal to	(ii) 2
(C) $\sin^{-1}x + 4\cos^{-1}x = 2\pi$, then x is equal to	(iii) 1
(D) In $\triangle ABC$, $2\cos A \sin C = \sin B$, then $\frac{2a}{c}$ is equal to	(iv) 0

508. Match the following:

List –I	List-II
(A) Number of solutions of the equation $\sin^{-1}x + \cos^{-1}x^2 = \frac{\pi}{2}$	(i) 1
(B) The number of ordered pairs (x, y) satisfying $\frac{\sin^{-1}x}{x} + \frac{\sin^{-1}y}{y} = 2$ is	(ii) 2
(C) Number of solutions of the equation cos(cosx) = sin(sinx) is	(iii) 0
(D) Number of solutions of the equation $\tan\left(x + \frac{\pi}{6}\right) = 2\tan x$ is	(iv) 3

509. Match the following

(A) Number of solutions of the equation $\sin^{-1}x + \cos^{-1}x^2 = \frac{\pi}{2}$ (1) 1

(B) The number of ordered pairs (x, y) satisfying $\frac{\sin^{-1}x}{x} + \frac{\sin^{-1}y}{y} = 2$ is (2) 2

(C) Number of solutions of the equation cos(cosx) = sin(sinx) is (3) 0

(D) Number of solutions of the equation $\tan\left(x + \frac{\pi}{6}\right) = 2\tan x$ is (4) 3

510. Match the following pair of curves with their angle of intersections :

	Column – I		Column – II
(A)	$x^2 + y^2 = 2\pi^2$ and $y = \sin^{-1}\left[x^2+\frac{1}{2}\right]+\cos^{-1}\left[x^2-\frac{1}{2}\right]$ (where [x] = greatest integer function)	(P)	$\frac{\pi}{4}$
(B)	$y^2 = 2x$ and $y = [\lvert\sin x\rvert + \lvert\cos x\rvert]$ where [x] = greatest integer function	(Q)	$\cot^{-1}(1/3)$
(C)	$x^2 = 4ay$, $y = \frac{8a^3}{x^2+4a^2}$	(R)	$\frac{3\pi}{4}$
(D)	$y^2 = \frac{2x}{\pi}$, $y = \sin x$	(S)	$\cot^{-1}(\pi)$

Section-5 : (INTEGER type)

511. The total number of positive integral solution of $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)+\cos^{-1}\left(\frac{y}{\sqrt{1+y^2}}\right)=\sin^{-1}\left(\frac{3}{\sqrt{10}}\right)$ is _____________

512. If circum radius of ΔABC is 3 cm and its area is 6 cm^2 and DEF is triangle formed by foot of perpendicular drawn from A, B, C on sides BC, CA, AB respectively then perimeter of ΔDEF in cm is ________.

513. The greatest and least values of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$ are I_{max} and I_{min} then $\frac{I_{max}}{I_{min}}$ is ____________

514. In a ΔABC, $b \cot B + c \cot C = 2(r + R)$. If the base AC = 3 units and $\angle A = 60°$, BC is _____________

515. If in a ΔABC, a = 2, b = 3, c = 4, then the value of $a^3 \cos(B - C) + b^3 (C - A) + c^3 \cos(A - B)$ is ____________

516. In a right angled triangle ΔABC with C as a right angle, a perpendicular CD is drawn to AB. The radii of the circles inscribed into the triangles ACD and BCD are equal to 3 and 4 respectively. Then the radius of the circle inscribed into the ΔABC is _____________

517. In ΔABC, $\frac{\sum a\sin\frac{A}{2}\cos\left(\frac{B-C}{2}\right)}{\sum \sin A} = nR$ where R is the radius of circumcircle, then n is equal to

518. In the quadrilateral, the length AC and BD are x and y respectively, AB = 5, BC = 7, CD = 6, AD = 8 and if angle between OD and OC is ω, where O is the point of intersection of two diagonals then, the value of $2xy \cos\omega$ is _______________

519. In an acute angled triangle the minimum value of secA secB secC(1 + secA)(1 + secB)(1 + secC) is——— .

520. P is any point and O being the origin. On the circle with OP as diameter two point Q and R are on same side of OP such that $\angle POQ = \angle QOR = \theta$. Let P, Q, R be z_1, z_2, z_3 such that $2\sqrt{3}z_2^2 = (2+\sqrt{3})z_1z_3$. Then the degree measure of θ is________ .

521. In a triangle ABC, side AB = 20, AC = 11, BC = 13, then the diameter of the semicircle inscribed in triangle ABC, whose diameter lies on AB and is touching AC and BC is _________

522. If $x^2 + y^2 \le 1$, then $\min\left\{\frac{kx^2}{y^2}+\frac{1}{k}\left(\frac{y+kx^2}{x^2}\right)\right\}$ (where k is positive) is ____________

523. The number of solutions that the equation sin(cos(sinx)) = cos(sin(cosx)) has in $\left[0, \frac{\pi}{2}\right]$ is _________.

524. If $\sin^{-1}x \in \left(0, \frac{\pi}{2}\right)$, then the value of $\tan\left[\frac{\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)+\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)}{2}\right]$ is ________

525. If $\sqrt{4\sin^4\theta+\sin^2 2\theta}+4\cos^2\left(\frac{\pi}{4}-\frac{\theta}{2}\right)=k$, when θ lies in third quadrant, then k is equal to

526. The smallest positive integral value of p for which the equation tan (p sin x) = cot (p cos x) in x has a solution in $[0, 2\pi]$ is :

527. Let $A_1, A_2, ..., A_n$ be the vertices of an n–sided regular polygon such that; $\frac{1}{A_1A_2} = \frac{1}{A_1A_3} + \frac{1}{A_1A_4}$.

Find the value of n.

528. Find the value of $\operatorname{cosec}\frac{4\pi}{15} + \operatorname{cosec}\frac{8\pi}{15} + \operatorname{cosec}\frac{16\pi}{15} + \operatorname{cosec}\frac{32\pi}{15}$

529. In any ΔABC, then minimum value of $\frac{r_1r_2r_3}{r^3}$ is equal to :

530. The radii r_1, r_2, r_3 of escribed circles of a triangle ABC are in harmonic progression. If its area is 24 sq. cm and its perimeter is 24 cm, find the sum of squares of lengths of its sides.

CO-ORDINATE GEOMETRY

EXERCISE # 01

SECTION-1 : (ONE OPTION CORRECT TYPE)

531. The co-ordinates of the point on the parabola $y^2 = 8x$, which is at minimum distance from the circle $x^2 + (y + 6)^2 = 1$ are

(A) $(-2, 4)$ (B) $(2, -4)$ (C) $(2, 4)$ (D) none of these

532. The values of k for which the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 4\lambda x + 8 = 0$ have two common tangents is

(A) $\left(-\frac{9}{4}, \frac{9}{4}\right)$ (B) $\left(-\infty, -\frac{9}{4}\right) \cup \left(\frac{9}{4}, \infty\right)$

(C) $\left(-\infty, -\frac{9}{4}\right] \cup \left[\frac{9}{4}, \infty\right)$ (D) None of these

533. The chords of the hyperbola $x^2 - y^2 = a^2$ touches parabola $y^2 = 4ax$, then the locus of their mid-point is

(A) $y^2 (a - x) = x^3$ (B) $x^2 (a + x) = y^3$ (C) $y^2 (x - a) = x^3$ (D) $y^2 (x + a) = x^3$

534. Consider a triangle ABC, where B and C are $(-a, 0)$ and $(a, 0)$ respectively. A be any point (h, k). P, Q, R divides the sides of this triangle in same ratio. Then centroid of ΔPQR is

(A) $(0, 0)$ (B) $\left(\frac{h}{3}, 0\right)$ (C) $\left(0, \frac{k}{3}\right)$ (D) $\left(\frac{h}{3}, \frac{k}{3}\right)$

535. If a focal chord of the parabola $y^2 = 4ax$ be at a distance d from the vertex, then its length is equal to

(A) $\frac{a^2}{d^2}$ (B) $\frac{d^2}{a}$ (C) $\frac{4a^3}{d^2}$ (D) $\frac{2a^2}{d}$

536. An ellipse slides between two perpendicular straight lines, then the locus of its centre is

(A) circle (B) parabola (C) ellipse (D) hyperbola

537. If the normal to the parabola $y^2 = 4ax$ at the point $P(at^2, 2at)$ cuts the parabola again at $Q(aT^2, 2aT)$, then

(A) $-2 \le T \le 2$ (B) $T \in (-\infty, -8) \cup (8, \infty)$

(C) $T^2 < 8$ (D) $T^2 \ge 8$

538. If the tangents at P and Q on a parabola meet in R and S be its focus. If p = SP, q = SR and r = SQ, the roots of the equation $px^2 + 2qx + r = 0$ are

(A) rational (B) real and equal (C) imaginary (D) real and unequal

539. Equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is

(A) $3y = 9x + 2$ (B) $y = 2x + 1$

(C) $2y = x + 8$ (D) $y = x + 2$

540. The tangent drawn from (α, β) to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ touches the circle $x^2 + y^2 = c^2$, then the locus of (α, β) is

(A) an ellipse (B) a circle (C) a parabola (D) none of these

541. Consider a point P(at^2, 2at) on the parabola y^2 = 4ax. A focal chord PS (S being focus) is drawn to meet parabola again at Q. From Q, a normal is drawn to meet parabola again at R. From R, a tangent is drawn to the parabola to meet focal chord PSQ (extended) at T. The area of ΔQRT is

(A) $8a^2\left(t^2+\frac{1}{t^2}\right)^3$ (B) $a^2\left(t+\frac{1}{t}\right)^4$ (C) $\frac{8a^2}{3}\left(t+\frac{1}{t}\right)^3$ (D) None of these

542. Two pair of straight lines have the equations $y^2 + xy - 20 = 0$ and $ax^2 + 2hxy + by^2 = 0$. One line will be common among them if

(A) a = 8(h – 2b)or a = –5(2h + 5b) (B) a = 4(h – 2b) or a = –3(2h + 5b)

(C) a = 8(h – 2b) or a = 5(2h + 5b) (D) None of these

543. Let PQ, RS are the tangents at the extremities of a diameters PR of a circle of radius r such that PS, RQ intersect at a point X on the circumference of the circle, then 2r equals

(A) $\frac{PQ+RS}{2}$ (B) $\sqrt{PQ\cdot RS}$ (C) $\frac{2PQ\cdot RS}{PQ+RS}$ (D) $\sqrt{\frac{PQ^2+RS^2}{2}}$

544. If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that OPQ is an equilateral triangle. O being the centre of the hyperbola, then the eccentricity e of the hyperbola satisfies

(A) $1 < e < \frac{2}{\sqrt{3}}$ (B) $e = \frac{2}{\sqrt{3}}$ (C) $e = \frac{\sqrt{3}}{2}$ (D) $e > \frac{2}{\sqrt{3}}$

545. An equilateral triangle is inscribed in the circle $x^2 + y^2 = a^2$ with vertex (a, 0) the equation of the side opposite to the vertex is

(A) 2x – a = 0 (B) x + a = 0

(C) 2x + a = 0 (D) 3x – 2a = 0

546. If the normal at any point P on an ellipse meets major and minor axis at G and G′ and OF be the perpendicular drawn from centre O to this normal then PF. PG must be equal to

(A) b^2 (B) a^2

(C) ab (D) None of these

547. The diameter of the smallest circle which touches the line y = 3x – 3 and passes through a point on the parabola $y = x^2 + 7x + 2$

(A) $\frac{1}{2\sqrt{5}}$ (B) $\frac{1}{\sqrt{10}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{10}$

548. If four distinct points of the curve $y = 2x^4 + 7x^3 + 3x - 5$ are collinear, then the A.M. of the x-co-ordinate of the four points is

(A) $-\frac{7}{8}$ (B) $\frac{3}{4}$ (C) $\frac{7}{8}$ (D) $-\frac{3}{4}$

549. Given a fixed circle C and a line L through the centre O of C. Take a variable point P on L and let K be the circle centre P through O. Let T be the point where a common tangent to C and K meets K. The locus of T is

(A) a circle (B) a parabola

(C) a pair of straight lines (D) None of these

550. If an ellipse slides between two perpendicular straight lines, then the point of intersection of these two lines lies on

(A) auxiliary circle of ellipse
(B) director circle of ellipse
(C) a line passes through centre of ellipse and perpendicular to axis of ellipse
(D) none of these

551. If a variable straight line x cosα + y sinα = p which is a chord of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (b > a) subtends a right angle at the centre of the hyperbola then it always touches a fixed circle whose radius is

(A) $\frac{a^2b^2}{\sqrt{b^2-a^2}}$ (B) $\frac{ab}{\sqrt{a^2-b^2}}$ (C) $\frac{ab}{\sqrt{b^2-a^2}}$ (D) $\frac{ab}{b^2-a^2}$

552. Equation of the circle of which the points (1, 2) and (2, 3) are the ends of a chord of segment containing an angle 45^0 is

(A) $x^2 + y^2 - 4x - 4y + 7 = 0$
(B) $x^2 + y^2 - 4x - 4y - 14 = 0$
(C) $x^2 + y^2 + 4x + 4y + 7 = 0$
(D) none of these

553. The locus of the mid point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix

(A) $x = -a$ (B) $x = \frac{a}{2}$ (C) $x = -\frac{a}{2}$ (D) $x = 0$

554. If the two circles $(x - 1)^2 + (y - 3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points, then

(A) $r < 2$ (B) $r = 2$ (C) $r > 2$ (D) $2 < r < 8$

555. A line $3x - 4y + 4 = 0$ is tangent to a circle whose radius is 3/4. If another straight line $3x - 4y + \lambda = 0$ is also tangent to same circle, then value of λ is

(A) $\frac{3}{4}$ (B) $-\frac{7}{2}$ (C) $-\frac{4}{3}$ (D) $-\frac{2}{7}$

SECTION–2 : (MORE THAN ONE OPTION CORRECT TYPE)

556. If the circle $x^2 + y^2 - 9 = 0$ and $x^2 + y^2 + 2\alpha x + 2y + 1 = 0$ touch each other, then α is

(A) $-\frac{4}{3}$ (B) 0 (C) 1 (D) $\frac{4}{3}$

557. If the ellipse $\frac{x^2}{4}+y^2 = 1$ meets the ellipse $x^2+\frac{y^2}{a^2}=1$ in four distinct points and $a = b^2 - 5b + 7$ then b can take values

(A) $(-\infty, 0)$
(B) [4, 5]
(C) [2, 3]
(D) $(0, \infty)$

558. The straight line $x+y=0, 3x+y-4=0$ and $x+3y-4=0$ form a triangle which is

(A) isosceles
(B) right -angled
(C) obtuse - angled
(D) equilateral

559. The point P(α, α + 1) will lie inside the ΔABC whose vertices are A(0, 3), B(–2, 0) and C(6, 1) if

(A) $\alpha = -1$ (B) $\alpha = -\frac{1}{2}$ (C) $\alpha = \frac{1}{2}$ (D) $-\frac{6}{7} < \alpha < \frac{3}{2}$

560. Tangents are drawn from any point with eccentric angle θ on the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ to the circle $x^2 + y^2 = 16$. If (x_1, y_1) is midpoint of chord of contact, then θ is equal to

(A) $\sec^{-1}\frac{4x_1}{\left(x_1^2 + y_1^2\right)}$ (B) $\sec^{-1}\frac{16x_1^2}{\left(x_1^2 + y_1^2\right)}$

(C) $\tan^{-1}\frac{16y_1}{3\left(x_1^2 + y_1^2\right)}$ (D) $\tan^{-1}\frac{4y_1^2}{\left(x_1^2 + y_1^2\right)}$

561. Consider a circle with its centre lying on the focus of the parabola $y^2 = 2px$ such that it touches the directrix of the parabola, then a point of intersection of the circle and the parabola is

(A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(\frac{p}{4}, p\right)$ (D) $\left(\frac{p}{4}, -p\right)$

562. The equation of the circle which touches the axes of coordinates and the line $\frac{x}{3} + \frac{y}{4} = 1$ and whose centre lies in the first quadrant is $x^2 + y^2 - 2cx - 2cy + c^2 = 0$ where c is

(A) 1 (B) 2 (C) 3 (D) 6

563. The points one line x = 2 from which tangents drawn to circle $x^2 + y^2 = 16$ are at right angles is (are)

(A) $\left(2, 2\sqrt{7}\right)$ (B) $\left(2, 2\sqrt{5}\right)$

(C) $\left(2, -2\sqrt{7}\right)$ (D) $\left(2, -2\sqrt{5}\right)$

564. The line(s) tangents to the curve $y = x^2 - x$

(A) $x - y = 0$ (B) $x + y = 0$ (C) $x - y = 1$ (D) $x + y = 1$

565. The area of a triangle formed by tangent at any point on the curve and co-ordinate axes is constant, then the curve may be

(A) a straight line (B) a circle

(C) a parabola (D) a hyperbola

566. The equation of a circle is $S_1 \equiv x^2 + y^2 = 1$. The orthogonal tangents to S_1 meet at another circle S_2 and orthogonal tangents to S_2 meet at the third circle S_3, then

(A) radius of S_2 and S_3 are in the ratio $1 : \sqrt{2}$ (B) radius of S_2 and S_3 are in the ratio 1 : 2

(C) the circles S_1, S_2 and S_3 are concentric (D) None of these

567. If the area of the quadrilateral formed by the tangents from the origin to the circle $x^2 + y^2 + 6x - 8y + \lambda = 0$ and the pair of radii at the point of contact of these tangent to the circle is $2\sqrt{6}$ sq. units, then the value of λ must be

(A) $4\sqrt{6}$ (B) 24 (C) 1 (D) $12\sqrt{6}$

568. The locus of the point of intersection of the tangents at the extremities of the chord of the circle $x^2 + y^2 = a^2$ which touches the circle $x^2 + y^2 - 2ax = 0$ passes through the point

(A) $\left(\frac{a}{2}, 0\right)$ (B) $\left(0, \frac{a}{2}\right)$

(C) (0, a) (D) (a, 0)

569. The equation of the tangent to the hyperbola $3x^2 - y^2 = 3$ parallel to the line $y = 2x + 4$ is

(A) $y = 2x + 3$ (B) $y = 2x + 1$

(C) $y = 2x - 1$ (D) $y = 2x + 2$

570. From a point P, the chord of contact to the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = (a + b)$ …(1)

touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ …(2) then the locus of the point P is

(A) director circle of (1) (B) auxillary circle of (2)

(C) $x^2 + y^2 = (a + b)^2$ (D) $x^2 + y^2 = a^2 + b^2$

571. The point (1, – 3) is inside the circle S, $S \equiv x^2 + y^2 - 8x + 4y + k = 0$. What are the possible values of k if the circle S neither touches the axes nor cuts them?

(A) 5 (B) 6

(C) 7 (D) 8

572. AB and CD are two equal and parallel chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Tangents to the ellipse at A and B intersect at P and at C and D at Q. The line PQ

(A) passes through the origin (B) is bisected at the origin

(C) cannot pass through the origin (D) is not bisected at the origin

573. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2, then value of a is / are

(A) a = – 8 (B) a = 8

(C) a = 27 (D) a = – 27

574. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $(\sqrt{3}, 0)$ when associated with two asymptotes constitutes :

(A) isosceles triangle

(B) an equilateral triangle

(C) a triangles whose area is $\sqrt{3}$ sq. units

(D) a right isosceles triangle.

575. Variable chords of the parabola $y^2 = 4ax$ subtend a right angle at the vertex. Then:

(A) locus of the feet of the perpendiculars from the vertex on these chords is a circle

(B) locus of the middle points of the chords is a parabola

(C) variable chords passes through a fixed point on the axis of the parabola

(D) none of these

SECTION - 3: (COMPREHENSION TYPE)

COMPREHENSION-1

Paragraph for Questions Nos. 576 to 578

The straight line(s) which passes through a given point (a, b) and make a given angle α with the given straight line y = mx + c where m = tan θ [$(0 < \theta < 90°)$ and $\alpha \neq 0, 90°$], then

576. How many such lines are possible

(A) one (B) two (C) infinite (D) None of these

577. If β is the angle made by the line L with positive direction of x-axis, then tan β is equal to

(A) $\frac{\tan\alpha + m}{1 - m\tan\alpha}$ (B) $\frac{\tan\alpha + m}{\tan\alpha - m}$ (C) $\frac{m - \tan\alpha}{m + \tan\alpha}$ (D) $\frac{m - \tan\alpha}{1 - m\tan\alpha}$

578. The equation of line L is

(A) $(y - b) = \frac{m + \tan\alpha}{1 + m\tan\alpha}(x - a)$

(B) $(y - b) = \frac{\tan\alpha + m}{\tan\alpha - m}(x - a)$

(C) $(y - b) = \frac{m - \tan\alpha}{m + \tan\alpha}(x - a)$

(D) $(y - b) = \frac{m - \tan\alpha}{1 + m\tan\alpha}(x - a)$

COMPREHENSION-2

Paragraph for Questions Nos. 579 to 581

A boy moving along the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at each and every point of it he is drawing a tangent and finding the area of triangle formed by it with co-ordinate axes at point P, Q, R and S starting from positive axis anticlockwise. He found that area of triangle is $\geq$ m and is m at P, Q, R and S.

579. The co-ordinate of point Q are

(A) $\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$

(B) $\left(-\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$

(C) $\left(\frac{a}{\sqrt{2}}, -\frac{b}{\sqrt{2}}\right)$

(D) $\left(-\frac{a}{\sqrt{2}}, -\frac{b}{\sqrt{2}}\right)$

580. The area of Δ formed by P, Q and S is

(A) ab

(B) πab

(C) $\sqrt{ab}$

(D) $\pi\sqrt{ab}$

581. Equation of normal to the ellipse at S is

(A) $ax + by = \frac{b^2}{\sqrt{2}}$

(B) $ax + by = \frac{a^2 - b^2}{\sqrt{2}}$

(C) $bx + ay = \frac{a^2}{2}$

(D) $bx + ay = \frac{b^2 - a^2}{\sqrt{2}}$

COMPREHENSION-3

Paragraph for Questions Nos. 582 to 584

Let an ellipse having major axis and minor axis parallel to x-axis and y-axis respectively. Its two foci S and S′ are (2, 1), (4, 1) and a line x + y = 9 is a tangent to this ellipse at point P.

582. Eccentricity of the ellipse is

(A) $\frac{1}{\sqrt{12}}$ (B) $\frac{1}{\sqrt{13}}$

(C) $\frac{1}{2}$ (D) None of these

583. Length of major axis

(A) $\sqrt{13}$ (B) $2\sqrt{11}$

(C) $\sqrt{52}$ (D) $2\sqrt{12}$

584. The latus rectum of ellipse

(A) $\frac{12}{13}$ (B) $\frac{12}{\sqrt{13}}$

(C) $\frac{24}{\sqrt{13}}$ (D) $\frac{25}{\sqrt{13}}$

COMPREHENSION-4

Paragraph for Questions Nos. 585 to 587

Perpendiculars are drawn from the focus S of the parabola $y = ax^2 + bx + c$ upon the tangents to the parabola at the points A(– 1, 0) and B(1, 2) meeting them at the point $C\left(-\frac{1}{4}, \frac{9}{4}\right)$ and $D\left(\frac{3}{4}, \frac{9}{4}\right)$ respectively.

585. The co-ordinates of the focus are

(A) $\left(\frac{1}{2}, \frac{9}{4}\right)$ (B) $\left(\frac{1}{2}, 2\right)$ (C) $\left(0, \frac{9}{4}\right)$ (D) (– 1, 2)

586. The normals at A and B intersect at a point P. The foot of the third normal through the point P is

(A) $\left(\frac{1}{2}, \frac{9}{4}\right)$ (B) $\left(-\frac{1}{2}, \frac{5}{4}\right)$ (C) (0, 2) (D) $\left(\frac{3}{2}, \frac{5}{4}\right)$

587. Area of the region bounded by the parabola and the x– axis is

(A) $\frac{5}{4}$ (B) 5

(C) $\frac{5}{2}$ (D) $\frac{9}{2}$

Comprehension-5

Paragraph for Questions Nos. 588 to 590

Consider an ellipse $\frac{x^2}{4}+y^2=\alpha$; (α is parameter > 0) and a parabola $y^2 = 8x$. If a common tangent to the ellipse and the parabola meets the co-ordinate axes at A and B respectively, then

588. Locus of mid point of AB is

(A) $y^2 = -2x$

(B) $y^2 = -x$

(C) $y^2 = -\frac{x}{2}$

(D) $\frac{x^2}{4}+\frac{y^2}{2}=1$

589. If the eccentric angle of a point on the ellipse where the common tangent meets it is $\left(\frac{2\pi}{3}\right)$, then α is equal to

(A) 4 (B) 5

(C) 26 (D) 36

590. If two of the three normals drawn from the point (h, 0) on the ellipse to the parabola $y^2 = 8x$ are perpendicular, then

(A) h = 2 (B) h = 3

(C) h = 4 (D) h = 6

Comprehension-6

Paragraph for Questions Nos. 591 to 593

Two straight lines rotate about two fixed points (– a, 0) and (a, 0). If they start from their position of coincidence such that one rotates at the rate double that of the other, then

591. The point (– a, 0) always lies

(A) inside the curve (B) Outside the curve

(C) on the curve (D) None of these

592. Locus of the curve is

(A) circle (B) straight line

(C) parabola (D) ellipse

593. Distance of the point (a, 0) from the variable point on the curve is

(A) 0 (B) 2a

(C) 3a (D) 4a

COMPREHENSION-7

Paragraph for Questions Nos. 594 to 596

Consider a point P on a parabola such that 2 of the normals drawn from it to the parabola are at right angles on parabola, then

594. If the equation of parabola is $y^2 = 8x$, then locus of P is

(A) $x^2 = 4(y - 6)$ (B) $y^2 = 2(x - 6)$
(C) $y^2 = 8(x - 6)$ (D) $2x^2 = (y - 6)$

595. The ratio of latus rectum of given parabola and that of made by locus of point P is

(A) 4 : 1 (B) 2 : 1
(C) 16 : 1 (D) 1 : 1

596. If $P \equiv (x_1, y_1)$, the slope of third normal is

(A) $\frac{y_1}{8}$ (B) $\frac{y_1}{2}$
(C) $-\frac{y_1}{8}$ (D) $-\frac{y_1}{2}$

COMPREHENSION-8

Paragraph for Questions Nos. 597 to 599

Read the following writeup carefully:

Observe the following facts for a parabola.

(i) Axis of the parabola is the only line which can be the perpendicular bisector of the two chords of the parabola.

(ii) If AB and CD are two parallel chords of the parabola and the normals at A and B intersect at P and the normals at C and D intersect at Q, then PQ is a normal to the parabola.

597. The vertex of the parabola passing through (0, 1), (– 1, 3), (3, 3) and (2, 1) is

(A) $\left(1, \frac{1}{3}\right)$ (B) $\left(\frac{1}{3}, 1\right)$
(C) (1, 3) (D) (3, 1)

598. The directrix of the parabola is

(A) $y - \frac{1}{24} = 0$ (B) $y + \frac{1}{24} = 0$
(C) $y + \frac{1}{12} = 0$ (D) $y - \frac{1}{12} = 0$

599. For the parabola $y^2 = 4x$, AB and CD are any two parallel chords having slope 1. C_1 is a circle passing through O, A and B and C_2 is a circle passing through O, C and D. C_1 and C_2 intersect at

(A) (4, – 4) (B) (– 4, 4)
(C) (4, 4) (D) (– 4, – 4)

COMPREHENSION-9

Paragraph for Questions Nos. 600 to 602

To the circle $x^2 + y^2 = 4$ two tangents are drawn from P(–4, 0), which touches the circle at T_1 and T_2 and a rhombus $PT_1 P'T_2$ is completed.

600. Circum centre of the triangle PT_1T_2 is at

(A) (–2, 0) (B) (2, 0)

(C) $\left(\frac{\sqrt{3}}{2}, 0\right)$ (D) None of these

601. Ratio of the area of triangle PT_1P' to that the $P'T_1T_2$ is

(A) 2 : 1 (B) 1 : 2

(C) $\sqrt{3}:2$ (D) None of these

602. If P is taken to be at (h, 0) such that P′ lies on the circle, the area of the rhombus is

(A) $6\sqrt{3}$ (B) $2\sqrt{3}$

(C) $3\sqrt{3}$ (D) None of these

COMPREHENSION-10

Paragraph for Questions Nos. 603 to 605

A circle C whose radius is 1 unit, touches the x–axis at point A. The centre Q of C lies in first quadrant. The tangent from origin O to the circle touches it at T and a point P lies on it such that ΔOAP is a right angled triangle at A and its perimeter is 8 units.

603. The length of QP is

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$

(C) $\frac{5}{3}$ (D) None of these.

604. Equation of circle C is

(A) $\{x-(2+\sqrt{3})\}^2+(y-1)^2=1$ (B) $\{x-(\sqrt{3}+\sqrt{2})\}^2+(y-1)^2=1$

(C) $(x-\sqrt{3})^2+(y-2)^2=1$ (D) None of these.

605. Equation of tangent OT is

(A) $x-\sqrt{3}y=0$ (B) $x-\sqrt{2}y=0$

(C) $y-\sqrt{3}x=0$ (D) None of these

COMPREHENSION-11

Paragraph for Questions Nos. 606 to 608

If a circle with centre C(α, β) intersects a rectangular hyperbola with centre L (h, k) at four points P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3) and S(x_4, y_4), then the mean of the four points P, Q, R, S is the mean of the points C and L. In other words, the mid-point of CL coincides with the mean point of P, Q, R, S. Analytically

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{\alpha + h}{2};\ \frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{\beta + k}{2}.$$

606. Five points are selected on a circle of radius a. The centres of the rectangular hyperbolas, each passing through four of these points lie on a circle of radius

(A) a (B) 2a (C) $\frac{a}{\sqrt{2}}$ (D) $\frac{a}{2}$

607. A, B, C and D are the points of intersection of a circle and a rectangular hyperbola which have different centres. If AB passes through the centre of the hyperbola, then CD passes through

(A) centre of the hyperbola (B) centre of the circle

(C) mid-point of the centres of the circle and hyperbola

(D) none of these

608. A circle with fixed centre (3h, 3k) and of variable radius cuts the rectangular hyperbola $x^2 - y^2 = 9a^2$ at the points A, B, C , D. The locus of the centroid of the triangle ABC is given by

(A) $(x - 2h)^2 - (y - 2k)^2 = a^2$ (B) $(x - h)^2 - (y - k)^2 = a^2$

(C) $\frac{x^2}{h^2} - \frac{y^2}{k^2} = a^2$ (D) $\frac{x^2}{h^2} + \frac{y^2}{k^2} = a^2$

COMPREHENSION-12

Paragraph for Questions Nos. 609 to 611

The vertices of a ΔABC lies on a rectangular hyperbola such that the orthocentre of the triangle is (3, 2) and the asymptotes of the rectangular hyperbola are parallel to the coordinate axis. If the two perpendicular tangents of the hyperbola intersect at the point (1, 1).

609. The equation of the asymptotes is

(A) xy – 1 = x – y (B) xy + 1 = x + y

(C) 2xy = x + y (D) None of these

610. Equation of the rectangular hyperbola is

(A) xy = 2x + y – 2

(B) 2xy = x + 2y + 5

(C) xy = x + y + 1

(D) None of these

611. Number of real normals that can drawn from the point (1, 1) to the rectangular hyperbola is

(A) 4 (B) 0

(C) 3 (D) 2

COMPREHENSION-13

Paragraph for Questions Nos. 612 to 614

Let ABCD be a parallelogram whose diagonals equations are AC: $x + 2y = 3$; BD: $2x + y = 3$. If length of diagonal AC = 4 units and area of ABCD = 8 sq. units.

612. The length of other diagonal BD is

(A) $\frac{10}{3}$ (B) 2 (C) $\frac{20}{3}$ (D) None of these

613. The length of side AB is equal to

(A) $\frac{2\sqrt{58}}{3}$ (B) $\frac{4\sqrt{58}}{9}$ (C) $\frac{3\sqrt{58}}{9}$ (D) $\frac{4\sqrt{58}}{9}$

614. The length of BC is equal to

(A) $\frac{2\sqrt{10}}{3}$ (B) $\frac{4\sqrt{10}}{3}$

(C) $\frac{8\sqrt{10}}{3}$ (D) None of these

COMPREHENSION-14

Paragraph for Questions Nos. 615 to 617

A coplanar beam of light emerging from a point source have equation $\lambda x - y + 2(1 + \lambda) = 0$, $\lambda \in R$. the rays of the beam strike an elliptical surface and get reflected. The reflected rays form another convergent beam having equation $\mu x - y + 2(1 - \mu) = 0$, $\mu \in R$. Further it is found that the foot of the perpendicular from the point (2, 2) upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$.

615. The eccentricity of the ellipse is equal to

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{2}{3}$ (D) $\frac{1}{2}$

616. The area of the largest triangle that an incident ray and the corresponding reflected ray can enclose with the axis of the ellipse is equal to

(A) $4\sqrt{5}$ (B) $2\sqrt{5}$

(C) $\sqrt{5}$ (D) None of these

617. Total distance travelled by an incident ray and the corresponding reflected ray is the least if the point of incidence coincides with

(A) an end of the minor axis (B) an end of the major axis

(C) an end of the latus rectum (D) None of these

COMPREHENSION-15

Paragraph for Questions Nos. 618 to 620

A circular arc of radius 2 units subtend an angle of x radians at the centre O such that $x \in \left(0, \frac{\pi}{2}\right)$. Tangents at the end points P and Q of the arc intersect at R. Let f(x) be the area of triangle PQR and let $\phi(x)$ be the area of region enclosed by the chord PQ and the arc PQ.

618. Length of OR is given by

(A) $\tan\frac{x}{2}$ (B) $2\tan\frac{x}{2}$ (C) $2\sec\frac{x}{2}$ (D) $2\cot\frac{x}{2}$

619. $\phi(x)$ will be given by

(A) $x - \sin x$ (B) $2(x - \sin x)$ (C) $\frac{x}{2} - \tan\frac{x}{2}$ (D) $\tan\frac{x}{2}\sec\frac{x}{2}$

620. $\lim\limits_{x\to 0}\frac{\phi(x)}{f(x)}$ is equal to

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

SECTION - 4 (MATRIX MATCH Type)

621. Match the following:

List – I	List – II
(A) The length of latus rectum of parabola $2y^2 + 3y - 4x - 3 = 0$ is	(i) 3/4
(B) The length of tangent from point (0, – 1) to the circle $\frac{(x-3)^2}{5^2} + \frac{(y-4)^2}{5^2} = 1$ is	(ii) 2
(C) The length of shortest focal chord of parabola $y^2 = 4x - 3$ is	(iii) 3
(D) Straight line with slope m represents the locus of middle points of chords of hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ parallel to y = 2x, then m is	(iv) 4

622. Match the following:

List – I	List – II
(A) The length of common chord of the ellipse $\frac{(x-1)^2}{9} + \frac{(y-2)^2}{4} = 1$ and circle $(x-1)^2 + (y-2)^2 = 1$ is	(i) 0
(B) $\text{cosec}^2 A \cdot \cot^2 A - \sec^2 A \cdot \tan^2 A - (\cot^2 A - \tan^2 A)(\sec^2 A\, \text{cosec}^2 A - 1)$ is	(ii) 1
(C) If $a \neq p, b \neq q, c \neq r$ $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-1}$ is equal to	(iii) 2
(D) The circle $x^2 + y^2 = 4x - 7y + 12 = 0$ cuts an intercept on y-axis of length	(iv) 3

623. Match the following:

List – I	List – I
(A) The value of 'a' for which the image of point (a, a, –1) w.r.t. the line mirror $3x + y = 6a$ is is the point $(a^2 + 1, a)$ is	(i) 3
(B) The normal chord at a point 't' on the parabola $16y^2 = x$ subtends right angle at the vertex. Then t^2 is equal to	(ii) 2
(C) If e and e′ be the eccentricity of a hyperbola and its conjugate. Then $\frac{1}{e^2} + \frac{1}{e'^2} =$	(iii) 5
(D) The shortest distance between parabola $y^2 = 4x$ and circle $x^2 + y^2 + 6x - 12y + 20 = 0$ is $4\sqrt{2}$ – A. The A is	(iv) 1

624. Match the following:

List – I	List – II
(A) Let $\alpha = \lim\limits_{m\to\infty}\lim\limits_{n\to\infty}\cos^{2m}\angle n\pi x$, where x is rational, $\beta = \lim\limits_{m\to\infty}\lim\limits_{n\to\infty}\cos^{2m}\angle n\pi x$, where x is irrational, then the area of the triangle having vertices (α, β), (– 2, 1) and (2, 1) is	(i) 56
(B) If the circumference of the circle $x^2 + y^2 + 8x + 8y - b = 0$ is bisected by the circle $x^2 + y^2 - 2x + 4y + a = 0$, then $\|a + b\| =$	(ii) 2
(C) Given a circle of radius 3, tangents are drawn from points A and B lying on one of its diameters which meet at a point P lying on another diameter perpendicular to the other diameter. The minimum area of triangle PAB is	(iii) 90
(D) If the radius of the circle $(x - 1)^2 + (y - 2)^2 = 1$ and $(x - 7)^2 + (y - 10)^2 = 4$ are increasing uniformly w.r.t. time as 0.3 and 0.4 unit/sec, if they touch each other internally after t sec. then t is equal to	(iv) 18

625. Match the following:
ABC be a triangle with a = 3, b = 4, c = 5

List– I	List–II
(A) Distance between circumcentre and orthocentre	(i) 1/3
(B) Distance between centroid and circumcentre	(ii) 5/2
(C) Distance between centroid and incentre	(iii) 5/6
(D) Distance between centroid and orthocentre	(iv) 5/3

626. Match the following curve with their corresponding orthogonal trajectory

List – I	List – II
(A) $ay^2 = x^3$	(i) $y = kx$
(B) $y = ax^2$	(ii) $y^{2/3} - x^{2/3} = c$
(C) $x^{2/3} + y^{2/3} = a^{2/3}$	(iii) $y^{2/3} + x^{2/3} = c$
(D) $x^2 + y^2 = a^2$	(iv) $2y^2 + x^2 = c$
	(v) $3y^2 + 2x^2 = c$

627. Match the following

List – I		List – II	
(A)	For a rectangular hyperbola $xy = c^2$ (c is purely imaginary) if a point on it is (a, α) where 'a' is a positive number, α can take value	(i)	3
(B)	If sides of a triangle are in A.P. and $(a - b + c)s = kb^2$ (where s is the semi perimeter) then k is equal to	(ii)	$\frac{1-\sqrt{3}}{2}$
(C)	Radius of the circle having centre (3, 4) and touching the circle $x^2 + y^2 = 4$ can be	(iii)	17
(D)	Maximum distance of any point on the circle $(x-7)^2 + \left(y - 2\sqrt{30}\right)^2 = 16$ from the centre of the ellipse $25x^2 + 16y^2 = 400$ is	(iv)	$\frac{3}{2}$

628. Match the following

(A) The number of rational points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is — (1) ∞

(B) Number of integral points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is — (2) 4

(C) Number of rational points on the curve $\frac{x^2}{3} + y^2 = 1$ is — (3) 0

(D) Number of integral points on the curve $\frac{x^2}{3} + y^2 = 1$ is — (4) 2

629. Match the following:

List – I	List – II
(A) The triangle PQR is inscribed in the circle $x^2 + y^2 = 169$. If Q and R have coordinates (5, 12) and (–12, 5) respectively find $\angle QPR$	(i) $\frac{\pi}{4}$
(B) What is the angle between the line joining origin to the point of intersection of the line $4x + 3y = 24$ with circle $(x - 3)^2 + (y - 4)^2 = 25$	(ii) $\frac{\pi}{2}$
(C) Two parallel tangents drawn to given circle are cut by a third tangent. The angle subtended by the third tangent at the centre is	(iii) π
(D) For a circle if a chord is drawn along the point of contact of tangents drawn from a point P. If the chord subtends an angel $\frac{\pi}{2}$ then find the angle at P	(iv) $\frac{\pi}{6}$

630. Match the following:

List – I	List – II
(A) The normal at an end of a latusrectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through an end of the minor axis if e^4 is equal to	(i) $\frac{1}{9}$
(B) PQ is a double ordinate of a parabola $y^2 = 4ax$. If the locus of its points of trisection is another parabola length of whose latus rectum is k times the length of the latus rectum of the given parabola then k is equal to	(ii) $\frac{1}{a}$
(C) If e and e′ are the distances of the extremities of any focal chord from the focus f of the parabola $y^2 = 4ax$, then $\frac{1}{e} + \frac{1}{e'}$ is equal to	(iii) 1

(D) If e and e′ be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2}+\frac{1}{e'^2}$ is equal to	(iv) $1 - e^2$

631. Match the following:

	List – I	List – II	
(A)	The equation of tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ which cuts off equal intercepts on axes is x – y = a where a equal to	(i)	$\sqrt{2}$
(B)	The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if m equals to	(ii)	$\sqrt{3}$
(C)	The equation of the common tangent to parabola $y^2 = 4x$ and $x^2 = 4y$ is $x + y + \frac{k}{\sqrt{3}} = 0$, then k is equal to	(iii)	$\sqrt{8}$
(D)	An equation of common tangent to parabola $y^2 = 8x$ and the hyperbola $3x^2 - y^2 = 3$ is $4x - 2y + \frac{k}{\sqrt{2}} = 0$, then k is equal to	(iv)	$\sqrt{41}$

632. The parabola $y^2 = 4ax$ has a chord AB joining points A $\left(at_1^2, 2at_1\right)$ and B$\left(at_2^2, 2at_2\right)$. Then match the following

List – I	List – II
(A) AB is a normal chord	(i) $t_2 = -t_1 - \frac{1}{2}$
(B) AB is a focal chord	(ii) $t_2 = -\frac{4}{t_1}$
(C) AB subtends 90° at point (0, 0)	(iii) $t_2 = -\frac{1}{t_1}$
(D) AB is inclined at 45° to the axis of parabola	(iv) $t_2 = -t_1 - \frac{2}{t_1}$

Section-5 : (INTEGER type)

633. A point P moves in such a way that $\frac{AP}{PB} = 3$ where A(1, 2) and B(3, 5), then maximum distance of P from A is __________

634. The line L has intercepts 1 and 1/2 on the co-ordinate axes. When keeping the origin fixed, the co-ordinate axes are rotated through a fixed angle, if the same line has intercepts p and q on the rotates axes. Then $\frac{1}{p^2}+\frac{1}{q^2}$ is ____________

635. If PSQ is the focal chord of the parabola $y^2 = 8x$ such that SP = 6, then the length SQ is ______________

636. If the ellipse $x^2 + 81y^2 - 81 = 0$ and the circle $x^2 + y^2 - 9 = 0$ intersect an angle θ in first quadrant, then the value of $(-3 \tan\theta)$ is ____________

637. Let (x_i, y_i) where i = 1, 2, 3, 4 are the integral solutions of equation $2x^2y^2 + y^2 - 6x^2 - 12 = 0$. The area of quadrilateral whose vertices are (x_i, y_i), i = 1, 2, 3, 4 is _______

638. Square of diameter of the circle having tangent at (1, 1) as $x + y - 2 = 0$ and passing through (2, 2) is ________

639. If P be a point on ellipse $4x^2 + y^2 = 8$ with eccentric angle $\frac{\pi}{4}$. Tangent and normal at P intersects the axes at A, B, A′ and B′ respectively, then the ratio of area of $\Delta APA'$ and area of $\Delta BPB'$ is ___________ .

640. P is the positive extremity of the latus rectum of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and A is the positive major vertex and B is the positive minor vertex then the 10 times of the area bounded by BPA and chords BP and AP is ___________

641. The minimum of the distances from the point (0, 1) to the points of intersection of the lines $(3\cos\theta + 4\sin\theta)x + (2\cos\theta + 2\sin\theta)y - (5\cos\theta + 6\sin\theta) = 0$, where different values of θ gives different lines, is ____________

642. A line passes through the point P (2, 3) and makes an angle θ with positive direction of x-axis. If it meets the lines represented by $x^2 - 2xy - y^2 = 0$ at the points A and B. If $PA \cdot PB = 17$, then the value of θ in degrees is ________________

643. Six points (x_i, y_i), i = 1, 2, …, 6 are taken on the circle $x^2 + y^2 = 4$ such that $\sum_{i=1}^{6} x_i = 8$ and $\sum_{i=1}^{6} y_i = 4$. The line segment joining orthocentre of a triangle made by any three points and the centroid of the triangle made by other three points passes through a fixed points (h, k), then h + k is ________

644. The square of the length of the intercept on the normal at the point P(18, 12) of the parabola $y^2 = 8x$ made by the circle on the line joining the focus and P as diameter, is ___________

645. The angle between the straight lines $x\cos\alpha + y\sin\alpha = p$ and $ax + by + p = 0$ is $\frac{\pi}{4}$. They meet the straight line $x\sin\alpha - y\cos\alpha = 0$ in the same point, then the value of $a^2 + b^2$ is ___________

646. Area of the rectangle formed by a asymptotes of the hyperbola $xy - 3y - 2x = 0$ and co–ordinate axes is

647. The tangent at the point A(12, 6) to a parabola intersects its directrix at the point B(– 1, 2). The focus of the parabola lies on x-axis. The number of such parabolas is

648. The circle $x^2 + y^2 = 4$ cuts the circle $x^2 + y^2 - 2x - 4 = 0$ at the point A and B. If the circle $x^2 + y^2 - 4x - k = 0$ passes through A and B, then the value of k is

649. If G is the centroid of ΔABC with vertices A(a, 0), B(– a, 0) and C(b, c) then $\frac{AB^2+BC^2+CA^2}{GA^2+GB^2+GC^2} =$

650. A man running round a race course notes that the sum of the distance of two flag posts from him is always 10 meter and distance between flag posts is 8 m. The area of the path, he encloses (in square meters) is $k\pi$. What is the value of k?

PART # 03

CALCULUS

EXERCISE # 01

SECTION-1 : (ONE OPTION CORRECT TYPE)

651. $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x}dx = \frac{f(x)}{(\sin x)^7} + C$, then f(x) is equal to

(A) sin x (B) cos x (C) tan x (D) cot x

652. Let f(x) = (x + 1) (x + 2) (x + 3) . . . (x + 100) and g(x) = f(x). $f''(x) - (f'(x))^2$, then g(x) = 0, has

(A) no solution (B) exactly one solution
(C) exactly two solutions (D) minimum three solutions

653. Let f(x) = min $(\ell n(\tan x), \ell n(\cot x))$, which of the following statement are incorrect

(A) f(x) is continuous for x $\in \left(0, \frac{\pi}{2}\right)$

(B) Lagrange's mean value theorem is applicable on f(x) for x $\in \left[\frac{\pi}{8}, \frac{\pi}{4}\right]$

(C) Rolle's theorem is not applicable on f(x) for x $\in \left[\frac{\pi}{4}, \frac{3\pi}{8}\right]$

(D) Rolle's theorem is applicable on f(x) for x $\in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$

654. Let 'n' be the number of elements in the Domain set of the function f(x) = $\left|\ln\sqrt{{}^{x^2+4x}C_{2x^2+3}}\right|$ and 'Y' be the global maximum value of f(x), then $[n + [Y]]$ is (where [·] = Greatest Integer function)

(A) 4 (B) 5 (C) 6 (D) 7

655. The value of the integral $\int_0^\infty e^{-2\theta}(\sin 2\theta + \cos 2\theta)d\theta$ is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) does not exist (D) none of these

656. The value of $\int_0^{\pi/3}[\sqrt{3}\tan x]dx$ (where [.] denotes the greatest integer function) is

(A) $\cot^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (B) $\frac{5\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (C) $\frac{\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) none of these

657. The value of $\lim_{x\to\infty}\left(\frac{a_1^{1/x} + a_2^{1/x} + ... + a_n^{1/x}}{n}\right)^{nx}$; $a_i > 0$, i = 1, 2, …, n is

(A) $a_1 + a_2 + … + a_n$ (B) $e^{a_1+a_2+...a_n}$ (C) $\frac{a_1 + a_2 + ...a_n}{n}$ (D) $a_1a_2a_3…a_n$

658. The value of integral $\int \cot^{-1}\left(\frac{\sqrt{4x-x^2}}{x-2}\right)dx$ is equal to

(A) $\frac{x-2}{2}\sin^{-1}\frac{x-2}{2}+\sqrt{4x-x^2}+c$ (B) $(x-2)\sin^{-1}\left(\frac{x-2}{2}\right)+\sqrt{4x-x^2}+c$

(C) $\frac{x-2}{2}\sin^{-1}\left(\frac{x-2}{2}\right)-\sqrt{\frac{4x-x^2}{2}}+c$ (D) none of these

659. Let f: [– 2, 2] → R, where f(x) = $x^3+\sin x+\left[\frac{x^2+1}{a}\right]$ be an odd function, then

(A) a < 3 (B) a > 5 (C) a < 1 (D) a < – 2

660. $\lim_{n\to\infty}\left(\frac{3n+8}{3n+5}\right)^{5n+9}$ is equal to

(A) $3e^5$ (B) e^5 (C) e^3 (D) None of these

661. The differential equation of the orthogonal trajectories of the system of curves x + $\tan^{-1}$(y/x) = c, is

(A) xdx + ydy = $x^2 + y^2$ (B) xdy + ydx = $x^2 + y^2$

(C) xdx + ydy = $(x^2 + y^2)$ dy (D) xdx + ydy = $(x^2 – y^2)$ dy

662. The area bounded by the curves f (x) = x^2 – 2x + 2 and its inverse i.e. f^{-1} is given by

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{5}{3}$ (D) $\frac{4}{3}$

663. The value of the series $\frac{C_0}{5}-\frac{C_1}{6}+\frac{C_2}{7}-\frac{C_3}{8}+\ldots+(-1)\frac{{}^nC_n}{n+4}$

(A) $\int_0^1 x^2(1-x)^n\,dx$ (B) $\int_0^1 x^3(1-x)^n\,dx$ (C) $\int_0^1 x^4(1-x)^n$ dx (D) none of these

664. If f(n) = $\frac{1}{n}$[(n + 1) (n + 2) … (n + n)]$^{1/n}$ then $\lim_{n\to\infty} f(n)$ equals

(A) e (B) 1/e (C) 2/e (D) 4/e

665. If $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ where y ≠ 0 for all x, y ∈ R and f′(1) = 2. then the function f(x) is symmetric about

(A) x-axis (B) y-axis (C) origin (D) y = x

666. Let f : R → R be such that f(1) = 3, f′(1) = 0 and f′′(1) = 6, then $\lim_{x\to 0}\left[\frac{f(1+x)}{f(1)}\right]^{1/x^2}$ equal to

(A) e (B) $e^{1/2}$ (C) e^2 (D) e^3

667. Let f(x) and g(x) be differentiable for 0 ≤ x ≤ 1 such that f(0) = 2, g(0) = 0 f(1) = 6. Let there exists a real numbers x in [0, 1] such that f′(c) = 2g′(c) then the value of g(1) must be

(A) 1 (B) 2 (C) –1 (D) none of these

668. The area of the greatest circle inscribed in 2 |x| + 2 |y| = 4 is given by

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) π^2 (D) π

669. $\int_0^{\pi/2} \sin x \, \log(\sin x) dx$ is equal to

(A) $\log_e(e/2)$ (B) log2 – e (C) $\log_e((2/e)$ (D) loge – 2

670. f $I_n = \int_0^{\pi/4} \tan^n x dx$, then $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}, \frac{1}{I_5+I_7}$ are in

(A) A.P. (B) G.P. (C) H.P. (D) None of these

671. $\int \frac{\ln(1+x^{2/3}+2x^{1/3})}{x+x^{2/3}} dx$ is equal to

(A) $3\ln(1 + x^{1/3})^2 + c$ (B) $\ln(1 + x^{1/3}) + c$
(C) $\ln(x^{1/3} – 1) + c$ (D) none of these

672. $\lim_{x\to 0}\left[\frac{e^x-1}{x}\right]$ is equal to [.] represents G.I. function

(A) 1 (B) 0
(C) does not exist (D) none of these

673. The differential equation of all ellipses centred at origin is :

(A) $y_2 + xy_1^2 – yy_1 = 0$ (B) $xyy_2 + xy_1^2 – yy_1 = 0$
(C) $yy_2 + xy_1^2 – xy_1 = 0$ (D) none of these

674. The solution of the differential equation $2x\frac{dy}{dx} - y = 3$ represent

(A) straight lines (B) circles (C) parabolas (D) ellipses

675. If $y = e^{4x} + 2e^{-x}$ satisfies the relation $\frac{d^3y}{dx^3} + A\frac{dy}{dx} + By = 0$ then value of A and B respectively are:

(A) –13, 14 (B) –13, –12 (C) –13, 12 (D) 12, –13

676. A particle moves in a straight line with velocity given by $\frac{dx}{dt} = x+1$ (x being the distance described). The time taken by the particle to describe 99 metres is :

(A) $\log_{10}e$ (B) $2\log_e 10$ (C) $2\log_{10} e$ (D) $\frac{1}{2}\log_{10} e$

677. The acute angle between the curve $y = |x^2 – 1|$ and $y = |x^2 – 3|$ at their point of intersection is

(A) $\frac{\pi}{4}$ (B) $\tan^{-1}\left(\frac{4\sqrt{2}}{7}\right)$ (C) $\tan^{-1}(4\sqrt{7})$ (D) None of these

678. The range of the function $y = \sqrt{2\{x\} - \{x\}^2 - \frac{3}{4}}$ is (where {.} denotes fractional part)

(A) $\left[-\frac{1}{4}, \frac{1}{4}\right]$ (B) $\left[0, \frac{1}{2}\right]$ (C) $\left[0, \frac{1}{4}\right]$ (D) $\left[\frac{1}{4}, \frac{1}{2}\right]$

679. If $f(x + y) = f(x) – f(y) + 2xy – 1 \ \forall \ x, y \in R$. Also if f(x) is differentiable and $f'(0) = b$ also $f(x) > 0 \ \forall \ x$, then the set of values of b

(A) ϕ (B) {1} (C) {1, 2} (D) none of these

680. $\lim\limits_{k\to\infty}\int_0^{k[x]}(kx-[kx])^k\,dx$; k $\in$ N is equal to (where [.] denotes the greatest integer function)

(A) [kx] (B) [x] (C) $\left[\dfrac{x}{k}\right]$ (D) $[x^k]$

681. The equation of curve passing through (1, 1) in which the subtangent is always bisected at the origin is

(A) $y^2 = x$ (B) $2x^2 - y^2 = 1$ (C) $x^2 + y^2 = 2$ (D) $x + y = 2$

682. If f'(3) = 5 then $\lim\limits_{h\to 0}\dfrac{f\left(3+h^2\right)-f\left(3-h^2\right)}{2h^2}$ is :

(A) 5 (B) 1/5

(C) 2 (D) None of these

683. If f is twice differentiable function then $\lim\limits_{h\to 0}\dfrac{f(a+h)-2f(a)+f(a-h)}{h^2}$ is :

(A) 2f'(a) (B) f''(a)

(C) f'(a) (D) f'(a) + f''(a)

684. If f(x) = sinx, $g(x) = x^2$, h(x) = logx and F(x) = (hogof)(x) then F''(x) is :

(A) 2cosec^3x (B) $2\cot x^2 - 4x^2\text{cosec}^2x^2$

(C) $2x\cot x^2$ (D) -2cosec^2x

685. If $x = \sec\theta - \cos\theta,\ y = \sec^n\theta - \cos^n\theta$ then $\left(\dfrac{dy}{dx}\right)^2$ is equal to :

(A) $\dfrac{n^2\left(y^2+4\right)}{x^2+4}$ (B) $\dfrac{n^2\left(y^2-4\right)}{x^2}$ (C) $n\dfrac{y^2-4}{x^2-4}$ (D) $\left(\dfrac{ny}{x}\right)^2-4$

686. If $y = f\left(\dfrac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x^2$ then $\dfrac{dy}{dx}$ is:

(A) $\cos x^2.f'(x)$ (B) $-\cos x^2.f'(x)$

(C) $\dfrac{2\left(1+x-x^2\right)}{\left(x^2+1\right)^2}\sin\left(\dfrac{2x-1}{x^2+1}\right)^2$ (D) None of these

687. If $y^2 = p(x)$, a polynomial of degree 3 then $2\dfrac{d}{dx}\left(y^3\dfrac{d^2y}{dx^2}\right)$ is equal to:

(A) p'''(x) + p'(x) (B) p'''(x) + p''(x)

(C) p(x) p'''(x) (D) a constant

688. $I = \int\dfrac{(x+x^{\frac{2}{3}}+x^{\frac{1}{6}})}{x\,(1+x^{\frac{1}{3}})}dx$ is equal to

(A) $\dfrac{3}{2}x^{\frac{2}{3}}+6\tan^{-1}\left(x^{\frac{1}{6}}\right)+c$ (B) $\dfrac{3}{2}x^{\frac{2}{3}}-6\tan^{-1}\left(x^{\frac{1}{6}}\right)+c$

(C) $=\dfrac{3}{2}x^{\frac{2}{3}}+\tan^{-1}\left(x^{\frac{1}{6}}\right)+c$ (D) none of these

689. $\int \frac{\left(\sqrt{x^2+1}\right)\left\{\ln\left(x^2+1\right)-2\ln x\right\}}{x^4}$ dx is equal to :

(A) $\frac{(x^2+1)\sqrt{x^2+1}}{x^3}\left[2-3\ln\left(\frac{x^2+1}{x^2}\right)\right]$ + c

(B) $\frac{1}{9}\,\frac{(x^2+1)\sqrt{x^2+1}}{x^3}\left[2+3\ln\left(\frac{x^2+1}{x^2}\right)\right]$ + c

(C) $\frac{(x^2+1)\sqrt{x^2+1}}{x^3}\left[2+3\ln\left(\frac{x^2+1}{x^2}\right)\right]$ + c

(D) $\frac{1}{9}\,\frac{(x^2+1)\sqrt{x^2+1}}{x^3}\left[2-3\ln\left(\frac{x^2+1}{x^2}\right)\right]$ + c

690. If the positive number x and y are connected by the relation $x^2 - xy + y^2 = 12$, then maximum value of 2x + 3y, is

(A) $\frac{20}{\sqrt{19}}$ (B) $\frac{74}{\sqrt{19}}$ (C) $\frac{67}{\sqrt{19}}$ (D) $\frac{76}{\sqrt{19}}$

691. f(x) = Minimum {tanx, cot x} $\forall$ x $\in \left(0, \frac{\pi}{2}\right)$. Then $\int_0^{\pi/3} f(x)\,dx$ is equal to :

(A) $\ell n\left(\frac{\sqrt{3}}{2}\right)$ (B) $\ell n\left(\sqrt{\frac{3}{2}}\right)$ (C) $\ell n\left(\sqrt{2}\right)$ (D) $\ell n\left(\sqrt{3}\right)$

692. If f(x) is a function satisfying f $\left(\frac{1}{x}\right)$ + x^2 f(x) = 0 for all non-zero x, then $\int_{\sin\theta}^{\operatorname{cosec}\theta} f(x)\,dx$ equals

(A) sinθ + cosecθ (B) $\sin^2\theta$ (C) $\operatorname{cosec}^2\theta$ (D) none of these

693. If $\int_0^{100} f(x)\,dx$ = a, then $\sum_{r=1}^{100}\left(\int_0^1 f(r-1+x)\,dx\right)$ =

(A) 100 a (B) a (C) 0 (D) 10 a

694. The area bounded by the curves y = x (1 – ln x); x = e^{-1} and a positive X–axis between x = e^{-1} and x = e is :

(A) $\left(\frac{e^2-4e^{-2}}{5}\right)$ (B) $\left(\frac{e^2-5e^{-2}}{4}\right)$ (C) $\left(\frac{4e^2-e^{-2}}{5}\right)$ (D) $\left(\frac{5e^2-e^{-2}}{4}\right)$

695. The area bounded by y = x^2 , y = [x + 1], x ≤ 1 and the y-axis is :

(A) 1/3 (B) 2/3 (C) 1 (D) 7/3

696. The area bounded by the curve y = f(x), x-axis and the ordinates x = 1 and x = b is (b – 1) sin (3b + 4), $\forall$ b $\in$ R, then f(x) =

(A) (x – 1) cos (3x + 4)
(B) sin (3x + 4)
(C) sin (3x + 4) + 3(x – 1) cos (3x + 4)
(D) None of these

697. The areas of the figure into which curve $y^2 = 6x$ divides the circle $x^2 + y^2 = 16$ are in the ratio

(A) $\frac{2}{3}$ (B) $\frac{4\pi-\sqrt{3}}{8\pi+\sqrt{3}}$ (C) $\frac{4\pi+\sqrt{3}}{8\pi-\sqrt{3}}$ (D) none of these

698. If $\sqrt{(x^2+y^2)} = ae^{\tan^{-1}(y/x)}$, a > 0. Then y″(0), equals

(A) $\frac{a}{2}e^{\pi/2}$
(B) $ae^{\pi/2}$
(C) $-\frac{2}{a}e^{-\pi/2}$
(D) $\frac{a}{2}e^{-\pi/2}$

699. The function $f(\theta) = \frac{d}{d\theta}\int_0^{\theta}\frac{dx}{1-\cos\theta\cos x}$ satisfies the differential equation

(A) $\frac{df}{d\theta} + 2f(\theta)\cot\theta = 0$ (B) $\frac{df}{d\theta} - 2f(\theta)\cot\theta = 0$

(C) $\frac{df}{d\theta} + 2f(\theta) = 0$ (D) $\frac{df}{d\theta} - 2f(\theta) = 0$

700. The solution of the differential equation

$$(x^2 \sin^3 y - y^2 \cos x)\,dx + (x^3 \cos y \sin^2 y - 2y \sin x)\,dy = 0 \text{ is :}$$

(A) $x^3 \sin^3 y = 3y^2 \sin x + C$ (B) $x^3 \sin^3 y + 3y^2 \sin x = C$

(C) $x^2 \sin^3 y + y^3 \sin x = C$ (D) $2x^2 \sin y + y^2 \sin x = C$

SECTION-2 (MORE THAN ONE OPTION CORRECT TYPE)

701. A curve that passes through (2, 4) and having subnormal of constant length of 8 units can be;

(A) $y^2 = 16x - 8$ (B) $y^2 = -16x + 24$

(C) $x^2 = 16y - 60$ (D) $x^2 = -16y + 68$

702. Let $f : R \to R$, such that $f''(x) - 2f'(x) + f(x) = 2e^x$ and $f'(x) > 0\ \forall\ x \in R$, then which of the following can be correct

(A) $|f(x)| = -f(x), \forall x \in R$ (B) $|f(x)| = f(x), \forall x \in R$

(C) $f(3) = -5$ (D) $f(3) = 7$

703. Let $|f(x)| \le \sin^2 x,\ \forall\ x \in R$, then

(A) f(x) is continuous at x = 0

(B) f(x) is differentiable at x = 0

(C) f(x) is continuous but not differentiable at x = 0

(D) f(0) = 0

704. $f(x) = x^3 + x^2 f'(1) + x f''(2) = f'''(3)\quad \forall\ x \in R$, then

(A) f (0) + f (2) = f (1) (B) f (0) + f (3) = 0

(C) f (1) + f (3) = f (2) (D) none of these

705. The function f (x) = 9 + |sin x| is

(A) continuous every where (B) continuous nowhere

(C) differentiable at infinite number of points (D) not differentiable at infinite number of points

706. If $\lim_{x\to 0}\frac{x(1+a\cos x) - b\sin x}{x^3} = 1$, then

(A) $a = -\frac{5}{2}$ (B) $a = -\frac{3}{2}$

(C) $a = -\frac{7}{2}$ (D) $b = -\frac{3}{2}$

707. Let h (x) = min $\{x^2, x^4\}$ for every real number of a, then

(A) h is not differentiable at two points (B) h is differentiable $\forall$ x

(C) h is continuous $\forall$ x (D) none of these

708. The value of the integral $\int_0^{\pi} xf(\sin x)dx$ is

(A) $\pi\int_0^{\pi/2} f(\sin x)dx$ (B) $\frac{\pi}{2}\int_0^{\pi} f(\sin x)dx$ (C) 0 (D) none of these

709. If f(x) = $|\log_{10}x|$, then at x = 1

(A) f(x) is continuous and $f'(1^-) = -\log_e 10$ (B) f(x) is continuous and $f'(1^-) = -1$

(C) f(x) is differentiable on R – {1} (D) f(x) is differentiable on R.

710. $\int_0^{\pi/2}\left(\frac{\pi}{2}-x\right)\sec x dx$ is equal to

(A) $-2\int_0^1 \frac{\cot^{-1}x}{x}dx$ (B) $\int_0^1 \tan^{-1}x dx$ (C) $2\int_0^1 \frac{\tan^{-1}x}{x}dx$ (D) $\int_0^1 \cot^{-1}x dx$

711. For f(x) = $\int_0^x 2|t|\,dt$, then tangent parallel to bisector of positive co-ordinate axes are

(A) $y = x - \frac{1}{4}$ (B) $y = x + \frac{1}{4}$ (C) $y = x - \frac{3}{2}$ (D) $y = x + \frac{3}{2}$

712. Let f(x) = $\begin{cases} -2, & -3 \le x \le 0 \\ |x-2|, & 0 \le x \le 3 \end{cases}$ and g(x) = $\int_{-3}^{x} f(t)dt$, then

(A) g(1) = – 3 (B) g(2) = – 4 (C) g′(1) = 1 (D) g′(2) does not exist

713. The domain of the definition of the function f(x) = $\left([x]-|x-1|\right)^{-1/2} + \sec^{-1}[\cos x]$, in the region [–π, 2π] where [.] denotes greater integer function lies in the internal

(A) $\left(\frac{\pi}{2}, \pi\right)$ (B) $\left[-\pi, -\frac{\pi}{2}\right) \cup \{2\pi\}$ (C) $\left[\pi, \frac{3\pi}{2}\right) \cup \{2\pi\}$ (D) [1, 2π]

714. If f(x) = sinπ(x – [x]) (where [.] denotes the greatest integer function), then

(A) f(x) has period 1 (B) f(x) is non-differentiable at x = 1, 2, 3

(C) $\int_0^{100} f(x)dx = \frac{200}{\pi}$ (D) $\int_0^{100} f(x)dx = 200\pi$

715. Let f : R → R be defined by f(x) = $3^{[x]} + 3^{-x}$, (where [.] denotes the greatest integer function) then which of the following statements are current ?

(A) f(x) is many-one (B) f(x) is into

(C) f(x) is bijective (D) neither even nor odd

716. Let f(x) = (x + |x|)|x| then for all x

(A) f is continuous (B) f′ is differentiable for all x

(C) f′ is continuous (D) f″ is continuous

717. If $\int \log\left(\sqrt{1-x}+\sqrt{1+x}\right)dx$ = x f(x) + Ax + B $\sin^{-1}$x + c

(A) f(x) = $\log\left(\sqrt{1-x}+\sqrt{1+x}\right)$ (B) A = $-\frac{1}{2}$

(C) B = $\frac{2}{3}$ (D) B = $-\frac{1}{2}$

718. If f(x) = [x(x – 1)] + |2x – 1|, then f(x) is (where [.] denotes the greatest integer function)
(A) continuous at x = 10 (B) differentiable at x = 10
(C) discontinuous at x = 10 (D) nondifferentiable at x = 10

719. If $\lim_{x\to\infty}\left(1+\frac{\lambda}{x}+\frac{\mu}{x^2}\right)^{2x}$ = e^2, then
(A) λ = 1, μ = 2 (B) λ = 2, μ = 1 (C) λ = 1, μ = any R (D) λ = μ = 1

720. In which of the following intervals $2x^3 - 24x + 5$ increases
(A) (– 2, 2) (B) (2, ∞) (C) (– ∞, – 2) (D) None of these

721. If f(x) = $\int_1^x \frac{\ln t}{1+t}dt$, then
(A) $f\left(\frac{1}{x}\right)=-\int_1^x \frac{\ln t}{t(1+t)}dt$ (B) $f\left(\frac{1}{x}\right)=\int_1^x \frac{\ln t}{t(1+t)}dt$
(C) $f(x)+f\left(\frac{1}{x}\right)=0$ (D) $f(x)+f\left(\frac{1}{x}\right)=\frac{1}{2}(\ln x)^2$

722. Let f and g be functions from the interval [0, ∞) to the interval [0, ∞) f being an increasing function and g being a decreasing function. If f {g (0)} = 0, then
(A) f {g (x)} ≥ f {g (0)} (B) g {f (x)} ≤ g {f (0)} (C) f {g (1)} = 0 (D) none of these

723. If the line ax + by + c = 0 is a normal to the curve xy = 1, then
(A) a > 0, b > 0 (B) a > 0, b < 0 (C) a < 0, b > 0 (D) a < 0, b < 0

724. If y = $x\log\left(\frac{x}{a+bx}\right)$, $x^3\frac{d^2y}{dx^2}$ =
(A) $\left(x\frac{dy}{dx}-y\right)^2$ (B) $\frac{a^2x^2}{(a+bx)^2}$ (C) $\left(\frac{dy}{dx}-y\right)^2$ (D) $\left(x\frac{dy}{dx}+y\right)^2$

725. On the interval I = [–2, 2] the function $f(x)=\begin{cases}(x+1)e^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x\neq 0\\ 0, & x=0\end{cases}$
(A) is continuous for all x ∈ I – {0} (B) is continuous for all x ∈ I
(C) assumes all intermediate values for f(–2) to f(2)
(D) has a maximum values equal to 3/e

726. If I = $\int_{-\pi/3}^{\pi/3}\frac{e^{\sec x}|\sin x|\sec^2 x}{(1+e^{\operatorname{cosec} x})}dx$, then
(A) I can be evaluated using the substitution secx = t
(B) I is irrational number
(C) I = e^2 – e (D) I = e – 1

727. Let f : R → R be a function defined by $f(x+1)=\frac{f(x)-5}{f(x)-3}$ ∀ x ∈ R. Then which of the following statement(s) is/are true
(A) f(2008) = f(2004) (B) f(2006) = f(2010)
(C) f(2006) = f(2002) (D) f(2006) = f(2018)

728. If $\lim_{x\to 0}\dfrac{ae^{2x}-b\cos 2x+ce^{-2x}-x\sin x}{x\sin x}=1$ and f (t) = (a + b)t^2 + (a – b)t + c, then

(A) a + b + c = 1 (B) a + b + c = 2 (C) f(1) = $\dfrac{3}{4}$ (D) f (1) = 1

729. If f(x) = $\sec^{-1}\left(\dfrac{x+2}{2x-3}\right)+\sin^{-1}\left(\dfrac{2x-3}{x+2}\right)$ then in their domain of definition

(A) f is non decreasing (B) f is non increasing
(C) f′(1) = 10 (D) f′(0) = 0

730. If f : R → R is decreasing and g : R → R is increasing then which of the following function is increasing

(A) f o f (B) g o g (C) f o g (D) g o f

731. If f : R → R^- (set of all negative reals) is decreasing and g : R → R^- is increasing then which of the following is decreasing

(A) fof (B) gof (C) f^2 (D) g^2

732. If f′(sin x) = $\cos^2$x for all x and f(1) = 1 then

(A) f is increasing (B) f is injective (C) f(0) = 1/3 (D) f(–1) = –1/3

733. If f(x + 1/x) = x^3 + 1/x^3 (x ≠ 0) then

(A) f(x) is increasing function (B) f(x) has a local maximum at x = –1
(C) f(x) is injective in its domain of definition
(D) The equation f(x) = 3 has a unique real root

734. The function f(x) = 4x^2 – 1/x increases over the interval

(A) (0, ∞) (B) (–∞, –1/2)
(C) (–1/2, 0) (D) (1, ∞)

735. The function f(x) = 2 ln |x| – x |x| decreases over the interval

(A) (1, ∞) (B) (–∞, –1) (C) (0, 1) (D) (–1, 0)

736. The function f(x) = 2|x| + 1/x^2 is increasing in the interval

(A) (–∞, –1) (B) (–1, 0) (C) (0, 1) (D) (1, ∞)

737. Which of the following is/are true

(A) e^{π}/π^{e} (B) $(1+\sin\pi/3)^{1+\cos\pi/3} > (1+\cos\pi/3)^{1+\sin\pi/3}$
(C) $101^{202} > 202^{101}$ (D) $(4/3)^{9/4} > (9/4)^{4/3}$

738. If f is differentiable at x = a ; then which of the following is FALSE

(A) If f(a) is an extreme value of f(x), then f′(a) = 0
(B) If f′(a) = 0, then f(a) is an extreme value of f(x)
(C) If f(a) = is not an extreme value of f(x) then f′(a) ≠ 0
(D) only one of these statement is false

739. If f(x) = $\dfrac{x}{\sin x}$ *and* $g(x)=\dfrac{x}{\tan x}$ where 0≤ x ≤ 1, then in this interval

(A) both f(x) and g(x) are increasing function (B) both f(x) and f(x) are decreasing function
(C) f(x) is an increasing function (D) g(x) is a decreasing function

740. Which of the following DOESNOT hold Rolle's theorem in [–1, 1]

(A) f(x) = |x| (B) f(x) = x^2 – 1 (C) f(x) = x^2 + x + 1 (D) f(x) = ln|x|

741. Let f(x) = f(x) + f(1 – x) and f″(x) < 0, 0 ≤ x ≤ 1. Then

(A) $g(x)$ *increases on* $\left[\frac{1}{2},1\right]$ (B) $g(x)$ *decreases on* $\left[\frac{1}{2},1\right]$

(C) $g(x)$ *decreases on* $\left[0,\frac{1}{2}\right]$ (D) $g(x)$ *increases on* $\left[0,\frac{1}{2}\right]$

742. If f′(x) = g(x) $(x – a)^2$, where g(a) ≠ 0 and g is continuous at x = a, then f is

(A) increasing in the nbd. of a if g(a) > 0 (B) increasing in the nbd. of a if g(a) < 0

(C) decreasing in the nbd. of a if g(a) > 0 (D) decreasing in the nbd. of a if g(a) < 0

743. Let $f(x) = 1 + 2^2x^2 + 3^2x^4 + 4^2x^6 + \ldots\ldots + n^2 x^{2n-2}$ then f(x) has

(A) exactly one critical point (B) at least one maximum

(C) exactly one minimum (D) None of these

744. Let f(x) = $\left|x^2 - 3x - 4\right|, -1 \le x \le 4$. Then

(A) f(x) is m.i. in $\left[-1,\frac{3}{2}\right]$ (B) f(x) is m.d. in $\left(\frac{3}{2},4\right)$

(C) the maximum value of f(x) is 25/4 (D) the minimum value of f(x) is 0

745. A particle is moving in a straight line such that its distance at any time t is $S = \frac{t^4}{4} - 2t^3 + 4t^2 + 7$, then

(A) velocity is max at t = $\frac{\left(6-2\sqrt{3}\right)}{3}$ (B) acceleration is min at t = 2

(C) the distance is min at t = 0, 4 (D) None of these

746. Let f : R → (–1, 1) defined by f(x) = $\frac{e^{x^3} + e^{-x^3}}{e^{x^3} - e^{-x^3}}$, then f is

(A) a one – one function (B) an increasing function

(C) a decreasing function (D) onto function

747. Let f(x) = $\frac{x^2+2}{[x]}$, 1 ≤ x ≤ 4.9, where [x] denotes the integral part of x. Then

(A) f(x) is m.i. in [1, 4.9] (B) least value of f(x) = 3

(C) greatest value of f(x) = 6.0075 (D) f(x) is m.d. in [1, 4.9]

748. Let $f(x) = ax^3 + bx^2 + cx + 1$ have extrema at x = α, β such that αβ < 0 and f(α). f(β) < 0 then the equation f(x) = 0 has

(A) three equal real roots (B) three distinct real roots

(C) one positive root if f(α) < 0 & f(β) > 0 (D) one negative root if f(α) > 0 & f(β) < 0

749. Let $h(x) = \{f(x)\}^3 + \{f(x)\}^2 + 10f(x)$. Then

(A) h increases as f increases (B) h decreases as f decreases

(C) h increases as f decreases (D) None of these

750. Let $h(x) = f(x) – \{f(x)\}^2 + \{f(x)\}^3$ for all real values of x. Then

(A) h is increasing if f(x) is increasing (B) h is increasing if f′(x) < 0

(C) h is decreasing if f is decreasing

(D) nothing can be said in general

751. Let f(x) = cos x sin 2x then

(A) $\min\limits_{x\in(-\pi,\pi)} f(x) > -\frac{7}{9}$ (B) $\min\limits_{x\in(-\pi,\pi)} f(x) > -\frac{9}{7}$

(C) $\min\limits_{x\in[-\pi,\pi]} f(x) > -\frac{1}{9}$ (D) $\min\limits_{x\in[-\pi,\pi]} f(x) > -\frac{2}{9}$

752. If OT and ON are perpendiculars dropped from the origin to the tangent and normal to the curve $x = a\sin^3 t$, $y = a\cos^3 t$ at an arbitrary point, then

(A) $4OT^2 + ON^2 = a^2$

(B) the length of the tangent = $\left|\frac{y}{\cos t}\right|$

(C) the length of the normal = $\left|\frac{y}{\sin t}\right|$

(D) None of these

753. If F(x) = f(x) g(x) and $f'(x)\,g'(x) = c$, then

(A) $F' = c\left[\frac{f}{f'} + \frac{g}{g'}\right]$ (B) $\frac{F''}{F} = \frac{f''}{f} + \frac{g''}{g} + \frac{2c}{fg}$

(C) $\frac{F'''}{F} = \frac{f'''}{f} + \frac{g'''}{g}$ (D) $\frac{F'''}{F''} = \frac{f'''}{f''} + \frac{g'''}{g''}$

754. Let $x^{\cos y} + y^{\cos x} = 5$. Then

(A) at x = 0, y = 0, y′ = 0 (B) at x = 0, y = 1, y′ = 0

(C) at x = y, y = 1, y′ = 1 (D) at x = 1, y = 0, y′ = 1

755. Let f(x) = (ax + b) cos x + (cx + d) sin x and f′(x) = x cos x be an identify in x, then

(A) a = 0 (B) b = 1

(C) c = 1 (D) d = 0

756. The function f(x) = max{(1 – x), (1 + x), 2}, x ∈ (–∞,∞) is

(A) continuous for all x

(B) differentiable for all x

(C) except x = 1 and x = –1 differentiable for all x

(D) None of these

757. If $f_n(x) = e^{f_{n-1}(x)}$ for all n ∈ N and $f_0(x) = x$ then $\frac{d}{dx}\{f_n(x)\}$ is equal to

(A) $f_n(x).\frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x).f_{n-1}(x)$

(C) $f_n(x).f_{n-1}(x)......f_2(x).f_1(x)$

(D) None of these

758. Let $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1).x^2 + xf'(x) + f''(x)$ then

(A) f′(1) + f′(2) = 0 (B) g′(2) = g′(1)

(C) g′′(2) + f′′(3) = 6 (D) None of these

759. Let f(t) = ln t. Then $\frac{d}{dx}\left\{\int_{x^2}^{x^3} f(t)\,dt\right\}$

(A) has a value 0 when x = 0

(B) has a value 0 when x = 1, x = $\frac{4}{9}$

(C) has a value $9e^2 - 4e$ when x = e

(D) has a d.c. 27e – 8 when x = e

760. If –1, f(x) = $\begin{vmatrix} \sec\theta & \tan^2\theta & 1 \\ \theta\sec x & \tan x & x \\ 1 & \tan x - \tan\theta & 0 \end{vmatrix}$ and 6^n $\sin\left(2x+\frac{n\pi}{2}\right)\cos\left(\frac{3x+n\pi}{2}\right)$ are in AP for all x, y, and y_n, then

(A) 0

(B) $y = \int_0^x f(t)\sin\{k(x-t)\}\,dt$

(C) $y = \sin^2\alpha + \cos^2(\alpha+\beta) + 2\sin\alpha\sin\beta\cos(\alpha+\beta)$

(D) $\frac{1}{\sqrt{1+9y^2}}$

761. If $f(x) = \cos^{-1}\cos(x - \pi/4)$ then

(A) $f'\left(\frac{\pi}{2}\right) = 1$ (B) $f'(0) = -1$

(C) $f'(\pi) = 0$ (D) $f'\left(\frac{\pi}{4}\right) = 0$

762. If f(x) = $|\sin x - \cos x|$ then

(A) $f'\left(\frac{\pi}{2}\right) = 1$ (B) $f(\pi) = -1$

(C) $f'\left(\frac{3\pi}{4}\right) = -1$ (D) $f'(0) = -1$

763. If f(x) = (ax + b) sin x – (cx + d) cosx and f′(x) = x sin x then

(A) a = d = 0 (B) b = 1, c = –1

(C) b = c = 1 (D) a = 0, d = –1

764. If f(x + y) = f(x) + f(y) for all x, y $\in$ R and f′(0) exists then

(A) f′(x) = f′(0)

(B) f(x) = kx

(C) f(x) = xf(1)

(D) f(x) is an even as well as periodic function

765. If f(x – y), f(x) f(y) and f(x + y) are in A.P. for all x, y $\in$ R and f(0) $\neq$ 0, then

(A) f(x) is an even function (B) f′(1) + f′(–1) = 0

(C) f′(2) – f′(–2) = 0 (D) f(3) + f(–3) = 0

766. For the function $f(x) = \ell n \quad (\sin^{-1} \ell og_2 x)$,

(A) Domain is $\left[\frac{1}{2}, 2\right]$ (B) Range is $\left(-\infty, \ell n\frac{\pi}{2}\right]$

(C) Domain is (1, 2] (D) Range is R

767. A function 'f' from the set of natural numbers to integers defined by,

$$f(n) = \begin{cases} \frac{n-1}{2} & , \text{ when } n \text{ is odd} \\ -\frac{n}{2} & , \text{ when } n \text{ is even} \end{cases} \text{ is:}$$

(A) one-one (B) many-one (C) onto (D) into

768. If $F(x) = \frac{\sin \pi [x]}{\{x\}}$, then F (x) is:

(A) periodic with fundamental period 1

(B) even

(C) range is singleton

(D) identical to $\text{sgn}\left(\text{sgn}\frac{\{x\}}{\sqrt{\{x\}}}\right) - 1$, where {x} denotes fractional part function and [.] denotes greatest integer function and sgn (x) is a signum function.

769. $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective.

(A) $f(x) = x^2$ (B) $g(x) = x^3$ (C) $h(x) = \sin 2x$ (D) $k(x) = \sin(\pi x/2)$

770. If $f(x) = \sqrt{x}$ and $g(x) = x - 1$, then

(A) fog is continuous on $[0, \infty)$ (B) gof is continuous on $[0, \infty)$

(C) fog is continuous on $[1, \infty)$ (D) none of these

771. The function $f(x) = \begin{cases} x^m \sin\frac{1}{x} & , \quad x > 0 \\ 0 & , \quad x = 0 \end{cases}$ is continuous at x = 0 if

(A) $m \geq 0$ (B) $m > 0$ (C) $m < 1$ (D) $m \geq 1$

772. Let $f(x) = \frac{1}{[\sin x]}$ ([.] denotes the greatest integer function) then

(A) domain of f(x) is $(2n\pi + \pi, 2n\pi + 2\pi) \cup \{2n\pi + \pi/2\}$

(B) f(x) is continuous when $x \in (2n\pi + \pi, 2n\pi + 2\pi)$

(C) f(x) is continuous at $x = 2n\pi + \pi/2$ (D) f(x) has the period 2π

773. Let $f(x) = [x] + \sqrt{x - [x]}$, where [x] denotes the greatest integer function. Then

(A) f(x) is continuous on R^+ (B) f(x) is continuous on R

(C) f(x) is continuous on $R - I$ (D) discontinuous at x = 1

774. Let f(x) and g(x) be defined by f(x) = [x] and $g(x) = \begin{cases} 0 & , \quad x \in I \\ x^2 & , \quad x \in R - I \end{cases}$ (where [.] denotes the greatest integer function) then

(A) $\lim_{x \to 1} g(x)$ exists, but g is not continuous at x = 1

(B) $\lim_{x \to 1} f(x)$ does not exist and f is not continuous at x = 1

(C) gof is continuous for all x

(D) fog is continuous for all x

775. Which of the following function(s) defined below has/have single point continuity.

(A) $f(x) = \begin{bmatrix} 1 & \text{if } x \in Q \\ 0 & \text{if } x \notin Q \end{bmatrix}$ (B) $g(x) = \begin{bmatrix} x & \text{if } x \in Q \\ 1-x & \text{if } x \notin Q \end{bmatrix}$

(C) $h(x) = \begin{bmatrix} x & \text{if } x \in Q \\ 0 & \text{if } x \notin Q \end{bmatrix}$ (D) $k(x) = \begin{bmatrix} x & \text{if } x \in Q \\ -x & \text{if } x \notin Q \end{bmatrix}$

776. Two functions f & g have first & second derivatives at x = 0 & satisfy the relations,

$f(0) = \frac{2}{g(0)}$, $f'(0) = 2g'(0) = 4g(0)$, $g''(0) = 5f''(0) = 6f(0) = 3$ then:

(A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$ (B) if $k(x) = f(x).\, g(x) \sin x$ then $k'(0) = 2$

(C) $\underset{x\to 0}{\text{Limit}} \frac{g'(x)}{f'(x)} = \frac{1}{2}$ (D) none

777. If $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in N$ and $f_o(x) = x$, then $\frac{d}{dx}\{f_n(x)\}$ is equal to:

(A) $f_n(x).\frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x).\, f_{n-1}(x)$

(C) $f_n(x).\, f_{n-1}(x)........ f_2(x).\, f_1(x)$

(D) none of these

778. If f is twice differentiable such that $f''(x) = -f(x)$ and $f'(x) = g(x)$. If h(x) is a twice differentiable function such that $h'(x) = [f(x)]^2 + [g(x)]^2$. If h(0) = 2, h(1) = 4, then the equation y = h(x) represents:

(A) a curve of degree 2

(B) a curve passing through the origin

(C) a straight line with slope 2

(D) a straight line with y intercept equal to 2.

779. Given $f(x) = -\frac{x^3}{3} + x^2 \sin 1.5\, a - x \sin a.\, \sin 2a - 5 \sin^{-1}(a^2 - 8a + 17)$ then:

(A) $f'(x) = -x^2 + 2x \sin 6 - \sin 4 \sin 8$ (B) $f'(\sin 8) > 0$

(C) $f'(x)$ is not defined at $x = \sin 8$ (D) $f'(\sin 8) < 0$

780. $P(x)$ is a fourth degree polynomial such that

(a) $P(-x) = P(x)$ (b) $P(-x) > 0 \;\; \forall x \in R$ (c) $P(0) = 1$

(d) $P(x)$ has exactly two local minima at x_1 and x_2 such that $|x_1 - x_2| = 2$

The line $y = 1$ touches the curve at a certain point Q and the enclosed area between the line and the curve is $\frac{8\sqrt{2}}{15}$. Let $g(x) = Ax^2 + Bx + C \;\; (A \neq 0)$ such that $\lim_{x\to 0} \frac{P(x) - g(x) - g(-x)}{x^2}$ is finite and is equal to the slope of the tangent of $g(x)$ at $x = -1$. Also $P(x)$ and $g(x)$ have common tangent at Q parallel to x-axis, Then

(A) the value of A is $\frac{-1}{2}$ (B) the value of $B+C$ is $\frac{-1}{2}$

(C) the value of $A+C$ is 1 (D) the value of $A+B+C$ is

781. If f(x) = (ax + b) sin x + (cx + d) cos x, then the values of a, b, c and d such that $f'(x) = x \cos x$ for all x are
(A) a = d = 1 (B) b = 0 (C) c = 0 (D) b = c

782. If $f(x) = \sum_{k=0}^{n} a_k |x|^k$, where a_i's are real constants, then f(x) is
(A) continuous at x = 0 for all a_i (B) differentiable at x = 0 for all $a_i \in R$
(C) differentiable at x = 0 for all $a_{2k+1} = 0$ (D) None of these

783. Consider the curve $f(x) = x^{1/3}$, then
(A) the equation of tangent at (0, 0) is x = 0
(B) the equation of normal at (0, 0) is y = 0
(C) normal to the curve does not exist at (0, 0)
(D) f(x) and its inverse meet at exactly 3 points.

784. The equation of normal to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ ($n \in N$) at the point with abscissa equal to 'a' can be:
(A) $ax + by = a^2 - b^2$ (B) $ax + by = a^2 + b^2$
(C) $ax - by = a^2 - b^2$ (D) $bx - ay = a^2 - b^2$

785. If the line, ax + by + c = 0 is a normal to the curve xy = 2, then:
(A) a < 0, b > 0 (B) a > 0, b < 0
(C) a > 0, b > 0 (D) a < 0, b < 0

786. In the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$, at point (2, – 1)
(A) length of subtangent is 7/6. (B) slope of tangent = 6/7
(C) length of tangent = $\sqrt{(85)}/6$ (D) None of these

787. If y = f(x) be the equation of a parabola which is touched by the line y = x at the point where x = 1. Then
(A) $f'(1) = 1$ (B) $f'(0) = f'(1)$
(C) $2f(0) = 1 - f'(0)$ (D) $f(0) + f'(0) + f''(0) = 1$

788. If the tangent to the curve $2y^3 = ax^2 + x^3$ at the point (a, a) cuts off intercepts α, β on co-ordinate axes, where $\alpha^2 + \beta^2 = 61$, then the value of 'a' is equal to:
(A) 20 (B) 25
(C) 30 (D) – 30

789. The curves $ax^2 + by^2 = 1$ and $Ax^2 + By^2 = 1$ intersect orthogonally, then
(A) $\frac{1}{a} + \frac{1}{A} = \frac{1}{b} + \frac{1}{B}$ (B) $\frac{1}{a} - \frac{1}{A} = \frac{1}{b} - \frac{1}{B}$
(C) $\frac{1}{a} + \frac{1}{b} = \frac{1}{B} - \frac{1}{A}$ (D) $\frac{1}{a} - \frac{1}{b} = \frac{1}{A} - \frac{1}{B}$

790. The set of values of a for which the function $f(x) = x^2 + ax + 1$ is an increasing function on [1, 2] is I_1 and decreasing in [1, 2] is I_2, then :

(A) $I_1 : a \in (2, \infty)$ (B) $I_2 : a \in (-\infty, -4)$

(C) $I_2 : a \in (-\infty, -4]$ (D) $I_1 ; a \in [-2, \infty)$

791. If f is an even function then

(A) f^2 increases on (a, b) (B) f cannot be monotonic

(C) f^2 need not increases on (a, b) (D) f has inverse

792. Let $g(x) = 2f(x/2) + f(1 - x)$ and $f''(x) < 0$ in $0 \le x \le 1$ then g(x) :

(A) decreases in $\left[0, \frac{2}{3}\right]$ (B) decreases $\left[\frac{2}{3}, 1\right]$

(C) increases in $\left[0, \frac{2}{3}\right]$ (D) increases in $\left[\frac{2}{3}, 1\right]$

793. On which of the following intervals, the function $x^{100} + \sin x - 1$ is strictly increasing

(A) $(-1, 1)$ (B) $[0, 1]$ (C) $[\pi/2, \pi]$ (D) $[0, \pi/2]$

794. The function $y = \frac{2x-1}{x-2}$ $(x \ne 2)$:

(A) is its own inverse (B) decreases for all values of x

(C) has a graph entirely above x-axis (D) is bound for all x.

795. Let f and g be two functions defined on an interval I such that $f(x) \ge 0$ and $g(x) \le 0$ for all $x \in I$ and f is strictly decreasing on I while g is strictly increasing on I then

(A) the product function fg is strictly increasing on I

(B) the product function fg is strictly decreasing on I

(C) fog(x) is monotonically increasing on I

(D) fog (x) is monotonically decreasing on I

796. Let $f(x) = 40/(3x^4 + 8x^3 - 18x^2 + 60)$, consider the following statement about f(x).

(A) f(x) has local minima at x = 0

(B) f(x) has local maxima at x = 0

(C) absolute maximum value of f(x) is not defined

(D) f(x) is local maxima at x = – 3, x = 1

797. Maximum and minimum values of the function,

$$f(x) = \frac{2-x}{\pi}\cos \pi (x + 3) + \frac{1}{\pi^2}\sin \pi (x + 3) \quad 0 < x < 4 \text{ occur at :}$$

(A) x = 1 (B) x = 2 (C) x = 3 (D) $x = \pi$

798. If $\lim_{x\to a} f(x) = \lim_{x\to a} [f(x)]$ ([.] denotes the greater integer function) and f(x) is non-constant continuous function, then

(A) $\lim_{x\to a} f(x)$ is integer (B) $\lim_{x\to a} f(x)$ is non-integer

(C) f (x) has local maximum at x = a (D) f (x) has local minima at x = a

799. If the derivative of an odd cubic polynomial vanishes at two different values of 'x' then

(A) coefficient of x^3 & x in the polynomial must be same in sign

(B) coefficient of x^3 & x in the polynomial must be different in sign

(C) the values of 'x' where derivative vanishes are closer to origin as compared to the respective roots on either side of origin.

(D) the values of 'x' where derivative vanishes are far from origin as compared to the respective roots on either side of origin.

800. Let $f(x) = \ln(2x - x^2) + \sin\dfrac{\pi x}{2}$. Then

(A) graph of f is symmetrical about the line x = 1

(B) graph of f is symmetrical about the line x = 2

(C) maximum value of f is 1

(D) minimum value of f does not exist

801. The curve $y = \dfrac{x+1}{x^2+1}$ has:

(A) x = 1, the point of inflection

(B) $x = -2 + \sqrt{3}$, the point of inflection

(C) x = – 1, the point of minimum

(D) $x = -2 - \sqrt{3}$, the point of inflection

802. If the function y = f (x) is represented as, $x = \phi(t) = t^3 - 5t^2 - 20t + 7$
$y = \psi(t) = 4t^3 - 3t^2 - 18t + 3$ $(-2 < t < 2)$, then:

(A) $y_{max} = 12$ (B) $y_{max} = 14$ (C) $y_{min} = -67/4$ (D) $y_{min} = -69/4$

803. The maximum and minimum values of $y = \dfrac{ax^2 + 2bx + c}{Ax^2 + 2Bx + C}$ are those for which

(A) $ax^2 + 2bx + c - y(Ax^2 + 2Bx + C)$ is equal to zero

(B) $ax^2 + 2bx + c - y(Ax^2 + 2Bx + C)$ is a perfect square

(C) $\dfrac{dy}{dx} = 0$ and $\dfrac{d^2y}{dx^2} \neq 0$

(D) $ax^2 + 2bx + c - y(Ax^2 + 2Bx + C)$ is not a perfect square

804. If $\displaystyle\int \frac{(x-1)\,dx}{x^2\sqrt{2x^2 - 2x + 1}}$ is equal to $\dfrac{\sqrt{f(x)}}{g(x)} + c$ then

(A) $f(x) = 2x^2 - 2x + 1$

(B) g(x) = x + 1

(C) g(x) = x

(D) $f(x) = \sqrt{2x^2 - 2x}$

805. $\displaystyle\int \frac{dx}{5 + 4\cos x} = I\tan^{-1}\left(m\tan\frac{x}{2}\right) + C$ then:

(A) I = 2/3 (B) m = 1/3 (C) I = 1/3 (D) m = 2/3

806. If $\displaystyle\int \frac{3\cot 3x - \cot x}{\tan x - 3\tan 3x}\,dx = p\,f(x) + q\,g(x) + c$ where 'c' is a constant of integration, then

(A) $p = 1$; $q = \dfrac{1}{\sqrt{3}}$; $f(x) = x$; $g(x) = \ell n\left|\dfrac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x}\right|$

(B) $p = 1$; $q = -\dfrac{1}{\sqrt{3}}$; $f(x) = x$; $g(x) = \ell n\left|\dfrac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x}\right|$

(C) $p = 1$; $q = -\dfrac{2}{\sqrt{3}}$; $f(x) = x$; $g(x) = \ell n\left|\dfrac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x}\right|$

(D) $p = 1$; $q = -\dfrac{1}{\sqrt{3}}$; $f(x) = x$; $g(x) = \ell n\left|\dfrac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x}\right|$

807. $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x}\, dx$ is equal to:

(A) $\cot^{-1}(\cot^2 x) + c$ (B) $-\cot^{-1}(\tan^2 x) + c$ (C) $\tan^{-1}(\tan^2 x) + c$ (D) $-\tan^{-1}(\cos 2x) + c$

808. If f(x) is integrable over [1, 2], then $\int_1^2 f(x)\, dx$ is equal to

(A) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r}{n}\right)$ (B) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=n+1}^{2n} f\left(\frac{r}{n}\right)$

(C) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r+n}{n}\right)$ (D) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{2n} f\left(\frac{r}{n}\right)$

809. If $f(x) = \int_0^x (\cos^4 t + \sin^4 t)\, dt$, $f(x + \pi)$ will be equal to

(A) $f(x) + f(\pi)$ (B) $f(x) + 2f(\pi)$ (C) $f(x) + f\left(\frac{\pi}{2}\right)$ (D) $f(x) + 2f\left(\frac{\pi}{2}\right)$

810. The value of $\int_0^1 \frac{2x^2+3x+3}{(x+1)\left(x^2+2x+2\right)}\, dx$ is:

(A) $\frac{\pi}{4} + 2\ln 2 - \tan^{-1} 2$ (B) $\frac{\pi}{4} + 2\ln 2 - \tan^{-1}\frac{1}{3}$

(C) $2\ln 2 - \cot^{-1} 3$ (D) $-\frac{\pi}{4} + \ln 4 + \cot^{-1} 2$

811. The differential equation of the curve for which the initial ordinate of any tangent is equal to the corresponding subnormal

(A) is linear (B) is homogeneous

(C) has separable variables (D) is none of these

812. The solution of $x^2 y_1^2 + xy\, y_1 - 6y^2 = 0$ are

(A) $y = Cx^2$ (B) $x^2 y = C$ (C) $\frac{1}{2}\log y = C + \log x$ (D) $x^3 y = C$

813. The orthogonal trajectories of the system of curves $\left(\frac{dy}{dx}\right)^2 = a/x$ are

(A) $9a(y + c) = 4x^3$ (B) $y + C = \frac{-2}{3\sqrt{a}} x^{3/2}$ (C) $y + C = \frac{2}{3\sqrt{a}} x^{3/2}$ (D) None

814. The solution of $\left(\frac{dy}{dx}\right)(x^2 y^3 + xy) = 1$ is

(A) $1/x = 2 - y^2 + C e^{-y^2}/2$

(B) the solution of an equation which is reducible to linear equation.

(C) $2/x = 1 - y^2 + e^{-y}/2$ (D) $\frac{1-2x}{x} = -y^2 + Ce^{-y^2}/2$

SECTION-3 (COMPREHENSION TYPE)

COMPREHENSION-1

Paragraph for Questions Nos. 815 to 817

Let f : R $\rightarrow$ R be a continuous function such that $f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{4}\right) = x^2$

815. f(3) is equal to

(A) f(0) (B) 4 + f(0)

(C) 9 + f(0) (D) 16 + f(0)

816. The equation f(x) – x – f(0) = 0 have exactly

(A) No solution (B) One solution

(C) Two solution (D) infinite solution

817. $f'(0)$ is equal to

(A) 0 (B) 1

(C) f(0) (D) - f(0)

COMPREHENSION-2

Paragraph for Questions Nos. 818 to 820

If f(x) = max. ($|x^2 - 1|$, $|x - 1|$) and g(x) = $\int_a^x f(t)\,dt$, x $\in$ R.

818. The value of f(x) is

(A) $f(x) = \begin{cases} x^2 - 1, & x \le -2 \\ 1 - x, & -2 < x \le 0 \\ 1 - x^2, & 0 < x < 1 \\ x^2 - 1, & x > 1 \end{cases}$

(B) $f(x) = \begin{cases} x^2 - 1, & x \le -2 \\ 1 - x^2, & -2 < x \le 0 \\ 1 - x, & 0 < x \le 1 \\ x - 1, & x > 1 \end{cases}$

(C) $f(x) = \begin{cases} x^2 - 1, & x \le 1 \\ x - 1, & x > 1 \end{cases}$

(D) $f(x) = \begin{cases} x - 1, & x \le 1 \\ x^2 - 1, & x > 1 \end{cases}$

819. The function f(x) is continuous for x belongs to

(A) R – {0, 1} (B) R – {–2, 0, 1}

(C) R (D) none of these

820. The function g(x) is differentiable for

(A) R – {0, 1} (B) R – {–2, 0, 1}

(C) R (D) None of these

COMPREHENSION-3

Paragraph for Questions Nos. 821 to 823

Let $f(x) = \log_{\{x\}}[x]$
$g(x) = \log_{[x]}\{x\}$
$h(x) = \log_{\{x\}}\{x\}$
where [.], {.} denotes the greatest integer function and fractional part.

821. For $x \in (1, 5)$ the f(x) is not defined at how many points

(A) 5 (B) 4
(C) 3 (D) 2

822. If A = {x: x $\in$ domain of f(x)} and B = {x: x $\in$ domain of g(x)} then A – B will be

(A) (2, 3) (B) (1, 3)
(C) (1, 2) (D) none of these

823. Domain of h(x) is

(A) R (B) I
(C) R – I (D) $R^+ - I$

COMPREHENSION-4

Paragraph for Questions Nos. 824 to 826

Let a function f(x) satisfies the condition $f(x+y) = \dfrac{f(x)+f(y)}{f(x)}$ such that $f'(0) = 2$ and $f(x) \geq 0$. Using the above information answer the following:

824. The curve y = f(x) is

(A) $y = \sqrt{2(x+1)}$ (B) $y = 2\sqrt{(x+1)}$
(C) y = ln(x + 1) (D) y = ln(x – 1)

825. Area bounded between $y = f(|x|)$ *and* $y = 7 - |x|$ is

(A) $\dfrac{23}{6}$ sq. unit (B) $\dfrac{11}{6}$ sq. unit
(C) $\dfrac{86}{6}$ sq. unit (D) 7 sq. unit

826. The number of points where g(x) = max. {f(x), 6, 7 – |x|} is non differentiable $\forall$ x $\in$[–10, 10] are

(A) 5 (B) 6
(C) 7 (D) 8

COMPREHENSION-5

Paragraph for Questions Nos. 827 to 829

Let f : [2, ∞) → [1, ∞) defined by f(x) = $2^{x^4-4x^2}$ and $g:\left[\frac{\pi}{2},\pi\right]\to A$ defined by $g(x)=\frac{\sin x+4}{\sin x-2}$ be two invertible functions, then

827. $f^{-1}(x)$ is equal to

(A) $-\sqrt{2+\sqrt{4+\log_2 x}}$ (B) $\sqrt{2+\sqrt{4+\log_2 x}}$

(C) $\sqrt{2-\sqrt{4+\log_2 x}}$ (D) None of these

828. The set A is equal to

(A) [–5, –2] (B) [2, 5] (C) [–5, 2] (D) [–3, –2]

829. The domain of $f^{-1}g^{-1}(x)$ is

(A) [–5, sin 1] (B) $\left[-5,\frac{\sin 1}{2-\sin 1}\right]$ (C) $\left[-5,-\frac{(4+\sin 1)}{2-\sin 1}\right]$ (D) $\left[-\frac{(4+\sin 1)}{2-\sin 1},-2\right]$

COMPREHENSION-6

Paragraph for Questions Nos. 830 to 832

A function is defined as the approaching value of the expression $\frac{1+2(x)^{2n}}{1+x^{2n}}$ as x approaches to infinity.

830. The domain and range of the function is

(A) $(-\infty, 1)\cup(1,\infty), \left\{1,-1,\frac{3}{2},-\frac{3}{2},2,-2\right\}$ (B) $(-\infty,\infty), \left\{1,\frac{3}{2},2\right\}$

(C) (1, ∞), {1, – 1} (D) None of these

831. The points of discontinuity of the function are

(A) 1, – 1 (B) 1, 0, – 1 (C) 1, 3 (D) None of these

832. The composition of the function with y = |x| is

(A) $\begin{cases}1, & |x|<1\\ \frac{3}{2}, & |x|=1\\ 2, & |x|>1\end{cases}$ (B) $\begin{cases}-2, & x\le -1\\ -\frac{3}{2}, & -1<x<0\\ 1, & 0\le x<1\\ 2, & x\ge 1\end{cases}$

(C) $\begin{cases}-2, & x\le -1\\ -\frac{3}{2}, & x=-1\\ -1, & -1<x<0\\ 1, & 0\le x<1\\ \frac{3}{2}, & x=1\\ 2, & x>1\end{cases}$ (D) None of these

COMPREHENSION-7

Paragraph for Questions Nos. 833 to 835

Let f(x) be a real valued function not identically zero, such that
$f(x + y^n) = f(x) + (f(y))^n \quad \forall x, y \in R$ where $n \in N$ $(n \neq 1)$ and $f'(0) \geq 0$.

833. The value of $f'(0)$ is

(A) 1 (B) 1 + n
(C) n (D) 2

834. The value of f(5) is

(A) 2 (B) 3
(C) 5n (D) 5

835. $\int_0^1 f(x)dx$ is equal to

(A) $\frac{1}{2n}$ (B) 2n
(C) $\frac{1}{2}$ (D) 2

COMPREHENSION-8

Paragraph for Questions Nos. 836 to 838

Let f: R $\to$ R be function defined as

$$f(x) = \begin{cases} 1-|x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

and $g(x) = f(x - 1) + f(x + 1) \ \forall x \in R$.

836. For $x \in [-1, 1]$, g(|x|) is equal to

(A) x (B) – |x|
(C) |x| (D) 2 + |x|

837. Value of $g\left(-\frac{3}{2}\right)$ is equal to

(A) 0 (B) $\frac{1}{2}$
(C) $\frac{7}{2}$ (D) $\frac{3}{2}$

838. The number of points at which y = |g(x)| is non differentiable is

(A) 3 (B) 4
(C) 5 (D) 6

COMPREHENSION-9

Paragraph for Questions Nos. 839 to 841

Let y = f(x) be a function continuous and differentiable every where also,
$g_1(x) = \min\{|f(x)|, |f(x-1)|\}$
$g_2(x) = f(|x|)$
$g_3(x) = -f(|x|)$
If f(x) = x – 1, then

839. The area bounded by $y = g_1(x)$, x-axis and lines x = 0 and x = 3 is equal to

(A) 1 (B) 2
(C) $\frac{5}{4}$ (D) None of these

840. The area bounded by $y = g_2(x)$ and $y = g_3(x)$ is equal to

(A) 2 (B) 4
(C) 1 (D) none of these

841. The area bounded by $y = g_3(x)$ and $y = \ln(|x|)$ is equal to

(A) 2 (B) 3
(C) 4 (D) None of these

COMPREHENSION-10

Paragraph for Questions Nos. 842 to 844

Let f be a polynomial function such that $f(x)\,f(y) + 2 = f(x) + f(y) + f(xy)\ \forall\ x, y \in R^+ \cup \{0\}$ and f(x) is one-one $\forall\ x \in R^+$ with f(0) = 1 and f′(1) = 2.

842. The function y = f(x) is given by

(A) $x^{1/3} - 1$ (B) $1 + \frac{2x^3}{3}$
(C) $1 + x^2$ (D) $1 - x^2$

843. Area bounded between the curve $y = x^2$ and y = g(x) where $g(x) = \frac{2}{f(x)}$ and x-axis is

(A) $\frac{\pi}{2} - \frac{1}{3}$ (B) $\pi - \frac{1}{3}$
(C) $\frac{\pi}{2} - \frac{1}{6}$ (D) $\pi - \frac{2}{3}$

844. If $h(x) = \min\left\{\frac{2}{f(x)},\ x^2,\ |1-|x||\right\}$, then the number of points of non-differentiability of h(x) is/are

(A) 3 (B) 4
(C) 5 (D) 6

COMPREHENSION-11

Paragraph for Questions Nos. 845 to 847

Let f(x) be a function such that its derivative $f'(x)$ is continuous in [a, b] and derivable in (a, b). Consider a function $\phi(x) = f(b) - f(x) - (b - x) f'(x) - (b - x)^2 A$. If Rolle's theorem is applicable to $\phi(x)$ on [a, b], answer following questions

845. If there exist some number c (a < c < b) such that $\phi'(c) = 0$ and $f(b) = f(a) + (b - a) f'(a) + \lambda (b - a)^2 f''(c)$, then λ is

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

846. Let $f(x) = x^3 - 3x + 3$, a = 1 and b = 1 + h. If there exists $c \in (1, 1 + h)$ such that $\phi'(c) = 0$ and $\frac{f(1+h)-f(1)}{h^2} = \lambda c$, then λ =

(A) $\frac{1}{2}$ (B) 2 (C) 3 (D) does not exists

847. Let f(x) = sin x , a = α and b = α + h. If there exists a real number t such that 0 < t < 1, $\phi'(\alpha + th)$ = 0 and $\frac{\sin(\alpha+h)-\sin\alpha-h\cos\alpha}{h^2} = \lambda \sin(\alpha + th)$, then λ =

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$

COMPREHENSION-12

Paragraph for Questions Nos. 848 to 850

Sometimes we can find the sum of series by use of differentiation. If we know the sum of a series

e.g. if $f(x) = f_1(x) + f_2(x) + \ldots\ldots\ldots$

$f'(x) = f_1'(x) + f_2'(x) + \ldots\ldots\ldots$

e.g. $(1 - x)^{-1} = 1 + x + x^2 + x^3 \ldots\ldots\ldots$ $|x| < 1$

Hence the sum of the AGP

$1 + 2x + 3x^2 + \ldots\ldots = (1 - x)^{-2}$ (By differentiation both the sides)

Now answer the question that follows

848. The sum of the series $\frac{2^2}{1!} + \frac{3^2}{2!} + \frac{4^2}{3!} + \ldots\ldots\ldots$ upto ∞ is

(A) 4e – 1 (B) 5e (C) 5e – 1 (D) 4e

849. Sum of the series $1 - \frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \ldots\ldots$ upto ∞ is

(A) $\frac{1}{2} - \ell n2$ (B) $1 - \ell n2$ (C) ∞ (D) $\frac{3}{2} - \ln 2$

850. Sum of the series $1 + 1 + \frac{3}{4} + \frac{4}{8} + \frac{5}{16} + \ldots\ldots$ upto infinite terms, is

(A) 4 (B) 2 (C) 1 (D) $\frac{1}{4}$

COMPREHENSION-13

Paragraph for Questions Nos. 851 to 853

If $y = \int_{u(x)}^{v(x)} f(t)\,dt$, let us define $\frac{dy}{dx}$ in a different manner as $\frac{dy}{dx} = v'(x) f^2(v(x)) - u'(x) f^2(u(x))$ and the equation of the tangent at (a, b) as $y - b = \left(\frac{dy}{dx}\right)_{(a,b)} (x - a)$

851. If $y = \int_{x}^{x^2} t^2\,dt$, then equation of tangent at x = 1 is

(A) $y = x + 1$ (B) $x + y = 1$ (C) $y = x - 1$ (D) $y = x$

852. If $F(x) = \int_{1}^{x} e^{t^2/2} (1 - t^2)\,dt$, then $\frac{d}{dx} F(x)$ at x = 1 is

(A) 0 (B) 1 (C) 2 (D) – 1

853. If $y = \int_{x^3}^{x^4} \ell nt\,dt$, then $\lim_{x \to 0^+} \frac{dy}{dx}$ is

(A) 0 (B) 1 (C) 2 (D) – 1

COMPREHENSION-14

Paragraph for Questions Nos. 854 to 856

Let f be a function defined so that every element of the codomain has at most two pre-images and there is at least one element in the co-domain which has exactly two pre-images we shall call this function as "two-one" function. A two-one function is definitely a many one function but vice-versa is not true. For example, $y = |e^x - 1|$ is a "two-one" function. $y = x^3 - x$ is a many one function but not a "two-one" function. In the light of above definition answer the following questions:

854. In the following functions which one is a "two-one" function :-

(A) $y = |\ell n|x||$ (B) $y = x^2 \sin x$
(C) $y = x^3 + 3x + 1$ (D) $y = x^4 - x + 1$

855. Let f(x) = {x} be the fractional part function. For what domain is the function "two-one"?

(A) $\left[\frac{1}{2}, \frac{5}{2}\right]$ (B) $\left[-\frac{1}{2}, \frac{3}{2}\right)$
(C) [1, 2) (D) None of these

856. A continuous "two-one" function defined for $x \in (a, b)$ has

(A) atmost one point of extremum
(B) atleast two points of extrema
(C) exactly one point of extremum
(D) none of these

COMPREHENSION-15

Paragraph for Questions Nos. 857 to 859

Continuous Probability Distributions. A continuous distribution is one in which the variate may take any value between certain limits a and b, a < b. Suppose that the probability of the variate X falling in the infinitesimal interval $x - \frac{1}{2}dx$ to $x + \frac{1}{2}dx$ is expressible as f(x) dx, where f(x) is a continuous function of x.

Symbolically, $P(x - \frac{1}{2}dx \le X \le x + \frac{1}{2}dx) = f(x)\,dx$

where f(x) is called the probability density function (abbreviated as p.d.f.) or simply density function. The continuous curve y = f(x) is called probability curve ; and when this is symmetrical, the distribution is said to be symmetrical. Clearly, the probability density function possesses the following properties:

(i) $f(x) \ge 0$ for every x in the interval [a, b], a < b

(ii) $\int_a^b f(x)\,dx = 1$, a, b > 0

since the total area under the curve is unity.

(iii) Furthermore, we define for any [c, d], where c, d ∈ [a, b], c < d;

$$P(c \le X \le d) = \int_c^d f(x)\,dx \qquad \text{...............(i)}$$

We define F(x), the cumulative distribution function (abbreviated as c.d.f.) of the random variate X where $F(x) = P(X \le x)$

or $$F(x) = \int_a^x f(x)\,dx\,. \qquad \text{...............(ii)}$$

857. If $f(x) = \begin{cases} 2x & ; \; 0 \le x \le 1 \\ 0 & ; \quad x > 1 \end{cases}$ then the probability that $x \le \frac{1}{2}$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) $\frac{1}{8}$

858. In Q. No. 857, probability that $x \ge \frac{3}{4}$ given $x \ge \frac{1}{2}$ is

(A) $\frac{7}{16}$ (B) $\frac{3}{4}$ (C) $\frac{3}{7}$ (D) $\frac{7}{12}$

859. Suppose the life in hours (x) of a certain kind of radio tube has the probability density function $f(x) = \frac{100}{x^2}$ when x > 100 and zero when x < 100. Then the probability that none of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation, is

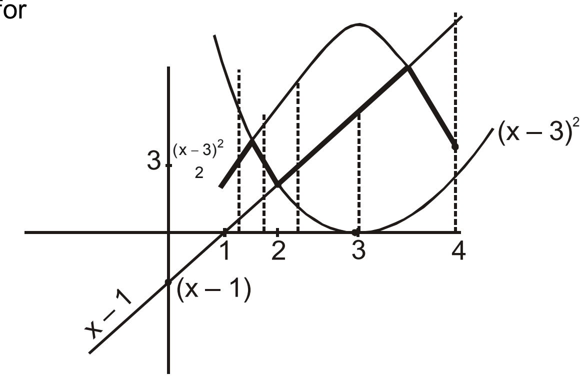
(A) $\frac{1}{27}$ (B) $\frac{8}{27}$ (C) $\frac{1}{225}$ (D) $\frac{26}{27}$

COMPREHENSION-16

Paragraph for Questions Nos. 860 to 862

If f(x) = Mid {g(x), h(x), p(x)} means the function which will be second in order when values of the three function at a particular value of x are arranged, then for

$$f(x) = \text{Mid}\left\{x-1, (x-3)^2, 3-\frac{(x-3)^2}{2}\right\},\ x \in [1, 4]$$



860. Numerical value of difference between the LHD and RHD at the point x = 2 for f(x) in $x \in [1, 4]$ will be
(A) 0 (B) 2 (C) 3 (D) 1

861. The greatest value of f(x) in [1, 4] will be
(A) $1 + \sqrt{3}$ (B) $2 + \sqrt{3}$ (C) $3 + \sqrt{3}$ (D) N.O.T.

862. Rate of change of x w.r.t. f(x) at x = 3 will be
(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $-\frac{3}{2}$

COMPREHENSION-17

Paragraph for Questions Nos. 863 to 865

A function f(x) having the following properties;
(i) f(x) is continuous except at x = 3
(ii) f(x) is differentiable except at x = – 2 and x = 3
(iii) $f(0) = 0$, $\lim_{x\to3} f(x) \to -\infty$, $\lim_{x\to-\infty} f(x) = 3$, $\lim_{x\to\infty} f(x) = 0$
(iv) $f'(x) > 0\ \forall x \in (-\infty, -2) \cup (3, \infty)$ and $f'(x) \le 0\ \forall x \in (-2, 3)$
(v) $f''(x) > 0\ \forall x \in (-\infty, -2) \cup (-2, 0)$ and $f''(x) < 0\ \forall x \in (0, 3) \cup (3, \infty)$
then answer the following questions

863. Range of f(x) is
(A) $(-\infty, \infty)$ (B) $(-\infty, 3]$
(C) $(-\infty, 3)$ (D) $(-\infty, f(-2)]$

864. Graph of function y = f (– | x |) is
(A) differentiable for all x, if $f'(0) = 0$
(B) continuous but not differentiable at two points, if $f'(0) = 0$
(C) continuous but not differentiable at one points, if $f'(0) = 0$
(D) discontinuous at two points, if $f'(0) = 0$

865. f(x) + 3x = 0 has five solutions if
(A) $f(-2) > 6$ (B) $f'(0) < -3$ and $f(-2) > 6$
(C) $f'(0) > -3$ (D) $f'(0) > -3$ and $f(-2) > 6$

Comprehension-18

Paragraph for Questions Nos. 866 to 868

l, m, n are real numbers and x_0 be an arbitrary real number in [p, q] and f is a real valued function such that

$l^2 [f(a-x) - f(a+x)] + 4l [f(x) + f(-x)] + \{\underset{x \to x_0}{\text{Lt}} f(x) - f(x_0)\} = 0,$

$m^2 [f(a-x) - f(a+x)] + 4m [f(x) + f(-x)] + \{\underset{x \to x_0}{\text{Lt}} f(x) - f(x_0)\} = 0,$

& $n^2 [f(a-x) - f(a+x)] + 4n[f(x) + f(-x)] + \{\underset{x \to x_0}{\text{Lt}} f(x) - f(x_0)\} = 0,$

866. The function f is
(A) periodic with period 'a' (B) periodic with period '4a'
(C) periodic with a period 2a (D) non periodic

867. $x_1, x_2, \ldots\ldots x_n \in (p, q)$ and for some $\xi \in (p, q)$, $(f(\xi) \neq 0) =$
(A) n (B) n + 1 (C) n – 1 (D) 2n

868. If f(x) > 0 $\forall$ x [0, 2a] then $\dfrac{\int_0^{2a} f(x)\,dx}{\int_0^{a} f(x)\,dx} =$

(A) 1 (B) 2 (C) 3 (D) 4

Comprehension-19

Paragraph for Questions Nos. 869 to 872

Let $f(x) = x^3+ax^2+bx+c$ be a cubic polynomial where a, b, c $\in R$. Now
$f'(x) = 3x^2+2ax+b$ and let $D = 4a^2-12b$ be the discriminant of the equation $f'(x) = 0$. If d > 0, $f'(x) = 0$ has two real roots. $\alpha, \beta(\alpha < \beta)$, then $x = \alpha$ will be point of local maxima and $x = \beta$ will be a point of local minima of f(x), also

If $f(\alpha).f(\beta)>0$, then f(x) = 0 would have just one real root.

$f(\alpha).f(\beta)<0$, then f(x) = 0 would have three real and distinct roots.

$f(\alpha).f(\beta)=0$, then f(x) = 0 would have three real roots.

869. If the function $f(x) = x^3-9x^2+24x+k$ has three real and distinct roots x_1,x_2,x_3 where $x_1<x_2<x_3$. Then the possible value of k will be
(A) k < -20 (B) k > 20 (C) 16 < k < 20 (D) -20<k<-16

870. In the question No. 869, $[x_1]+[x_3]$ is equal to {where [x] is greatest integer function}
(A) 2 (B) 3 (C) 4 (D) 5

871. In the question No. 869, x_2 lies in the interval
(A) (-2, 0) (B) (0, 2) (C) (2, 4) (D) none of these

872. If $f(x) = ax^3+bx^2+cx+d$ has it non-zero local minimum and maximum values at x = 2 and x = 1 respectively. If a be the root of the equation $x^2-2x-15 = 0$, then a is equal to
(A) -3 (B) 5 (C) both (a) and (b) (D) none of these

COMPREHENSION-20

Paragraph for Questions Nos. 873 to 875

One of the most famous functions in calculus is the Dirichlet's function, viz.

D(x) = $\begin{cases} 1, & x \in Q \\ 0, & x \notin Q \end{cases}$. This function is one of the rare functions whose graph cannot be drawn. A number of functions were later defined by imitating Dirichlet's function.

Let f(x) = $\begin{cases} x^3 + 2x^2 & , \; x \in Q \\ -x^3 + 2x^2 + ax, & x \notin Q \end{cases}$

873. The value of a so that this function is differentiable at x = 0 is
(A) 1 (B) –1 (C) 0 (D) none of these

874. For the value of a obtained in above question. f(x) is
(A) one-one and onto (B) many-one and onto
(C) one-one and into (D) many one and into

875. $\lim\limits_{x \to 0} |f'(x)|$
(A) equals 1 (B) equals 2 (C) equals 3 (D) does not exist

COMPREHENSION-21

Paragraph for Questions Nos. 876 to 878

Let $f(x) = \begin{cases} e^{\{x^2\}} - 1 & , \; x > 0 \\ \dfrac{\sin x - \tan x + \cos x - 1}{2x^2 + \ell n(2 + x) + \tan x} & , \; x < 0, \\ 0 & , \; x = 0 \end{cases}$

where { } represents fractional part function. Lines L_1 and L_2 represent tangent and normal to curve y = f(x) at x = 0. Consider the family of circles touching both the lines L_1 and L_2

876. Ratio of radii of two circles belonging to this family cutting each other orthogonally is
(A) 2 + $\sqrt{3}$ (B) $\sqrt{3}$
(C) 2 + $\sqrt{2}$ (D) 2 – $\sqrt{2}$

877. A circle having radius unity is inscribed in the triangle formed by L_1 and L_2 and a tangent to it. Then the minimum area of the triangle possible is
(A) 3 + $\sqrt{2}$ (B) 3 – $\sqrt{2}$
(C) 3 + $2\sqrt{2}$ (D) 3 – $2\sqrt{2}$

878. If centers of circles belonging to family having equal radii 'r' are joined, the area of figure formed is
(A) $2r^2$ (B) $4r^2$
(C) $8r^2$ (D) r^2

COMPREHENSION-22

Paragraph for Questions Nos. 879 to 881

While finding the Sine of a certain angle *x*, an absent minded professor failed to notice that his calculator was not in the correct angular mode. Howevery he was lucky to get the right answer. The two least positive values of *x* for which the Sine of *x* degrees is the same as the Sine of x radians were found by him as $\frac{m\pi}{n-\pi}$ and $\frac{p\pi}{q+\pi}$ where *m, n, p* and *q* are positive integers. Suppose $\frac{mn}{pq}$ be denoted by the quantity 'L'. Now answer the following questions.

879. The value of $(m + n + p + q)$ is equal to

(A) 720 (B) 900

(C) 1080 (D) 1260

880. If x is measured in radians and $\lim\limits_{x\to\infty}\left(\sqrt{Ax^2+Bx}-Cx\right)$ = L, the value of $\frac{BC}{A}$ equals (A,B,C $\in$R)

(A) 4 (B) 2 (C) $\frac{1}{2}$ (D) none

881. Assume that f is differentiable for all x. The sign of f' is as follows:

$f'(x) > 0$ on $(-\infty, -4)$

$f'(x) < 0$ on $(-4, 6)$

$f'(x) > 0$ on $(6, \infty)$

Let $g(x) = f(10 - 2x)$. The value of $g'(L)$ is

(A) Positive

(B) negative

(C) zero

(D) the function g is not differentiable at x = 5

COMPREHENSION-23

Paragraph for Questions Nos. 882 to 884

Consider a family of curves, hwere the ordinate is proportional to the cube of the abscissa and let A be a fixed point in the plane which has coordinates (a, b).

882. If tangents be drawn through A to the members of family of curves then the locus of the point of contact is

(A) $xy + bx - 3ay = 0$ (B) $xy - 4bx + 3ay = 0$

(C) $2xy + bx - 3ay = 0$ (D) $2xy - 4bx + 3ayu + 2 = 0$

883. If the tangent through A to a curve cuts the curve again at a point B then the locus of B is

(A) $xy + bx - 3ay = 0$ (B) $xy - 4bx + 3ay = 0$

(C) $x^2 - 3y^2 = ax - 3by$ (D) $x^2 + 3y^2 = ax + 3by$

884. If the tangent through A to a curve cuts the curve again at a point B then the locus of B is

(A) $xy - 4bx + 3ay = 0$ (B) $2xy + bx - 3ay = 0$

(C) $x^2 - 3y^2 = ax - 3by$ (D) $a^2x^2 + b^2y^2 = 1$

COMPREHENSION-24

Paragraph for Questions Nos. 885 to 887

A chemical manufacturing company has 1000 k*l* holding tank which it uses to control the release of pollutants into a sewage system. Initially the tank has 360 kl of water containing 2 kg of pollutant per kl. Water containing 3 kg of pollutant per kl enters the tank at the rate 80 kl per hour and is uniformly mixed with the water already in the tank. Simultaneously, water is released from the tank at the rate of 40 kl per hours.

885. If P(t) denotes the amount of pollutant at any given time 't' inside the tank, then the rate at which pollutant is leaving the tank is

(A) $\frac{P(t)}{9-t}$ (B) $\frac{P(t)}{9+t}$ (C) $\frac{P(t)}{10+t}$ (D) $\frac{P(t)}{10-t}$

886. The differential equation giving pollutant at any instant 't' is given by

(A) $\frac{dP}{dt}+\frac{P}{9+t}=240$ (B) $\frac{dP}{dt}-\frac{P}{9+t}=240$

(C) $\frac{dP}{dt}+\frac{P}{10+t}=240$ (D) $\frac{dP}{dt}-\frac{P}{10+t}=240$

887. The amount of pollutant at any time 't' is given by

(A) $P(t) = 120(9-t)-\frac{3240}{9+t}$ (B) $P(t) = 120(9+t)+\frac{3240}{9+t}$

(C) $P(t) = 120(10-t)+\frac{3240}{10-t}$ (D) $P(t) = 120(9+t)-\frac{3240}{9+t}$

COMPREHENSION-25

Paragraph for Questions Nos. 888 to 890

Let the derivative of f(x) be defined as $D^*f(x) = \lim_{h\to 0}\frac{f^2(x+h)-f^2(x)}{h}$, where $f^2(x) = \{f(x)\}^2$.

888. If u = f(x), v = g(x), then the value of D* (u . v) is

(A) $(D^*u)v + (D^*v)u$ (B) $u^2D^*v + v^2D^*u$

(C) $D^*u + D^*v$ (D) $uvD^*(u+v)$

889. If u = f(x), v = g(x) then the value of $D^*\left\{\frac{u}{v}\right\}$ is

(A) $\frac{u^2D^*v - v^2D^*u}{v^4}$ (B) $\frac{uD^*v - vD^*u}{v^2}$ (C) $\frac{v^2D^*u - u^2D^*v}{v^4}$ (D) $\frac{vD^*u - uD^*v}{v^2}$

890. The value of D* c, where c is constant, is

(A) non-zero constant (B) 2

(C) does not exist (D) zero

COMPREHENSION-26

Paragraph for Questions Nos. 891 to 893

Consider the implicit equation $x^2 + 5xy + y^2 - 2x + y - 6 = 0$ ....(i)

891. The value of $\frac{dy}{dx}$ at (1, 1) is

(A) $\frac{5}{8}$ (B) $-\frac{5}{8}$

(C) $\frac{8}{5}$ (D) $-\frac{8}{5}$

892. The value of $\frac{d^2y}{dx^2}$ at (1, 1) is

(A) $\frac{111}{256}$ (B) $-\frac{111}{256}$

(C) $\frac{256}{111}$ (D) $-\frac{256}{111}$

893. The equation of normal to the conic (i) at (1, 1) is

(A) $5x - 8y - 3 = 0$ (B) $8y - 5x - 3 = 0$

(C) $8x - 5y - 3 = 0$ (D) $8x - 5y + 3 = 0$

COMPREHENSION-27

Paragraph for Questions Nos. 894 to 896

If $f : [0, 2] \to [0, 2]$ is a bijective function defined by $f(x) = ax^2 + bx + c$, where a, b, c are non zero real numbers, then

894. f(2) is equal to

(A) 2 (B) α where $a \in (0, 2)$

(C) 0 (D) cannot be determined

895. Which of the following is one of the roots f(x) = 0 ?

(A) $\frac{1}{a}$ (B) $\frac{1}{b}$

(C) $\frac{1}{c}$ (D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

896. Which of the following is not a value of 'a'

(A) $a = -\frac{1}{4}$ (B) $a = \frac{1}{2}$

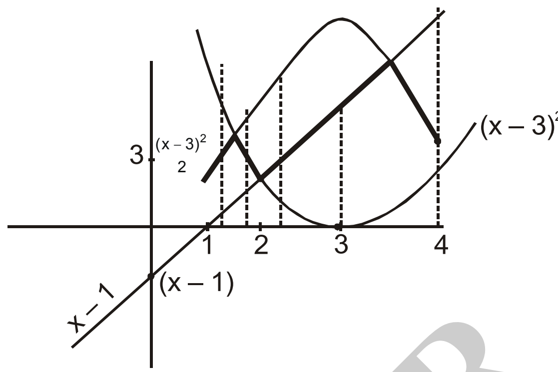
(C) $a = -\frac{1}{2}$ (D) $a = 1$

COMPREHENSION-28

Paragraph for Questions Nos. 897 to 899

If f(x) = Mid {g(x), h(x), p(x)} means the function which will be second in order when values of the three function at a particular x are arranged?

$$f(x) = \text{Mid}\left\{x-1, (x-3)^2, 3-\frac{(x-3)^2}{2}\right\}, x \in [1, 4]$$



897. Numerical value of difference between the LHD and RHD at the point x = 2 for f(x) in $x \in [1, 4]$ will be
(A) 0 (B) 2 (C) 3 (D) 1

898. The greatest value of f(x) in [1, 4] will be
(A) $1 + \sqrt{3}$ (B) $2 + \sqrt{3}$
(C) $3 + \sqrt{3}$ (D) N.O.T.

899. Rate of change of x w.r.t. f(x) at x = 3 will be
(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $-\frac{3}{2}$

COMPREHENSION-29

Paragraph for Questions Nos. 900 to 902

Let $f(x) = \begin{cases} 2x + a & : & x \geq -1 \\ bx^2 + 3 & : & x < -1 \end{cases}$

and $g(x) = \begin{cases} x + 4 & : & 0 \leq x \leq 4 \\ -3x - 2 & : & -2 < x < 0 \end{cases}$

functions

900. g(f(x)) is not defined if
(A) $a \in (10, \infty), b \in (5, \infty)$ (B) $a \in (4, 10), b \in (5, \infty)$
(C) $a \in (10, \infty), b \in (1, 5)$ (D) $a \in (4, 10), b \in (1, 5)$

901. If domain of g(f(x)) is [–1, 4], then
(A) a = 0, b > 5 (B) a = 2, b > 7
(C) a = 2, b > 10 (D) $a = 0, b \in R$

902. If a = 2 and b = 3 then range of g(f(x)) is
(A) (–2, 8] (B) (0, 8]
(C) [4, 8] (D) [–1, 8]

COMPREHENSION-30

Paragraph for Questions Nos. 903 to 905

Let $f : R \to R$ is a function satisfying $f(2 - x) = f(2 + x)$ and $f(20 - x) = f(x)$, $\forall x \in R$. For this function f answer the following.

903. If $f(0) = 5$, then minimum possible number of values of x satisfying $f(x) = 5$, for $x \in [0, 170]$, is
(A) 21 (B) 12 (C) 11 (D) 22

904. Graph of $y = f(x)$ is
(A) symmetrical about $x = 18$ (B) symmetrical about $x = 5$
(C) symmetrical about $x = 8$ (D) symmetrical about $x = 20$

905. If $f(2) \neq f(6)$, then
(A) fundamental period of f(x) is 1 (B) fundamental period of f(x) may be 1
(C) period of f(x) can't be 1 (D) fundamental period of f(x) is 8

COMPREHENSION-31

Paragraph for Questions Nos. 906 to 908

If $f : (0, \infty) \to (0, \infty)$ satisfy $f(xf(y)) = x^2y^a$ $(a \in R)$, then

906. Value of a is
(A) 4 (B) 2 (C) $\sqrt{2}$ (D) 1

907. $\sum_{r=1}^{n} f(r)\, {}^nC_r$ is
(A) $n.2^{n-1}$ (B) $n(n-1)\, 2^{n-2}$
(C) $n.2^{n-1} + n(n-1)\, 2^{n-2}$ (D) 0

908. Number of solutions of $2 f(x) = e^x$ is
(A) 1 (B) 2 (C) 3 (D) 4

COMPREHENSION-32

Paragraph for Questions Nos. 909 to 911

Consider two functions $f(x) = \lim_{n\to\infty} \left(\cos\frac{x}{\sqrt{n}}\right)^n$ and $g(x) = -\,x^{4b}$ where $b = \lim_{x\to\infty} \left(\sqrt{x^2+x+1} - \sqrt{x^2+1}\right)$.

Then

909. f(x)h is
(A) e^{-x^2} (B) $e^{\frac{-x^2}{2}}$ (C) e^{x^2} (D) $e^{\frac{x^2}{2}}$

910. g(x) is
(A) $-x^2$ (B) x^2 (C) x^4 (D) $-x^4$

911. Number of solutions of $f(x) + g(x) = 0$ is
(A) 2 (B) 4 (C) 0 (D) 1

COMPREHENSION-33

Paragraph for Questions Nos. 912 to 914

Let $f(x) = \lim_{n \to \infty} \left(\cos\sqrt{\frac{x}{n}}\right)^n$, $g(x) = \lim_{n\to\infty} \left(1 - x + x\sqrt[n]{e}\right)^n$. Now, consider the function $y = h(x)$, where $h(x) = \tan^{-1}(g^{-1} f^{-1}(x))$.

912. $\lim_{x\to 0} \frac{\ell n\,(f(x))}{\ell n\,(g(x))}$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) 1

913. Domain of the function y = h(x) is

(A) $(0, \infty)$ (B) R
(C) (0, 1) (D) [0, 1]

914. Range of the function y = h(x) is

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, 0\right)$ (C) R (D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

COMPREHENSION-34

Paragraph for Questions Nos. 915 to 917

If f(x) approaches to zero as x approaches to 'a' then

$\lim_{x\to a} \frac{\sin(f(x))}{f(x)} = 1$, $\lim_{x\to a} \frac{\tan(f(x))}{f(x)} = 1$, $\lim_{x\to a} \frac{\ell n(1+f(x))}{f(x)} = 1$

$\lim_{x\to a} \frac{K^{f(x)}-1}{f(x)} = \ell n(K)$, K > 0 (K is independent of x)

915. $\lim_{x\to 0} \frac{\tan(\sin x)}{\sin x}$ is

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) – 1

916. $\lim_{x\to\infty} x \sin\left(\frac{2}{x}\right)$ is

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) 0

917 $\lim_{x\to 0} \frac{\sin([x^2])}{x^2}$, where [.] denote the greatest integer function

(A) is 1 (B) is 0
(C) does not exist (D) none of these

SECTION - 4 (MATRIX MATCH Type)

918. Match the following:

List – I	List – II
(A) $\int_0^{\pi/2} \frac{dx}{1+\tan x}$	(i) $\frac{1}{117}$
(B) If $\int_0^{x^2(1+x^5+7x^{12})} f(t)dt$ = x, then f (3) is equal to	(ii) $\frac{\pi}{2}-\log 2$
(C) $\int_0^{\infty} e^{-2x}(\sin 2x+\cos 2x)dx$ is equal to	(iii) $\frac{1}{2\sqrt{2}}$
(D) $\int_0^1 \cot^{-1}(1+x^2-x)dx$ is equal to	(iv) $\frac{\pi}{4}$

919. Match the following:

List – I	List – II
(A) $\lim\limits_{x\to-\infty} \frac{x^4 \sin\left(\frac{1}{x}\right)+x^2}{1+\lvert x\rvert^3}$ is equal to	(i) – 24
(B) $\lim\limits_{x\to\infty}\left(\frac{x+8}{x+3}\right)^{x+6}$ is equal to	(ii) $\frac{1}{e}$
(C) $\lim\limits_{x\to\pi/3} \frac{\tan^3 x-3\tan x}{\cos\left(x+\frac{\pi}{6}\right)}$ is equal to	(iii) – 1
(D) $\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}$ is equal to	(iv) e^5

920. Match the following:

List – I	List – II
(A) $\int_{\pi/6}^{\pi/3} \frac{1}{1+\sqrt{\tan x}}dx$ =	(i) $\frac{3\pi}{2}$
(B) The value of α which satisfy $\int_{\pi/2}^{\alpha} \sin x dx = \sin 2\alpha$ is	(ii) $\frac{\pi}{2}$
(C) I = $\int_0^{\pi/2} \frac{f(x)}{f(x)+f\left(\frac{\pi}{2}-x\right)}dx$, then value of 2I is	(iii) $-\frac{\pi}{2}$
(D) $\int_{-1}^{1}\left\{\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right)\right\}dx$ =	(iv) $\frac{\pi}{12}$

921. Match the following:

List – I	List – II
(A) If $x \cdot e^{xy} = y + \sin^2 x$, then $\left[\frac{dy}{dx}\right]_{x=0}$ is equal to	(i) 4
(B) Derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ with respect to $\sqrt{1-x^2}$ at x = $\frac{1}{2}$ is	(ii) 0
(C) Let $f(x) = \begin{cases} -x-2, & for\ -3 \le x \le 0 \\ x-2, & for\ 0 < x \le 3 \end{cases}$, g(x) = |f(x)| + f(|x|), then the number of points of non-differentiability of g(x) is	(iii) 1
(D) Let F (x) = f (x) g (x) h (x) $\forall$ real x, where f, g and h are differentiable functions. At some point x_0, F′ (0) = 21 F (x_0), f′ (x_0) = 4f (x_0) g′ (x_0) = –7g (x_0) and h′ (x_0) = kh (x_0), then $\frac{k}{12}$ is equal to	(iv) 2

922. Match the following

List – I	List – II
(A) Domain of ${}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$	(i) $\left\{-1, 2, \frac{1}{2}\right\}$
(B) Range of f(x) = $\lim\limits_{n\to\infty} \frac{1+x^{2n}}{2x^{2n}-1}$	(ii) {–1, 1}
(C) Domain of $\sin^{-1}\left(\frac{1+x^2}{2x}\right)$	(iii) ϕ
(D) Domain of f(x) = $\frac{1}{\sqrt{x-\|x\|}}$	(iv) {2, 3}

923. If $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$, then match the following :

List – I	List – II
(A) $\int_0^\infty \frac{\sin 5x}{x} dx$	(i) 0
(B) $\int_0^\infty \frac{\sin ax \cos bx}{x} dx$ (a> b > 0)	(ii) $\frac{\pi}{2}$
(C) $\int_0^\infty \frac{\sin^2 x}{x^2} dx$	(iii) $\frac{\pi}{4}$
(D) $\int_0^\infty \frac{\sin^3 x}{x} dx$	(iv) π

924. Match the following :

List – I	List – II
(A) x^{100} + sinx – 1 is decreasing in	(i) $(-\infty, \infty)$
(B) Domain of $\log_4\log_5\log_3(18x - x^2 - 80)$	(ii) $\left[-4, \frac{3-\sqrt{21}}{2}\right] \cup (1, \infty)$
(C) Range of $x^3 + 3x^2 + 10x + 2\sin x$	(iii) (8, 10)
(D) $\left(\frac{\sqrt{a+4}}{1-a} - 1\right)x^5 - 3x + \log 5$ decreases for all R the set of values of a	(iv) none of these
	(v) $\left(\frac{\pi}{2}, \pi\right)$

925. Match the following:

List – I	List – II
(A) Let f : R → R be a periodic function such that $f(T + x) = 1 + [1 - 3f(x) + 3(f(x))^2 - (f(x)^3]^{1/3}$, where T is fixed positive number, then period of f(x) is AT, where A =	(i) 3
(B) The area between the curve $y = 2x^4 - x^2$, the x-axis and the ordinates of two minima of the curve is $\frac{B}{120}$ where B is	(ii) 2
(C) $\int_0^4 \frac{f(x)}{f(x)+f(4-x)}dx$ =	(iii) 7
(D) f(x) = $\begin{cases} \frac{1-\sin x}{(\pi-2x)^2}\frac{\log\sin x}{\log(1+\pi^2-4\pi x+4x^2)} \\ k, \quad x = \frac{\pi}{2} \end{cases}$ is continuous at x = $\frac{\pi}{2}$ then $-\frac{1}{k}$ equals to	(iv) 64

926. Match the list:

List – I	List – II
(A) $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\sqrt{\frac{n+r}{n-r}}$	(i) $\frac{\pi}{2}$
(B) $\lim_{x\to\infty}\left[\frac{1}{\sqrt{n^2-1}}+\frac{1}{n^2-2^2}+\cdots+\frac{1}{\sqrt{n^2-(n-1)^2}}\right]$	(ii) $\frac{\pi}{2}+1$
(C) $\lim_{x\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$	(iii) 2π
(D) $\int_0^{2\pi} e^{\cos x}\cos(\sin x)dx$	(iv) $\frac{1}{e}$
	(v) e

927. Match the following:

List I (Expression)		List II (Value)	
I.	If f(x + y) = f(x) f(y) (x, y are independent $\forall$ x, y $\in$ R and f(2) = f′(2) = 3 then f′(4) =	(A)	**0**
II.	If f(xy) = f(x) f(y) and f′(4) = 2f′(8), then f(2) =	**(B)**	**10**
III.	If f(x) is a diff. function such that f(xy) = f(x) + f(y) $\forall$ x, y $\in$ R then f(e) + f$\left(\frac{1}{e}\right)$ =	**(C)**	**9**
IV.	If f is a twice diff. function Such the f′(x) = –f(x) If h(x) = $(f(x)^2 + (g(x))^2$ And h(5) = 10. Then h(10) =	**(D)**	**1**

928. Match the following:

(A) $\frac{\sin 1}{\sin 2} - \frac{\sin 5}{\sin 6}$ (1) positive

(B) $\tan\frac{3}{2} - \frac{9}{4}$ (2) negative

(C) $\lim\limits_{x\to 0}\left[\frac{e^x - 1}{x}\right]$ (where [.] denotes the greatest integer function) (3) 1

(D) If f′(α) = 0 and f′(x) > 0 $\forall$ x $\in$ R – {α}, then f″(α) is (4) does not exist

(5) 0

929. If y = $\cos^{-1}\left(\frac{a\cos x + b}{a + b\cos x}\right) - 2\tan^{-1}\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right)$, then match the following :

Column– I		**Column – II**	
(A)	If a = 4, b = 3, then $\frac{dy}{dx}$ at x = 0	(P)	0
(B)	Number of points of local minima	(Q)	$\frac{4}{5}$
(C)	For a = 4, b = 3, value of y at x = 0	(R)	$\frac{3}{5}$
(D)	Number of tangents parallel to the y axis	(S)	2

930. Let f(x) = ax^2 + bx + c, Given that $f'(1) = 8$, f(2) + $f''(2) = 33$ and 2a + 3b + 6c = 14, then match the following

Column– I		**Column – II**	
(A)	Global maximum value of f(x)	(P)	Not defined
(B)	If global minimum value of f(x) = k then 28k is equal to	(Q)	48
(C)	Number of real roots of f(x) = 0	(R)	0

(D)	Number of real roots of f(x) = 3	(S)	2

931.

	Column I		Column II
(A)	The number of non-differentiability points on the curve $y = \|e^{\|x\|} - 3\|$ is/are	(p)	1
(B)	Length of the latus-rectum of the parabola defined by x = cos t – sint and y = sin 2t	(q)	0
(C)	The number of real solution of the equation $x^{2\log_x(x+3)} = 16$ is	(r)	3
(D)	If in a triangle $2R + r = r_1$, then $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right)$ is equal to	(s)	2

932.

(A)	Area of the rectangle formed by asymptotes of the hyperbola xy – 3y – 2x = 0 and co-ordinate axes is	(p)	32
(B)	Area bounded by min (|x|, |y|) = 1 and max (|x|, |y|) = 3 is,	(q)	2
(C)	The number of common tangents of the two circles $x^2 + y^2 - 10x - 6y + 9 = 0$ and $x^2 + y^2 - 4x + 2y - 11 = 0$ are	(r)	5
(D)	The greatest value of $f(x) = 2\cos(2xe^x + 7x^4 - \log(1 + x^2))$	(s)	6

933. **List I with List II and select the correct answers using the codes given below the lists :**

List I LIMIT	List II VALUE
1. $\lim\limits_{x\to 0}\frac{(x)}{\tan(x)}$	(A) $-\log_{16}e$
2. $\lim\limits_{x\to 0}\frac{\sqrt{1-x+x^2}-\sqrt{1+x^2}}{4^x-1}$	(B) e^{-1}
3. $\lim\limits_{x\to 0}\frac{2e^{\sin x}-(1+\sin x)^2}{2\left(\tan^{-1}(\sin x)\right)^2}$	(C) 1
4. $\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}$	(D) 0

934.

List I Limits	List II Value
I. $\lim\limits_{x\to 2}\frac{\sqrt{1+\sqrt{2+x}}-\sqrt{3}}{x-2}$	(A) $-\frac{4}{3}$
II. $\lim\limits_{x\to 2}\frac{3^x+3^{3-x}-12}{3^{3-x}-3^{x/2}}$	(B) $\frac{1}{2}$
III. $\lim\limits_{x\to 2}\left(\frac{x^3-4x}{x^3-8}\right)^{-1}-\left(\frac{x+\sqrt{2x}}{x-2}-\frac{\sqrt{2}}{\sqrt{x}-\sqrt{2}}\right)^{-1}$	(C) 8
IV. $\lim\limits_{x\to 2}\frac{2^x+2^{3/2}-6}{\sqrt{2^{-x}}+2^{1-x}}$	(D) $\frac{1}{4\left(3+\sqrt{3}\right)}$

935. **List I** **List II**

Function f(x) **The No.of points of discontinuity**

I. f(x) = $\dfrac{1+\cos 5x}{1-\cos 4x}$ in $0 \le x \le \pi$ (A) 0

II. f(x) = [cos x + sin x] $0 < x < 2\pi$ (B) Infinite

III. f(x) = 4x + 7[x] + 2log (1 + x) (C) 5

IV. f(x) = $\int_0^x t\sin\frac{1}{t}dt$ where $0 < x < \pi$ (D) 3

936. **List I** **List II**

(Function) **(Limit at x = 0)**

I. f(x) = $\left[\dfrac{x}{\sin(x)}\right]$ (A) $\underset{x\to 0}{Lt} f(x) = 0$

II. f(x) = $\dfrac{a^{[x]+x}-1}{[x]+x}$ (B) $\underset{x\to 0}{Lt} f(x) = e^{\sin^2 y}$

III. f(x) = $\dfrac{\sin[\cos x]}{1+[\cos x]}$ (C) $\underset{x\to 0-}{Lt} f(x) = 1-\dfrac{1}{a}$

IV. f(x) = $\dfrac{1}{x}\left(\int_y^a e^{\sin^2 t}dt - \int_{x+y}^a e^{\sin^2 t}dt\right)$ (D) $\underset{x\to 0}{Lt} f(x) = 1$

Where [.] denotes G.I.F..

937. **List I** **List II**

(Function) **(Derivative)**

I. f(x) = $\begin{cases} 1, & x \le 0 \\ 1+\sin x, & 0 \le x < \dfrac{\pi}{2} \end{cases}$ (A) L→' (0) = –1, Rf'(0) = 0

II. = x($\sqrt{x} - \sqrt{x+1}$) (B) f′(0) = 0

III. = x^3 sgn x (C) f′(0) = 1

IV. g(x) = $\begin{cases} -1, & -2 \le x \le 0 \\ x-1, & 0 < x \le 2 \end{cases}$

f(x) = g(|x|) + |g(x)| (D) f′(0) = –1 does not exist

938. Match the columns -

Column – I		Column - II	
(a)	The number of values of c for which $\int_0^1 \lvert(c-x)\rvert\, dx = \frac{1}{2}$ is	(P)	2
(b)	If $\int \frac{\operatorname{cosec}\left(2x-\frac{5\pi}{6}\right)}{\sin\left(2x-\frac{\pi}{6}\right)} dx = \frac{k}{\sqrt{3}} \ell n \left(\frac{\sin\left(2x-\frac{5\pi}{6}\right)}{\sin\left(2x-\frac{\pi}{6}\right)}\right)$ + c, then k =	(Q)	1
(c)	Area of the region bounded by the $x^2 + y^2 - 2x \le 0$, $x + y \le 1$, $y \ge 0$ is	(R)	$\frac{\pi}{8}$
(d)	$\frac{1}{\pi}\int_{-\pi}^{\pi} \frac{x\sin x}{e^x+1}$ dx is equal to	(S)	8

939. Match the following

Column – I		Column – II	
(a)	The number of solutions of the equation $x.2^x = x+1$ is	(P)	4
(b)	$\lim_{n\to\infty}\left(\frac{1+\sqrt[n]{4}}{2}\right)^n$ is equal to	(Q)	8
(c)	The number of points at which $g(x) = \frac{1}{1+\frac{2}{f(x)}}$ is not differentiable where $f(x) = \frac{1}{1+\frac{1}{x}}$, is :	(R)	2
(d)	$f\left(x+\frac{1}{2}\right)+f\left(x-\frac{1}{2}\right)=f(x)$ for all $x \in R$, then period of $f(x)$ is	(S)	3

940. Match the following

Column – I		Column – II	
(a)	The least positive integral solution of $x^2-4x > \cot^{-1} x$	(P)	1
(b)	The least positive integral value of x for which $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ is increasing	(Q)	2
(c)	Let $f: R \to R$ be such that $f(a) = 1, f'(a) = 2,$ then $\lim_{x\to 0}\left(\frac{f^2(a+x)}{f(a)}\right)^{\frac{1}{x}} = e^k,$ then $k =$	(R)	5
(d)	Number of integral values of x which satisfy equation $\sin^{-1}((3x-x)(x-1)) + \sin^{-1}(2-\lvert x\rvert) = \frac{\pi}{2}$ is/are	(S)	4

941. Consider f(x) = $t^{|x^2-4x+3|}$, where t is a real number greater than 1. Then

Column I		Column II
(A) f(x) increases in the interval	(P)	$(-\infty, 1)$
(B) f(x) decreases in the interval	(Q)	(0, 2)
(C) Local maxima of f(x) occurs in the interval	(R)	$(1, 2) \cup (3, \infty)$
(D) f(x) has a local minima in the interval	(S)	(1,3)

942. **Match the column**

	Column I		Column II
(A)	Let a, b, c be positive and a + b + c = abc the maximum value of square of least among a, b, c is	(P)	1
(B)	The fundamental period of the function $y = \sin^2\left(\dfrac{\sqrt{2}t+3}{6\pi}\right)$ is $\lambda\pi^2$ then the value of $\dfrac{\lambda}{\sqrt{2}}$ is	(Q)	2
(C)	If $(x + y)^m$ has three consecutive cofficients in A.P. ($m \in N$) for which the sum of first 'n' values of m is $an^3 + bn^2 + cn + d$. The value of greatest integer of $\left(\dfrac{a+b+c+d}{2}\right)$ is	(R)	3
(D)	If equation of tangent to the curve $y = \displaystyle\int_{x^2}^{x^3}\frac{dt}{\sqrt{1+t^2}}$ at x = 1 is $\sqrt{2}$ x = by + $\sqrt{2}$ then value of $\dfrac{b}{2}$ is	(S)	Number of solution of $\cos x + \cos\sqrt{2}\, x = 2$
		(T)	Number of values of x for which $f(x) = \dfrac{1}{\ell n\,|x|}$ is not defined

943. **Match the column**

	Column I		Column II
(A)	The area of the quadrilateral formed by the tangents from the point (4, 0) to the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ and the pair of radii through the points of contanct of the tangent is	(P)	1
(B)	The number of points at which the function $f(x) = \dfrac{1}{\ell n|x|}$ is discontinuous is	(Q)	3
(C)	Let f(x + y) = f(x). f(y) for all x, $y \in R$ if f (5) = 2 and $f'(0)$ = 3, then $f'(5)$ =	(R)	6
(D)	If the normal to the curve y = f (x) at the point (3, 4) makes an angle $\dfrac{3\pi}{4}$ with the positive x-axis, then $f'(3)$ =	(S)	4

944. **Match the following**

Column-I		**Column-II**
(a)	If the point (6, k) is closest to the curve $x^2 = 2y$ at (2, 2), then k =	(P) 0
(b)	If the curve $y = px^2 + qx + r$ passes through the point (1, 2) and touches the line y = x at the origin, then the value of p – q + r =	(Q) –2
(c)	Let $f(x) = kx^3 + 9kx^2 + 9x + 3$ be a strictly increasing function and has non stationary point. The greatest value of k is	(R) 1
(d)	Let $0 < a < b < \frac{\pi}{2}$. If $f(x) = \begin{bmatrix} \sin x & \sin a & \sin b \\ \cos x & \cos a & \cos b \\ \tan x & \tan a & \tan b \end{bmatrix}$, then minimum possiable number of roots of $f'(x) = 0$ lying in (a, b) is	(S) does not exist

945.

Column I		**Column II**
(A)	if $f(x) = \int_0^{g(x)} \frac{dt}{\sqrt{1+t^3}}$ where $g(x) = \int_0^{\cos x} (1 + \sin t^2)\,dt$ then the value of $f'(\pi/2)$	(P) 3
(B)	If $f(x)$ is a non zero differentiable function such that $\int_0^x f(t)dt = (f(x))^2$ for all x, then $f(2)$ equals	(Q) 2
(C)	If $\int_a^b (2 + x - x^2)\,dx$ is maximum then (a + b) is eual to	(R) 1
(D)	If $\lim\limits_{x\to 0} \left(\frac{\sin 2x}{x^3} + a + \frac{b}{x^2}\right) = 0$ then (3a + b) has the value equals to	(S) –1

946.

Column I		**Column II**
(A)	Number of integers which do not lie inthe range of the function $f(x) = \sec\left(2\sin^{-1}\frac{1}{x}\right)$	(P) 0
(B)	Let f : (0, ∞) onto (0, ∞) be a derivable function for which there exists its primitive F such that $2(F(x) - f(x)) = f^2(x)$ for any real positive x. Then $\lim\limits_{x\to\infty} \frac{f(x)}{x}$ equals	(Q) 1
(C)	How many of the following derivatives are correct (on their domains)? **I.** $\frac{d}{dx} \ln \lvert \sec x \rvert = \tan x$ **II.** $\frac{d}{dx} \ln(x + e^x) = 1 + \frac{1}{x}$ **III.** $\frac{d}{dx} x^{\ln x} = (\ln x)\, x^{\ln(x)-1}$	(R) 2
(D)	A differentiable function satisfies $f'(x) = f(x)\, 2e^x$ with initial conditions f (0) = 0. The area enclosed f (x) and the x-axis is	(S) 3

947. **Column I** **Column II**

(A) Suppose, $f(n) = \log_2(3) . \log_3(4).\log_4(5)......\log_{n-1}(n)$ then the sum $\sum_{k=2}^{100} f(2^k)$ equals — (P) 5010

(B) Let f(x) = $\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)(x+4)}}}$ then $\int_0^{100} f(x)\,dx$ is — (Q) 5050

(C) In an A.P. the series containing 99 terms, the sum of all the odd numbered terms is 2550. The sum of all the 99 terms of the A.P. is — (R) 5100

(D) $\lim_{x\to 0} \frac{\prod_{r=1}^{100}(1+rx)-1}{x}$ equals — (S) 5049

948. **Column - I** **Column - II**

(A) Let f be continuous and the function F is defined as — (P) 0

$F(x) = \int_0^x \left(t^2 . \int_1^t f(u)du \right) dt$ where f(1) = 3, then F′(1) + F″(1) has the value equal to — (Q) 1

(B) For each value of x a function f(x) is defined as — (R) 2

min {2x + 3, $\frac{(x+4)}{3}$, 3(6 – x)} Maximum value of f(x) is .

(C) $\lim_{x\to 1^+} (\ell n\, x)^{\frac{1}{(x-1)\tan x}}$ — (S) 3

(D) Exponent of 2 in the binomial coefficient ${}^{500}C_{212}$ is

949. **Column I** **Column II**

(A)	If three normals can be drawn to the curve $y^2 = x$ from the point (c, 0), then c can be equal to	(P) (Q)	1 0
(B)	Subnormal length to $xy = c^2$ at any point varies directly as	(R)	$\frac{5}{4}$
(C)	If the sides and angles of a plane triangle vary in such a way that its circum radius remains constant, then	(S) (T)	Cube of ordinate 2

$$\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} =$$

where da, db, dc are small increments in the sides a, b, c, respectively

950. **Column-I** **Column-II**

	Column-I		Column-II
(A)	$\int_{-\pi/2}^{\pi/2} \frac{\ell n(\cos x)}{1+e^x . e^{\sin x}} dx =$	(P)	$-2\pi\ \ell n\ 2$
(B)	$\int_0^{2\pi} \ell n\,(1+\sin x)\,dx =$	(Q)	$-\frac{\pi}{4}\ell n 2$
(C)	$\int_{-\pi/4}^{\pi/4} \ell n\sqrt{1+\sin 2x}\,dx =$	(R)	$-\pi\ \ell n 2$
(D)	$\int_{\infty}^{0} \frac{xe^{-x}}{\sqrt{1-e^{-2x}}} dx.$	(S)	$-\frac{\pi}{2}\ell n 2$

951. **Column I** **Column II**

	Column I		Column II
(A)	The function $f(x)-(x-[x])^2$, (where [x] is greatest integer function $\le x$) is	(P)	periodic
		(Q)	non - periodic
(B)	The function $f(x)=\log_a\left(x+\sqrt{(x^2+1)}\right)$; $a>0$, $a\neq 1$ is (assume it to be an onto)	(R)	one - one
		(S)	many one
(C)	The function f(x) = cos(5x + 2) is	(T)	invertible

952. **Column I** **Column II**

	Column I		Column II
(A)	The area bounded by the curve max $\{\lvert x\rvert, \lvert y\rvert\}=1$ is	(P)	0
(B)	If the point (a, a) lies between the lines |x + y| = 6, then [|a|] is (where [.] denotes the greatest integer function)	(Q)	1
		(R)	2
(C)	Number of integral values of b for which the origin and the point (1, 1) lie on the same side of the st. line $a^2x+aby+1=0$ for all $a\in R\sim\{0\}$	(S)	3
		(T)	4

953. **Column I** **Column II**

	Column I		Column II
(A)	The slope of the curve $2y^2=ax^2+b$ at $(1,-1)$ is -1, then	(P)	a - b = 2
(B)	If (a, b) be the point on the curve $9y^2=x^3$ where normal to the curve makes equal intercepts with the axes, then	(Q)	$a-b=\frac{7}{2}$
		(R)	$a-b=\frac{4}{3}$
(C)	If the tangent at any point (1, 2) on the curve $y=ax^2+bx+\frac{7}{2}$ be parallel to the normal at (-2, 2) on the curve $y = x^2 + 6x + 10$, then	(S)	$\alpha+\lvert b\rvert=\frac{20}{3}$
		(T)	5a + 2b = 0

954. Match the column

Column – I		Column – II	
(A)	The number of possible values of k if fundamental period of $\sin^{-1}(\sin kx)$ is $\frac{\pi}{2}$	(p)	1
(B)	Numbers of points in the domain of $f(x) = \tan^{-1}x + \sin^{-1}x + \sec^{-1}x$	(q)	2
(C)	$f(x) = \sin\left(\frac{\pi x}{2}\right) . \operatorname{cosec}\left(\frac{\pi x}{2}\right)$ is periodic with period	(r)	3
(D)	If range of the function $f(x) = \cos^{-1}[5x]$ where [.] denotes greatest integer, is {a, b, c}, then a + b + c is	(s)	4

955. Match the column

[.] and {.} represent the greatest integer and fractional part functions respectively.

Column – I		Column – II	
(A)	Number of solutions of $[x] = \cos^{-1}x$	(p)	3
(B)	Number of solutions of $\sin^{-1}x = \operatorname{sgn}(x)$	(q)	2
(C)	Number of solutions of $\{x\} = e^{x^2}$	(r)	1
(D)	Number of solutions of $\frac{\sin^{-1}x + \cos^{-1}x}{2} = \{x\}$	(s)	0

956. Match the column

Column – I		Column – II	
(A)	Smallest positive integral value of x for which $x^2 - x + \sin^{-1}(\sin 2) < 0$ is	(p)	$\frac{3\pi}{2}$
(B)	Number of solution of $2[x] = x + 2\{x\}$ is where [x], {x} are greatest integer and least integer functions respectively.	(q)	3
(C)	If $x^2 + y^2 = 1$, then maximum value of x + y is	(r)	1
(D)	$f\left(x+\frac{1}{2}\right) + f\left(x-\frac{1}{2}\right) = f(x)$ for all $x \in R$, then period of f(x) is	(s)	2

957. Match the column

Column – I		Column – II	
(A)	If function f(x) is defined in [–2, 2], then domain of f(|x| + 1) is	(p)	$\left[\frac{1}{4}, \frac{3}{4}\right]$
(B)	If range of the function f(x) = $\frac{\sin^{-1}x + \cos^{-1}x + \tan^{-1}x}{\pi}$ is	(q)	[–1, 1]
(C)	Range of the function f(x) = 3 |sin x| – 4 |cos x| is	(r)	[– 4, 3]
(D)	Range of f(x) = $(\sin^{-1}x)\sin x$ is	(s)	$\left[0, \frac{\pi}{2}\sin 1\right]$

958.

Column - I		Column - II	
(A)	Range of sgn {x} is (where {.} represents fractional part function)	(p)	{1}
(B)	Domain of $\sin^{-1}x + \sin^{-1}(1 - x)$ is	(q)	[0, 1)
(C)	Range of $\sqrt{\frac{2\tan^{-1}x}{\pi}}$ is	(r)	{0, 1}
(D)	Range of $\frac{2}{\pi}\sin^{-1}[x^2 + x + 1]$ is (where [.] represent greatest integer function)	(s)	[0, 1]

959.

Column – I		Column – II	
(A)	Domain of f(x) = $\sin^{-1}\left(\frac{2-x}{2x}\right)$ is	(p)	[–2, ∞)
(B)	Range of f(x) = $\frac{2x^2 - 2}{3x^2 + 1}$ is	(q)	(– ∞, –1] ∪ [1, ∞)
(C)	Set of all values of p for which the function f(x) = px + sin x is bijective is	(r)	(– ∞, –2] ∪ [2/3, ∞)
(D)	If f : (– ∞, 1] → A is defined by f(x) = $x^2 - 3x$, then set A for which f(x) becomes invertiable, is	(s)	[–2, 2/3)

960. Match the column

Column – I		**Column – II**	
(A)	$\lim\limits_{x\to\frac{\pi}{2}^+} \dfrac{\cos(\tan^{-1}(\tan x))}{x-\frac{\pi}{2}}$	(p)	$\dfrac{3\pi}{2}$
(B)	The number of solutions of the equation $2\cos x = \lvert \sin x \rvert$ $0 \le x \le 4\pi$ is	(q)	$-\dfrac{1}{\sqrt{2}}$
(C)	If $y = \tan^{-1}\sqrt{\dfrac{1-\sin x}{1+\sin x}}$, then the value of $\dfrac{dy}{dx}$ at $x = \dfrac{\pi}{6}$ is	(r)	1
(D)	If $f(x) = e^{\{2x\}} + \sin 2\pi x$, the period of f(x) is { } represents fractional part function	(s)	Does not exist

961. Match the column

Column – I		**Column – II**	
(A)	Let $f : R \to R$ be a differentiable function and $f(1) = 1$, $f'(1) = 3$. Then the value of $\lim\limits_{x\to 1} \displaystyle\int_1^{x^2} \frac{(f(t)-t)}{(x-1)^2}\,dt$ is	(p)	0
(B)	$\lim\limits_{n\to\infty}\left(\dfrac{1+\sqrt[n]{4}}{2}\right)^n$ is equal to	(q)	– 1
(C)	If $f(x) = \lim\limits_{n\to\infty} \dfrac{2x}{\pi}.\tan^{-1}(nx)$, $x > 0$ then $\lim\limits_{x\to 0^+} [f(x) - 1]$ is, {where [ ] represents greatest integer function}	(r)	2
(D)	$\lim\limits_{n\to\infty}\left[\displaystyle\sum_{r=1}^{n}\frac{1}{2^r}\right]$, where [ ] denotes the greatest integer function	(s)	4
		(t)	1

962. **Column - I** **Column - II**

(A) Number of points of discontinuity of $f(x) = \tan^2x - \sec^2x$ in $(0, 2\pi)$ is (p) 1

(B) Number of points at which $f(x) = \sin^{-1}x + \tan^{-1}x + \cot^{-1}x$ is non-differentiable in $(-1, 1)$ is (q) 2

(C) Number of points of discontinuity of $y = [\sin x]$, $x \in [0, 2\pi)$ where [.] represents greatest integer function (r) 0

(D) Number of points where $y = |(x-1)^3| + |(x-2)^5| + |x-3|$ is non-differentiable (s) 3

963. **Column – I** **Column – II**

For $x \in R$,

(A) $f(x) = \{\sin(\pi x)\}$ is discontinuous for $x \in$ (p) $[0, 1)$

(B) $g(x) = \left\{\dfrac{\sin x}{x}\right\}$ is discontinuous for $x \in$ (q) $\{1, 2\}$

(C) $h(x) = \dfrac{\{\sin x\}}{\{x\}}$ is non-differentiable for $x \in$ (r) $\{0\}$

(D) $u(x) = \dfrac{(\sin x)}{[x]}$ is discontinuous function for $x \in$ (s) $\left\{\dfrac{1}{2}\right\}$

964. **Column – I** **Column – II**

(A) Point of discontinuity of $y = \dfrac{1}{t^2 - t - 2}$ where $t = \dfrac{1}{x+1}$ (p) $-\dfrac{1}{2}$

(B) Points of continuity of $y = [x] + [-x]$ (q) -2

(C) $y = [\sin(\pi x)]$ is non differentiable at (r) -1

(D) $f(x) = |2x + 1| + |x + 2| - |x + 1| - |x - 4|$ is non differentiable at (s) 4

965. Match the column

Column – I		**Column – II**	
(A)	$\lim\limits_{x \to \frac{1}{\sqrt{2}}} \dfrac{x - \cos(\sin^{-1} x)}{1 - \tan(\sin^{-1} x)}$ is equal to	(p)	does not exist
(B)	If f(x) = $\log_{x^2}(\log x)$, then $f'\left(\dfrac{1}{2}\right)$ is equal to	(q)	0
(C)	For the function f(x) = $\ell n \tan\left(\dfrac{\pi}{4} + \dfrac{x}{2}\right)$ if $\dfrac{dy}{dx}$ = sec x + p, then p is equal to	(r)	28
(D)	$\lim\limits_{x \to 0} \dfrac{1}{x} \sqrt{\dfrac{1 - \cos 2x}{1 + \cos 2x}}$ is equal to	(s)	4

966. Match the column

Column – I		**Column – II**	
(A)	If y = $\cos^{-1}(\cos x)$, then y′ at x = 5 is equal to	(p)	–1
(B)	The value of $\dfrac{1}{2^{11}} \sum\limits_{0 \le i \le j \le 8} \sum i \,.\, {}^{8}C_j$ is	(q)	$-\dfrac{1}{2}$
(C)	The derivative of $\tan^{-1}\left(\dfrac{1+x}{1-x}\right)$ at x = 1 is	(r)	$\dfrac{1}{2}$
(D)	The derivative of $\dfrac{\log\lvert x\rvert}{x}$ at x = – 1 is	(s)	1

SECTION-5 (INTEGER TYPE)

967. Let f (x) = max$\{x^2, (1-x)^2, 2x(1-x)\}$, where 0 ≤ x ≤ 1. Then the integral part of area of the region bounded by the curves y = f (x), x-axis x = 0 and x = 1 is ___________

968. If g (x) = 2 + cos x cos $\left(x+\frac{\pi}{3}\right)-\left(\cos^2 x+\cos^2\left(x+\frac{\pi}{3}\right)\right)$ and $f\left(\frac{5}{4}\right)$ = 9, then the value of fog (x) is ______________

969. Let f (x) be a polynomial of degree 3 if the curve y = f (x) has relative extremities at x = $\pm\frac{2}{\sqrt{3}}$ and passes through (0, 0) and (1, – 2) dividing the circle $x^2 + y^2 = 4$ in two parts. Then the integral part of areas of these two parts is ______________

970. I = $\int_0^{1.5}\left[x^2\right]dx$ where [.] is greatest integer function then the value of [I] is ________

971. From a point A on the curve $x = 3y^2 - 2y + 7$, subnormal and subtangent are drawn. If they measure 1 unit each, distance of A from (4, 1) is ___________

972. The value of $\frac{8\sqrt{2}}{\pi}\int_0^1\left(\frac{1-x^2}{1+x^2}\right)\frac{dx}{\sqrt{1+x^4}}$ is _________

973. Let A be the area of the region bounded by the curve $a^4y^2 = (2a - x)x^5$ and B be the area of the circle whose radius is $\frac{a}{2}$, then $\frac{A}{B}$ is ___________

974. The area bounded by curves y = $\left[6+x\left[\frac{1}{x}\right]\right]$, $y^2 - 18x + 18 = 0$ and 6x – 5y – 6 = 0, (where [.] denotes the greatest integer function) is _________.

975. $\left[\int_{-1}^{\sqrt{3}}\frac{\sin^{-1}\frac{2x}{1+x^2}}{1+x^2}dx\right]$ = ____________; where [.] denotes G.I.F.

976. The value of $\lim_{x\to 0}\frac{16-16\cos(1-\cos x)}{x^4}$ is ______________

977. The altitude of a right circular cone of minimum volume circumscribed about a sphere of radius 2 is _________ .

978. $\int 7\left\{\frac{x^2(x^4-4x-3)}{(x^3-1)^2}\cos x-\frac{(x^4+1)}{(x^3-1)}\sin x\right\}dx$ = ___________.

979. The value of $\lim_{x\to\pi/2}12\tan^2 x\left[\sqrt{6+3\sin x-2\cos^2 x}-\sqrt{3+6\sin x-\cos^2 x}\right]$ is _______________

980. If $\int_0^{\pi/2}\frac{dx}{\left(\sqrt{\cos x}+\sqrt{\sin x}\right)^4}=A$. Then the values of 6A is ___________

981. A closed right circular cylinder has volume 2156 cubic units. The radius of its base so that its total surface area may be minimum is ___________.

982. If g(x) is a polynomial satisfying g(x) g(y) = g(x) + g(y) + g(xy) – 2 $\forall$ x, y $\in$ R and g(2) = 5, then the value of g(3) is ____________.

983. Let A be the area of the region in the first quadrant bounded by the x-axis the line 2y = x and the ellipse $\frac{x^2}{9}+\frac{y^2}{1}=1$. Let A′ be the area of the region in the first quadrant bounded by the y-axis, the line y = kx and the ellipse. The value of 9k such that A and A′ are equal is ____________

984. If A be the area bounded by y = f (x), y = f^{-1} (x) and line 4x + 4y – 5 = 0 where f (x) is a polynomial of 2nd degree passing through the origin and having maximum value of 1/4 at x = 1, then 96 A is equal to ______________

985. Let y = g(x) be the image of f(x) = x + sinx about the line x + y = 0. If the area bounded by y = g(x), x-axis, x = 0 and x = 2π is A, then $\frac{A}{\pi^2}$ is ___________

986. If $I_n = \int_0^\infty e^{-x}(\sin x)^n dx$ (n > 1), then the value of $\frac{101 I_{10}}{I_8}$ is equal to ___________

987. The number of solutions of the equation $\left[\sin^{-1}x\right] = x-[x]$, where [.] denotes the greatest integer function is

988. Find the value of a + c so that:

$$\lim_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2}-\sqrt{x^4+2x^3-cx^2+3x-d}\right)=4$$

989. Find the value of limits using expansion : $2\underset{x\to 0}{\text{Limit}}\left[\frac{\ell n(1+x)^{(1+x)}}{x^2}-\frac{1}{x}\right]$

990. Evaluate $3\underset{x\to 0^+}{\text{Limit}}\left\{\underset{n\to\infty}{\text{Limit}}\left(\frac{[1^2(\sin x)^x]+[2^2(\sin x)^x]+\ldots\ldots+[n^2(\sin x)^x]}{n^3}\right)\right\}$,

where [.] denotes the greatest integer function.

991. Evaluate the following limit $\underset{x\to\infty}{\text{Limit}}\log_{x-1}(x).\log_x(x+1).\log_{x+1}(x+2).\log_{x+2}(x+3).....\log_k(x^5)$;

where $k = x^5-1$.

992. Let $P_n = \frac{2^3-1}{2^3+1}\cdot\frac{3^3-1}{3^3+1}\cdot\frac{4^3-1}{4^3+1}\ldots\ldots\ldots\frac{n^3-1}{n^3+1}$. find the value of $\underset{n\to\infty}{\text{Limit}}\,6P_n$.

993. $\underset{n\to\infty}{\text{Limit}}\frac{(n+2)!+(n+1)!}{(n+3)!}$, $n\in N =$

994. If $f(x)=\begin{cases}x-1, & x\ge 1\\ 2x^2-2, & x<1\end{cases}$, $g(x)=\begin{cases}x+1, & x>0\\ -x^2+1, & x\le 0\end{cases}$, and $h(x)=|x|$ then find $\underset{x\to 0}{\text{Lim}}\,f(g(h(x)))$

995. Let [x] denote the greatest integer function & $f(x)$ be defined in a neighbourhood of 2 by

$$f(x) = \begin{cases} \dfrac{\exp\left\{(x+2)\dfrac{1}{4}[x+1]\ \ln 4\right\} - 16}{4^x - 16}, & x < 2 \\ A\dfrac{1-\cos(x-2)}{(x-2)\tan(x-2)}, & x > 2 \end{cases}.$$

Find the value of $A + 2f(2)$ in order that $f(x)$ may be continuous at $x = 2$.

996. Let the greatest and the least values of the function $f(x)$ be respectively a and b

$f(x) =$ minimum of $\{3t^4 - 8t^3 - 6t^2 + 24t; 1 \le t \le x\}, 1 \le x < 2.$

maximum of $\left\{3t + \dfrac{1}{4}\sin^2 \pi t + 2\,;\, 2 \le t \le x\right\}, 2 \le x \le 4$. Then find the value of $a + b$.

997. Find the area of the largest rectangle with lower base on the x-axis & upper vertices on the curve $y = 12 - x^2$.

998. The cosine of the angle at the vertex of an isosceles triangle having the greatest area for the given constant length ℓ of the median drawn to its lateral side is p. Find 100p.

999. The fuel charges for running a train are proportional to the square of the speed generated in m.p.h. & costs Rs. 48/- per hour at 16 mph. What is the most economical speed if the fixed charges i.e. salaries etc. amount to Rs. 300/- per hour.

1000. A figure is bounded by the curves, $y = x^2 + 1, y = 0, x = 0$ & $x = 1$. At a point (a, b), a tangent should be drawn to the curve, $y = x^2 + 1$ for it to cut off a trapezium of the greatest area from the figure. Find $2a + 12b$

JEE ADVANCED REVISION PACKAGE - Answer Key (MATHEMATICS)

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
1	A	51	AB	101	AD	151	A-(iii), B-(ii), C-(i), D-(iv)
2	C	52	AC	102	CD	152	A-(3), B-(2), C-(1), D-(4)
3	C	53	BD	103	ACD	153	A-(iii), B-(i), C-(ii), D-(iv)
4		54	ABCD	104	AB	154	A-(iii), B-(ii), C-(iv), D-(i)
5	C	55	BD	105	C	155	A-(R), B-(S), C-(Q), D-(P)
6	C	56	AC	106	C	156	A-(Q), B-(Q), C-(Q), D-(P)
7	C	57	ABC	107	B	157	A-(r), B-(s), C-(q), D-(p)
8	C	58	AB	108	C	158	A-(q), B-(p), C-(s), D-(r)
9	D	59	ABCD	109	B	159	A-(q), B-(p), C-(s), D-(r)
10	A	60	AB	110	A	160	A-(q), B-(p), C-(s), D-(r)
11	C	61	BD	111	C	161	A-(r), B-(p), C-(s), D-(q)
12	A	62	ABCD	112	D	162	A-(q), B-(r), C-(p)
13	C	63	CD	113	A	163	A-(p), B-(q), C-(r), D-(s)
14	A	64	ABCD	114	D	164	A-(q), B-(r), C-(s), D-(s)
15	D	65	AC	115	C	165	A-(q,s), B-(p,q,r), C-(p), D-(p,s)
16	B	66	ABC	116	D	166	A-(P), B-(S), C-(Q), D-(Q)
17	A	67	ABCD	117	B	167	A-(R), B-(Q), C-(S), D-(P)
18	B	68	ABC	118	C	168	A-(q), B-(p), C-(r), D-(r)
19	B	69	BC	119	C	169	A-(p), B-(p,q,r), C-(p,q,r,s), D-(p)
20	A	70	AC	120	D	170	A-(q), B-(s), C-(r), D-(p)
21	D	71	ABCD	121	A	171	15
22	C	72	AC	122	A	172	52
23	B	73	BC	123	A	173	50
24	D	74	BCD	124	C	174	3
25	C	75	ABD	125	D	175	1
26	B	76	ABC	126	D	176	729
27	A	77	ABC	127	A	177	4
28	B	78	ABC	128	C	178	7
29	C	79	BD	129	B	179	9
30	B	80		130	B	180	1
31	B	81	CD	131	C	181	210
32	A	82	ABCD	132	A	182	192
33	A	83	AB	133	A	183	142857
34	B	84	ACD	134	A	184	44
35	C	85	AC	135	C	185	0
36	B	86	AD	136	C	186	70
37	A	87	ABCD	137	A	187	(2, 1) or (2/5, -1/5)
38	A	88	ABCD	138	A	188	4
39	D	89	BC	139	B	189	16
40	B	90	AB	140	C	190	730
41	B	91	ABCD	141	A-(q), B-(q), C-(q), D-(p)	191	8315
42	C	92	ABCD	142	A-(ii), B-(iii), C-(iv), D-(i)	192	24
43	B	93	BC	143	A-(iii), B-(i), C-(iv), D-(ii)	193	40
44	D	94	AD	144	A-(ii), B-(i), C-(iv), D-(iii)	194	0
45	C	95	AD	145	A-(iii), B-(i), C-(ii), D-(iv)	195	1
46	D	96	AC	146	A-(ii), B-(iv), C-(ii), D-(iv)	196	n=9
47	C	97	BD	147	A-(v), B-(i), C-(ii), D-(i)	197	280
48		98	BCD	148	A-(iii), B-(ii), C-(iv), D-(i)	198	1260
49	C	99	CD	149	A-(iii), B-(iv), C-(i), D-(ii)	199	11
50	D	100	ABC	150	A-(I),(ii), B-(i), C-(iii), D-(iii)	200	744

Answer Key

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
201	A	251	AC	301	B	351	D
202	C	252	AB	302	C	352	C
203	B	253		303	D	353	D
204	B	254	AB	304	B	354	C
205	C	255	AB	305	B	355	A
206	C	256	BD	306	B	356	C
207	C	257	ABD	307	C	357	B
208	D	258	ABD	308	B	358	D
209	D	259	BD	309	A	359	B
210	C	260	ABD	310	A	360	A-(Q), B-(P), C-(S), D-(R)
211	D	261	ABCD	311	A	361	
212	B	262	AB	312	B	362	
213	D	263	CD	313	D	363	
214	A	264	BD	314	C	364	
215	A	265	AC	315	B	365	A-(S), B-(S), C-(Q), D-(P)
216	C	266	BC	316	B	366	A-(S), B-(R), C-(Q), D-(S)
217	C	267	ABC	317	B	367	A-(R), B-(Q), C-(S), D-(P)
218	B	268	BCD	318	D	368	A-(Q), B-(S), C-(R), D-(P)
219	A	269	AD	319	D	369	A-(P), B-(Q), C-(R), D-(P)
220	C	270	ABC	320	C	370	A-(q), B-(p), C-(s), D-(r)
221	C	271	AB	321	C	371	A-(s), B-(p), C-(q), D-(r)
222	A	272	BD	322	B	372	A-(r), B-(p), C-(q), D-(s)
223	A	273	BD	323	A	373	A-(s), B-(p), C-(q), D-(r)
224	B	274	ACD	324	C	374	A-(r), B-(p), C-(s), D-(q)
225	A	275	AD	325	C	375	A-(q), B-(r), C-(q), D-(q)
226	A	276	BC	326	B	376	2
227	A	277	ACD	327	C	377	1
228	B	278	BC	328	A	378	8
229	C	279	AD	329	D	379	13
230	B	280	AB	330	A	380	82
231	C	281	ABC	331	A	381	6
232	D	282	AB	332	B	382	98
233	A	283	AD	333	C	383	3
234	D	284	BC	334	C	384	156
235	C	285	ABCD	335	A	385	40
236	B	286	BC	336	A	386	18
237	A	287	AC	337	B	387	15
238	C	288	CD	338	A	388	55
239	A	289	AC	339	D	389	55
240	B	290	CD	340	B	390	8
241	A	291	ABCD	341	C	391	16
242	C	292	B	342	A	392	134
243	A	293	C	343	D	393	1
244	C	294	B	344	BC	394	2
245	B	295	B	345	A	395	3
246	ABC	296	B	346	A	396	27
247	ABCD	297	C	347	B	397	0
248	AB	298	D	348	D	398	8
249	BCD	299	B	349	B	399	1
250	AB	300	C	350	B	400	1

Answer Key

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
401	C	451	BC	501	A-(iii), B-(iii), C-(ii), D-(iv)	551	C
402		452	AC	502	A-(ii), B-(i), C-(iii), D-(iv)	552	A
403	B	453	BC	503	A-(ii), B-(iii), C-(ii)	553	D
404	C	454	BC	504	A-(iv), B-(v), C-(i), D-(ii)	554	D
405	C	455	BD	505	A-(ii), B-(iii), C-(i), D-(ii)	555	B
406	C	456	ABC	506	A-(ii), B-(i), C-(i), D-(iii)	556	AD
407	D	457	ABCD	507	A-(iv), B-(iii), C-(iv), D-(ii)	557	AB
408	B	458	ACD	508	A-(ii), B-(iii), C-(iii), D-(iii)	558	AC
409	C	459	ABCD	509	A-(2), B-(3), C-(3), D-(3)	559	D
410	A	460	ACD	510	A-(PR), B-(PR), C-(Q), D-(S)	560	AC
411	B	461	ACD	511	2	561	AB
412	C	462	BC	512	4	562	AD
413	C	463	AC	513	28	563	AC
414	B	464	BC	514	6	564	BC
415	C	465	AD	515	76	565	AD
416		466	A	516	5	566	AC
417	B	467	B	517	1	567	BC
418	A	468	C	518	52	568	AC
419	B	469	D	519	216	569	BC
420	A	470	D	520	15	570	AC
421	A	471	C	521	11	571	ABCD
422	D	472	C	522	3	572	AB
423	A	473	C	523	1	573	BD
424		474	D	524	1	574	BC
425		475	B	525	2	575	ABC
426	ABCD	476	D	526	2	576	B
427	ACD	477	D	527	7	577	A
428	AC	478	C	528	0	578	D
429	ABC	479	B	529	27	579	B
430	AD	480	B	530	200	580	A
431	AB	481	A	531	B	581	B
432	ABC	482	A	532	B	582	B
433	ABCD	483	B	533	C	583	C
434	ABC	484	A	534	D	584	C
435	AB	485	C	535	C	585	B
436	AB	486	B	536	A	586	D
437	ABC	487	A	537	D	587	D
438	ABD	488	A	538	B	588	
439	ABCD	489	B	539	D	589	
440	BC	490	A	540	A	590	
441	ABCD	491	B	541	C	591	C
442	AB	492	B	542	A	592	A
443	AC	493	B	543	B	593	B
444	BC	494	A	544	D	594	B
445	AB	495	C	545	C	595	A
446	AB	496	A-(iii), B-(iv), C-(i), D-(ii)	546	A	596	B
447	BCD	497	A-(iv), B-(iii), C-(i), D-(ii)	547	B	597	A
448	BD	498	A-(iv), B-(iii), C-(i), D-(ii)	548	A	598	B
449	AD	499	A-(iv), B-(iii), C-(ii), D-(i)	549	C	599	A
450	ABC	500	A-(iii), B-(iii), C-(ii), D-(iv)	550	B	600	A

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
601		651	C	701	AB	751	AB
602	A	652	A	702	BD	752	ABC
603	C	653	D	703	ABD	753	ABC
604	A	654	B	704	ABC	754	BC
605	C	655	A	705	AD	755	ABC
606	D	656	A	706	AD	756	AC
607	B	657	D	707	AC	757	AC
608	C	658	B	708	AB	758	ABC
609	B	659	B	709	AC	759	BC
610	C	660	B	710	AC	760	AC
611	D	661	C	711	AB	761	AB
612	C	662	A	712	BC	762	ABD
613	A	663	C	713	AC	763	AC
614	A	664	D	714	ABC	764	ABC
615	C	665	B	715	ABC	765	AB
616	B	666	A	716	AC	766	BC
617	D	667	B	717	AB	767	AC
618	C	668	A	718	CD	768	ABCD
619	B	669	C	719	ACD	769	BD
620	C	670	A	720	BC	770	BC
621	A-(ii), B-(iii), C-(iv), D-(i)	671	A	721	BD	771	BD
622	A-(i), B-(i), C-(iii), D-(ii)	672	C	722	AB	772	ABD
623	A-(ii), B-(ii), C-(iv), D-(iii)	673	B	723	AD	773	ABC
624	A-(ii), B-(i), C-(iv), D-(iii)	674	C	724	AB	774	ABC
625	A-(ii), B-(i), C-(iv), D-(iv)	675	B	725	ACD	775	BCD
626	A-(), B-(), C-(), D-()	676	B	726	ABC	776	ABC
627		677	B	727		777	AC
628	A-(1), B-(2), C-(1), D-(3)	678	B	728	AC	778	CD
629	A-(i), B-(ii), C-(ii), D-(ii)	679		729	AB	779	AD
630	A-(iv), B-(i), C-(ii), D-(iii)	680	B	730	AB	780	ABD
631		681	A	731	ABD	781	ABC
632	A-(iv), B-(iii), C-(ii), D-(i)	682	A	732	CD	782	AC
633		683	B	733	ACD	783	ABD
634	5	684	D	734	ACD	784	AC
635	3	685	A	735	ABD	785	AB
636	8	686	C	736	AD	786	ABC
637	16	687	C	737	ABC	787	AC
638	2	688	A	738	BC	788	CD
639	4	689	D	739	CD	789	BD
640	17	690	D	740	ACD	790	CD
641	1	691	D	741	BD	791	BC
642	90	692	D	742	AD	792	BC
643	3	693	B	743	AC	793	ABD
644	40	694	B	744	ABCD	794	AB
645	2	695	B	745	ABC	795	AD
646	6	696	C	746	ACD	796	ACD
647	4	697	C	747	BC	797	AC
648	4	698	C	748	BCD	798	AD
649	3	699	A	749	AB	799	BC
650	15	700	A	750	AC	800	ACD

Answer Key

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
801	ABD	851	C	901	A	951	A-(PS), B-(QRT), C-(PS)
802	BD	852	A	902	C	952	A-(T), B-(PQR), C-(S)
803	BD	853	A	903	A	953	A-(P), B-(R), C-(QT)
804	AC	854	D	904	A	954	A-(s), B-(q), C-(q), D-(s)
805	ABD	855	B	905	C	955	A-(s), B-(p), C-(s), D-(q)
806	AD	856	D	906	A	956	A-(q), B-(q), C-(q), D-(q)
807	ABCD	857	B	907	C	957	A-(q), B-(p), C-(r), D-(s)
808	BC	858	D	908	C	958	A-(r), B-(s), C-(q), D-(r)
809	AD	859	B	909	B	959	A-(r), B-(s), C-(q), D-(p)
810	AD	860	C	910	A	960	A-(r), B-(s), C-(p), D-(q)
811	AB	861	A	911	D	961	A-(s), B-(r), C-(q), D-(p)
812	ACD	862	A	912	B	962	A-(q), B-(r), C-(q), D-(s)
813	ABC	863	D	913	C	963	A-(q), B-(p), C-(s), D-(p)
814	ABD	864	B	914	D	964	A-(p,q,r), B-(p), C-(q,r,s), D-(p,q,r,s)
815	D	865	D	915	B	965	A-(q), B-(q), C-(r), D-(p)
816	C	866	B	916	A	966	A-(q), B-(p), C-(r), D-(s)
817	A	867	A	917	B	967	0
818	A	868	B	918	A-(iv), B-(i), C-(iii), D-(ii)	968	9
819	C	869	D	919	A-(iii), B-(iv), C-(i), D-(ii)	969	6
820	B	870	D	920	A-(iv), B-(iii), C-(ii), D-(iii)	970	0
821	C	871	C	921	A-(iii), B-(i), C-(iv), D-(iv)	971	4
822	C	872	A	922	A-(iv), B-(i), C-(ii), D-(iii)	972	2
823	C	873	C	923	A-(ii), B-(ii), C-(ii), D-(iii, i)	973	5
824	B	874		924	A-(v), B-(iv), C-(i), D-(ii)	974	11
825	C	875		925	A-(ii), B-(iii), C-(ii), D-(iv)	975	0
826	A	876		926	A-(ii), B-(i), C-(iv), D-(iii)	976	2
827	B	877	C	927	A-(C), B-(D), C-(A), D-(B)	977	8
828	A	878	B	928	A-(2), B-(1), C-(4), D-(5)	978	24
829	C	879	B	929	A-(P), B-(P), C-(P), D-(P)	979	1
830	B	880	A	930	A-(P), B-(Q), C-(R), D-(S)	980	2
831	A	881		931	A-(r), B-(p), C-(q), D-(s)	981	7
832	A	882	C	932	A-(s), B-(p), C-(q), D-(q)	982	10
833	A	883	D	933	A-(C), B-(A), C-(D), D-(B)	983	2
834	D	884	A	934	A-(D), B-(A), C-(B), D-(C)	984	34
835	C	885	B	935	A-(D), B-(C), C-(B), D-(A)	985	2
836	C	886	A	936	A-(D), B-(C), C-(A), D-(B)	986	90
837	B	887	D	937	A-(D), B-(C), C-(B), D-(A)	987	1
838	B	888	B	938	A-(P), B-(Q), C-(R), D-(Q)	988	7
839	C	889	C	939	A-(R), B-(R), C-(S), D-(S)	989	1
840	A	890	D	940	A-(R), B-(P), C-(S), D-(Q)	990	1
841	B	891	B	941	A-(R), B-(P), C-(S), D-(Q)	991	5
842	C	892	A	942	A-(RT), B-(RT), C-(RT), D-(PS)	992	4
843	D	893	C	943	A-(R), B-(Q), C-(R), D-(P)	993	0
844	D	894	C	944	A-(P), B-(P), C-(S), D-(R)	994	0
845	C	895	A	945	A-(S), B-(R), C-(R), D-(Q)	995	2
846	C	896	D	946	A-(R), B-(Q), C-(Q), D-(R)	996	22
847	B	897	C	947	A-(S), B-(R), C-(S), D-(Q)	997	32 sq.
848	C	898	A	948	A-(S), B-(S), C-(P), D-(P)	998	80
849		899	A	949	A-(P,R,T), B-(S), C-(Q)	999	40 mph
850	A	900		950	A-(S), B-(P), C-(Q), D-(S)	1000	16